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SCHOOL OF MECHANICAL AND AEROSPACE ENGINEERING (MAE)

MA4012 Mechatronics Engineering Design

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Group 2:

BALL FONDLERS

Submitted to:

DR JOHN HENG

Academic Year 23/24
Semester 2

Group Two

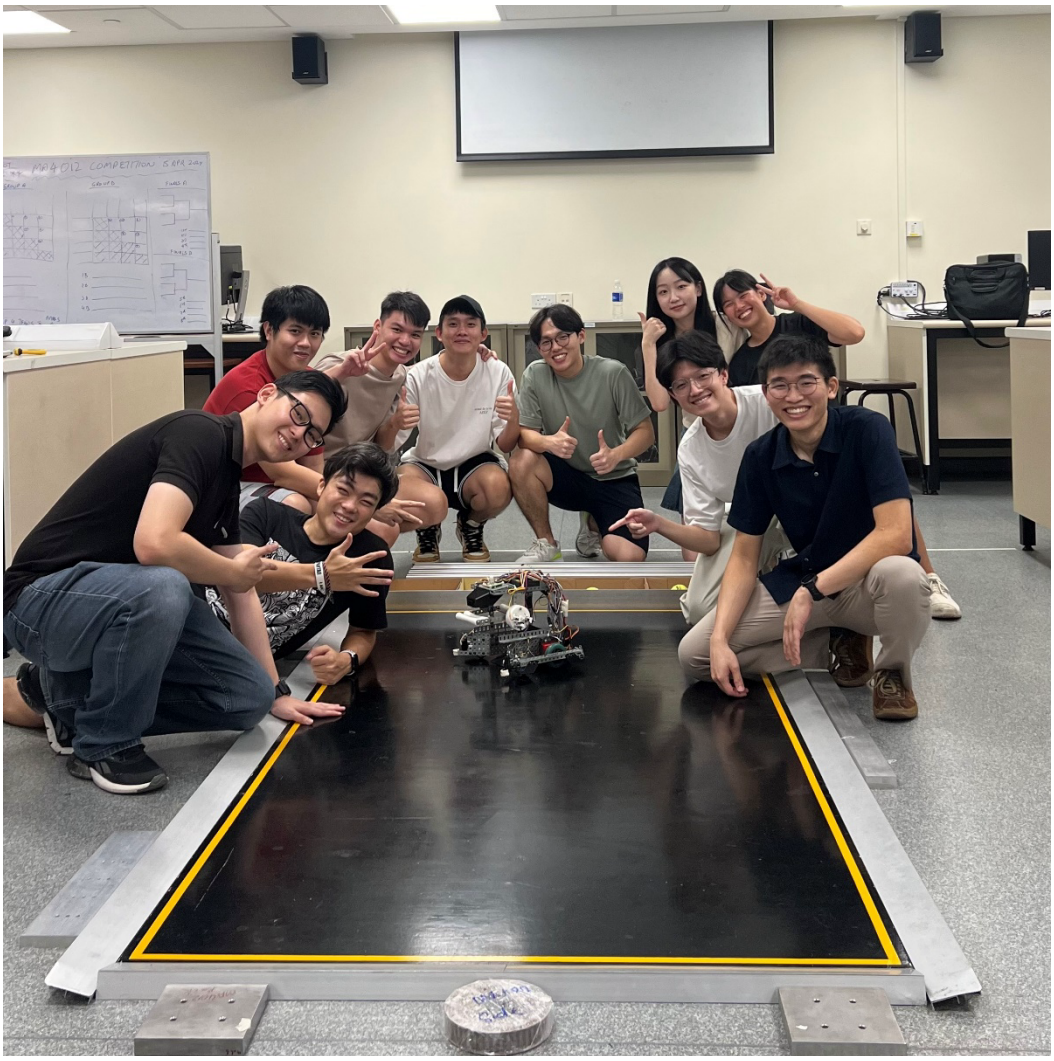

Figure 1. Group 2, from the left (Bryan, Lee Chen Hui, Wilson, Timothy Liu Yhin Sheng, Tan Hong Wei, Kelvin Lee, Fan Zifu, Eunice Ong Si Min, Wilsen Lotan, Benjamin Teh Jhen Hing)

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1. Introduction

1.1 Background information

Detection and retrieval systems are integral components of modern automation and robotics, enabling machines to interact with their surroundings. This project aims to construct an autonomous robot equipped with the ability to detect and retrieve a ball. The report showcased aims to illustrate a potential application scenario, whereas the concepts and overarching insights delineated can be extrapolated to broader use cases.

1.2 Objective

The primary objective of this project is to design, build, and program an autonomous robot capable of detecting and retrieving tennis balls within a designated arena. The vehicle should operate without human intervention, utilizing detection and retrieval mechanisms to navigate the playing field and collect balls within a specified time frame. The robot should ideally outperform its competitors by collecting and depositing a higher number of balls while navigating to avoid obstacles and remain within the playing field's boundaries.

1.3 Design Process

The design process adopted is shown in Figure 2below.

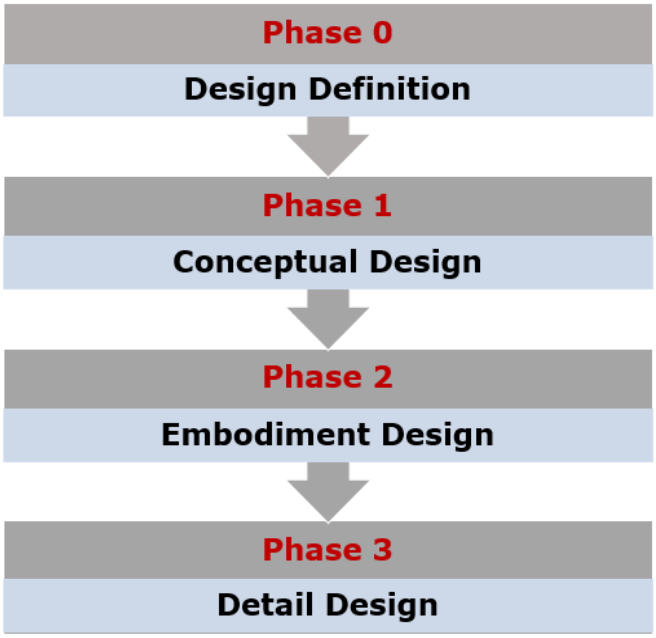


Figure 2: Design Process [1]

1.4 Project Timeline

Week (S2 23/24) **Date**	**Project Activity**
Week 1 15 Jan	-
Week 2 22 Jan	Forming groups Use of Miro Boards
Week 3 29 Jan	Mechanical Design Collection Design Strategy
Week 4 5 Feb	Mechanical Design Vehicle Design Drive System
Week 5 12 Feb	Celebrate CNY
Week 6 19 Feb	Electrical and Electronic Design Power Supply and sensor strategy
Week 7 26 Feb	Software Program Design Software Strategy for power, sensing, and navigation
Recess Week	Mechanical Assembly
Week 8 11 Mar	Sensor Assembly Mechanical Assembly Software Program
Week 9 18 Mar	Testing Software Program for motors and sensors
Week 10 25 Mar	Project Report Writing Software Program
Week 11 1 Apr	Testing Optimization Project Report Writing
Week 12 8 Apr	Trial Run Project Report Writing
Week 13 15 Apr	Competition
Week 13 21 Apr	Final Report Submission

Table 1: Project Schedule

1.5 Components Provided

S/N	Component	Qty
1	Vex ARM Cortex-based Microprocessor	1
2	Vex USB A-A Tether Cable	1
3	NiMH Battery Pack 7.2V, 3000 mAh	1
4	Battery Extension Cable	1
5	Vex Smart Charger and AC Power cord	1
6	Vex Gear Kit	1
7	High Strength Gear Kit, P/N: 276-2250	1
8	Wheel (Low friction)	4
9	Vex Metal and Hardware kit	1
10	Vex Continuous Rotation Motor (276-2163)	2
11	2-Wire Motor 393	2
12	Motor Controller 29	2
13	Sharp Distance Sensor (10 to 80 cm)	3
14	Sharp Distance Sensor (4 to 30 cm) 0A41SK	1
15	IR Line Tracking Module	4
16	Allen Keys (5/64’ and 3/32’) x4, 9mm Spanner x1, Screwdriver x1	6
17	Limit Switch	4
18	Digital Compass 1490	1
19	Arduino Mega + prototype board	1
20	Vex Shaft Encoder	2
21	Reflective Tape (Black Cardboard)	1

Table 2: List of Components

1.6 Component Visualization in Fusion

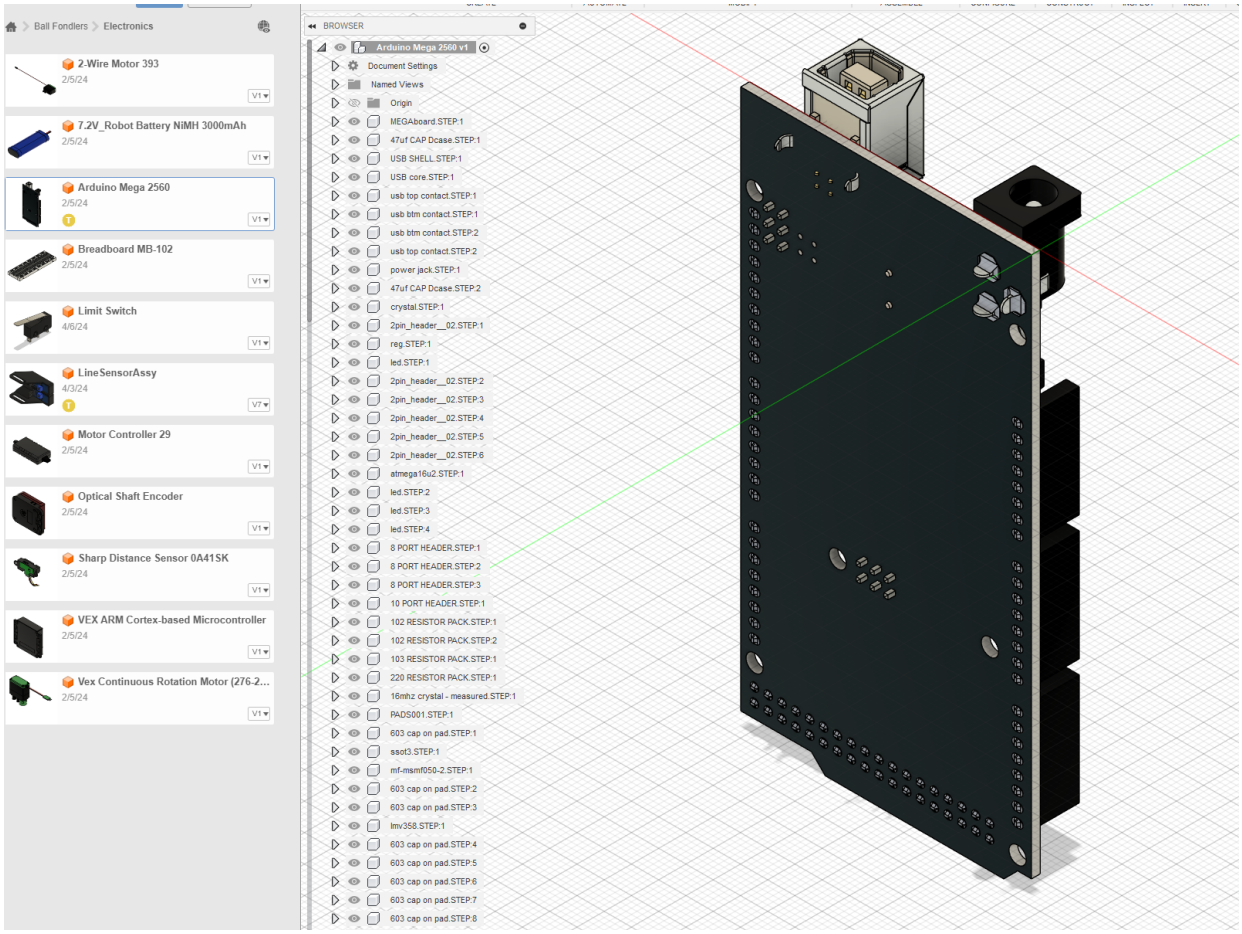


Figure 3 Screenshot of Fusion 360 sample model.

In our design process, we used Fusion 360 to import and utilize standard metal components, VEX robotics parts, and various electrical components to create the 3D model of our entire robot. This approach allowed us to visualize how each piece would fit together in the physical build, facilitating a cohesive and harmonized design. Fusion 360's cloud-based collaboration features enabled the team to work simultaneously from different locations, offering real-time updates and feedback on the design. Each component was positioned and examined within the virtual environment to ensure optimal fit, range of motion, and electrical integration. This digital assembly process not only improved the accuracy of the robot's design but also allowed us to conduct virtual testing and make immediate adjustments as needed, streamlining the development phase and significantly increasing our efficiency in arriving at the most optimal design that met all of our functional and performance requirements.

1.7 Competition Outcome

The Ball Fondlers Team achieved the 5th place out of 8 teams. This was due to issues during the round-robin phase.

In the first round the robot arm was not actuating to in a predictable manner. It was found that the housing of limit switch arm encoders was causing more friction than during testing due to misalignment. So, the controller of the swing arm was adjusted, and the housing of the limit switch arm encoder was reseated to fix the misalignment.

In the second round, again, the robot arm did not work to as it did during testing. The arm failed to swing correctly. It was found that one of the limit switch connectors was not seated properly on the VEX GPIO headers. This was fixed and the habit of checking if all the wires were seated in the GPIO headers is adopted for later rounds.

Then in the third and final round of round-robin, the robot prematurely released the ball when the opponent's robot triggered its two back limit switches. Subsequently, the condition to check for the back edge in addition to the triggering of the limit switch was added.

The culmination of these additions led to the team dominating the bottom league knock outs and securing the fifth place.

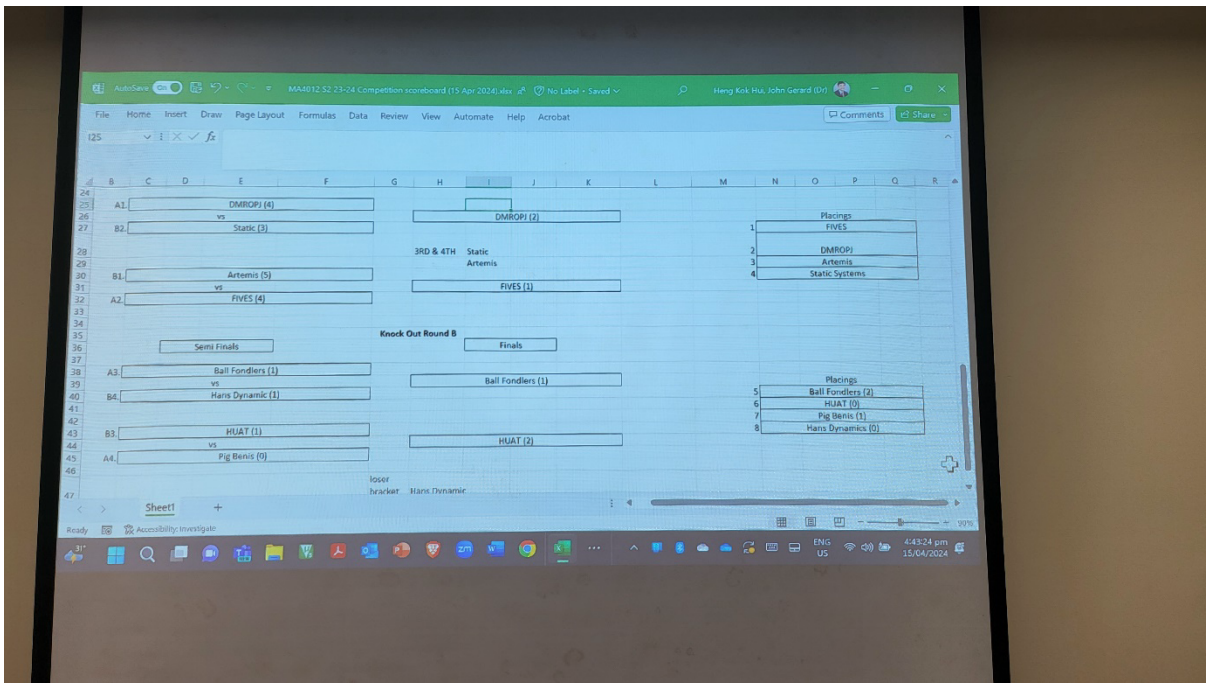


Figure 4 Competition Outcome

2. Design Definition

2.1 Stakeholders

Stakeholder Type	**Stakeholder**	**Implications of Design Consideration**
Internal	Mechanical Engineer	• Ease of assembly • Ease of repair, replacement, and adjustment of mechanical parts and robot structure • Feasibility of structure construction
	Electrical Engineer	• Ease of wiring • Ease of repair, replacement, and adjustment of wires and sensors • Ease of sensor testing
	Software Engineer	• Ease of programming • Ease of debugging and troubleshooting problem
	Robot Handler	• Ease of operation and charging
External	Judge	• Determine placement of tennis ball • Determine the design constraint • Determine the competition rules
	Opponent	• Reliability of robot during collision • Superiority of detection algorithm and pickup mechanism

2. 2 Design Specification

Task: Autonomous Ball Collector Vehicle	
Vehicle Description	An autonomous vehicle capable of performing main functions such as edge detection, target tracking, searching and retrieval.
Feature Description	- IR sensors deployed at the edges of vehicles to detect boundaries within 2 seconds. - Sharp sensors equipped to search for balls. - Collector mechanism available to retrieve the ball. - Releasing mechanism available to deposit the ball.
Vehicle Requirement	- Size smaller than 30cm x 30 cm x 30cm - Designed by more than half of the parts provided by the lab. - Power supply is limited to a 6-cell Ni-MH rechargeable

	battery pack. - Demonstrate good maneuverability. - No additional sensors, actuators or batteries are allowed. - No restriction on the quantity of limit switches mounted. - Not allowed to carry multiple balls at the same time
Additional Requirements	- A maximum of ± 3 cm drift over a 60cm vertical travel distance - Turning radius of vehicle is 12 cm. - Deposition of the height of the ball ranges from 5 - 10 cm - Size of the storage area must be larger than diameter of ball (7 cm) - Sufficient surface area for the collector mechanism to capture the ball. - Effective collection area for collector mechanism covers at least 20% of the robot base area. - Vehicle should run for at least 10 minutes continuously. - Ball at the yellow line should be collected instead of retreating the vehicle. - A minimum travelling speed of 5 cm/s - Ability to differentiate robot and ball within 60cm. - Ball deposition reliability at least 8 out of 10 attempts.

2.3 Design iterations

2.3.1 Ball Collecting Mechanism

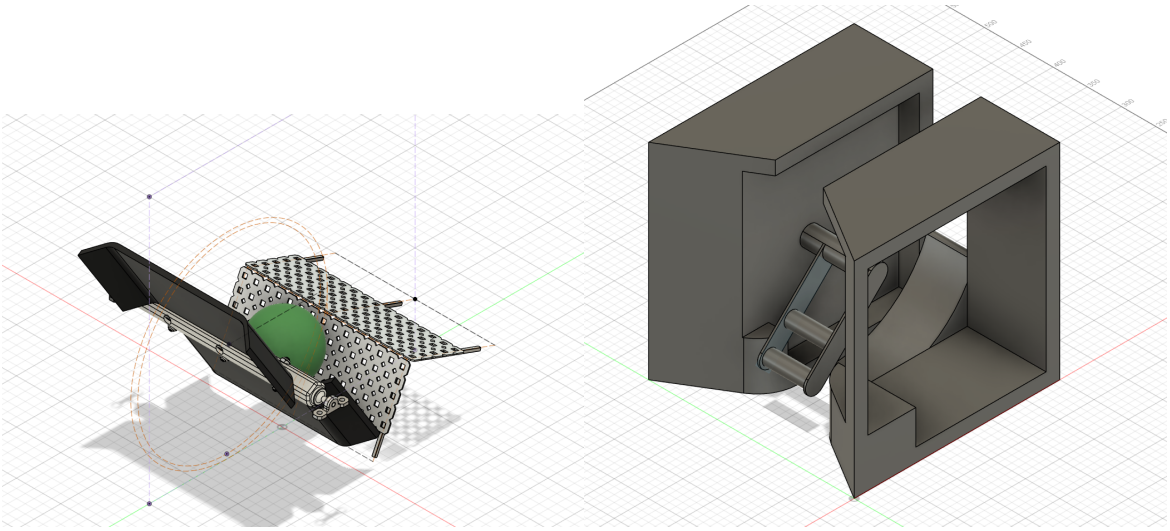


Figure 5: Ball collecting mechanism

To optimize our ball-collecting mechanism for efficiency and minimal weight, our team engaged in a comparative analysis of several designs, focusing on the quick and energy-efficient collection of tennis balls with the least possible strain on the servo motor. Using Fusion 360's robust simulation capabilities, each proposed design was evaluated for performance under dynamic conditions, which included the analysis of the energy consumption of the servo motor and the speed and reliability of the ball collection process. This process results in a democratic selection approach, where, armed with data and visual representations from the simulations, our group members voted on the design that best balanced speed, efficiency, and weight criteria, leading to a final, optimized ball collection mechanism.

2.3.2. Overall Design Visualization

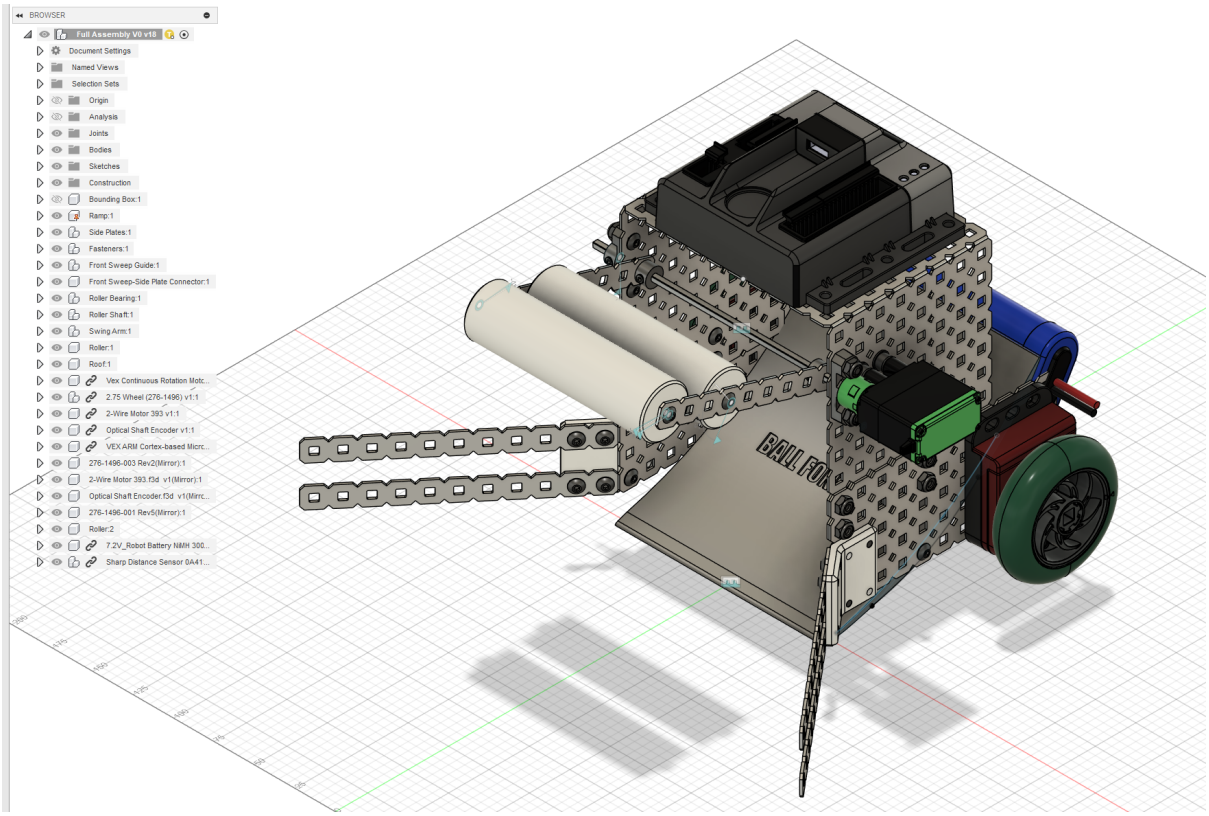


Figure 6: Overall robot design

Before finalizing the mechanical design of our robot, an initial visualization was rendered using Fusion 360, offering the team a comprehensive 3D model to evaluate form, function, and feasibility. This visualization served as a collaborative platform where team members could provide informed feedback, suggest tangible improvements, and collectively brainstorm innovative solutions to enhance the robot's design. Through this interactive review process, considerations such as accessibility for maintenance, structural integrity, component placement optimization, and weight distribution were meticulously scrutinized. The team's diverse expertise contributed to a multi-faceted improvement of the robot's mechanical systems, ensuring that the final design was not only technically sound but also aligned with our strategic objectives for performance and reliability.

3. Conceptual Design

3.1 Function analysis

3.1.1 Overall Functional Analysis

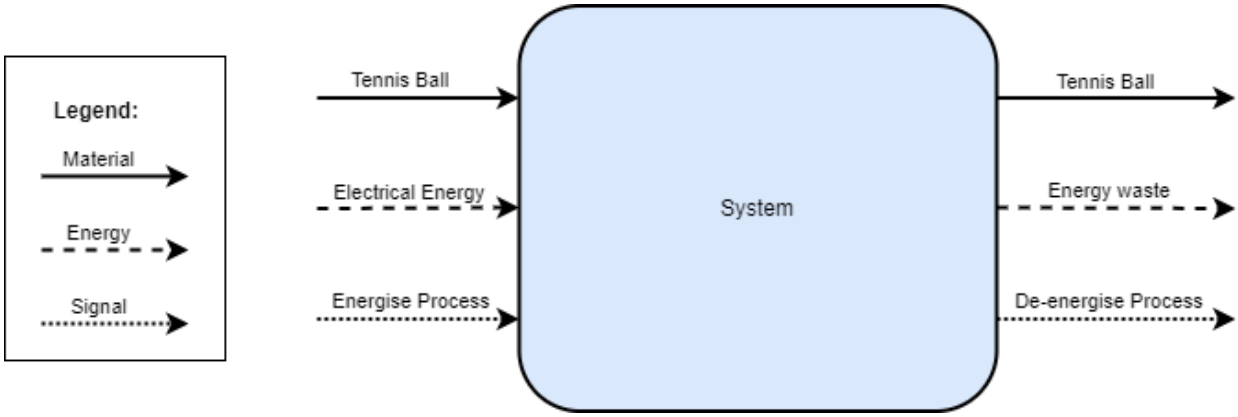


Figure 7: Functional Analysis Diagram

3.1.2 Mechanical Functional Analysis

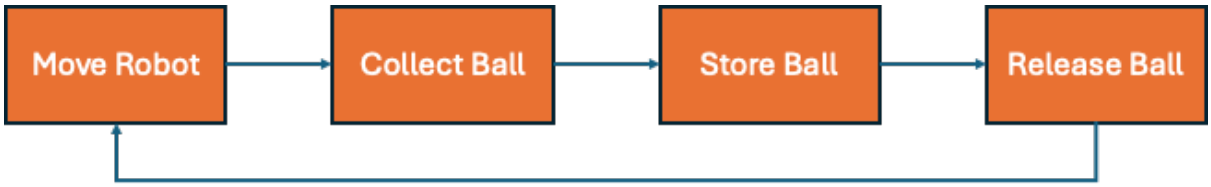


Figure 8: Mechanical function analysis diagram

Sub-Functions	**Explanation**
Move Robot	Robot will navigate to search for balls and subsequently retreat to the deposition area post-collection
Collect Ball	The collector mechanism will collect the ball
Store Ball	The ball will be stowed in the designated area until the robot arrives at the deposition site
Release Ball	The ball is discharged upon arrival at the deposition area

3.1.3 Software Functional Analysis

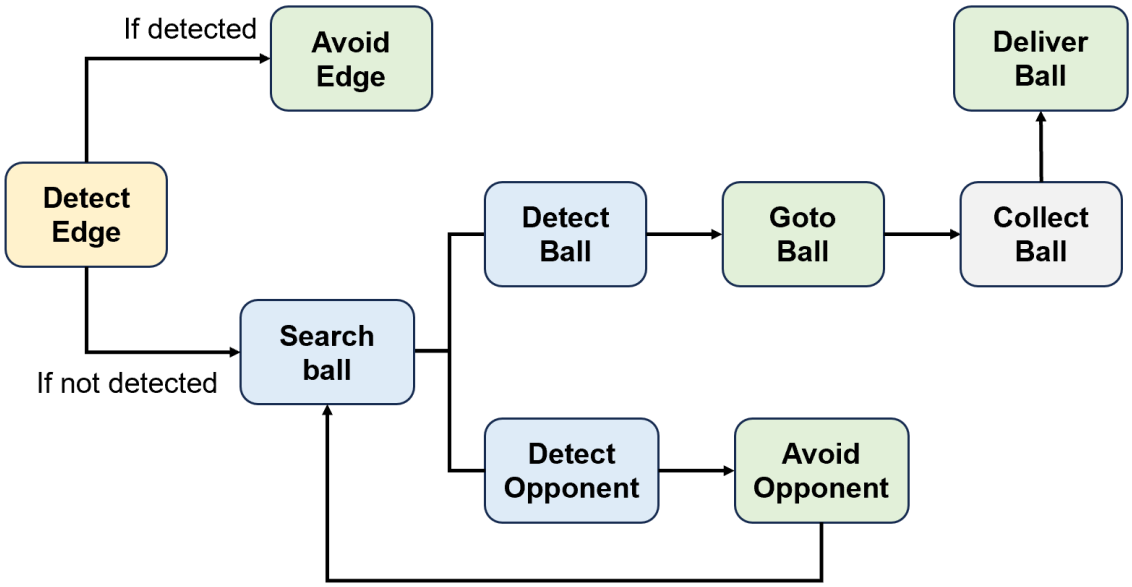
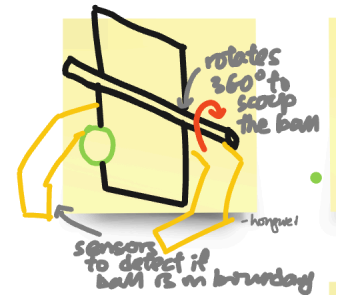
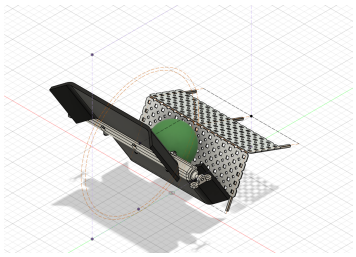
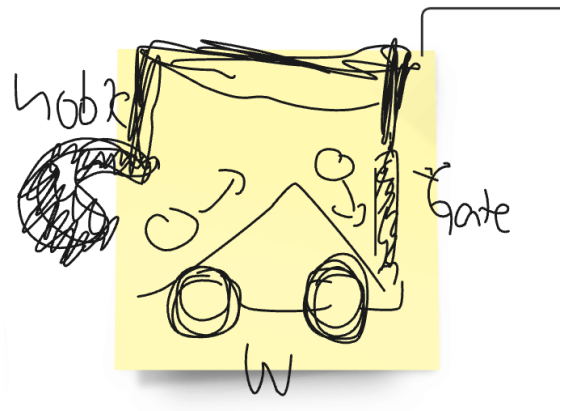


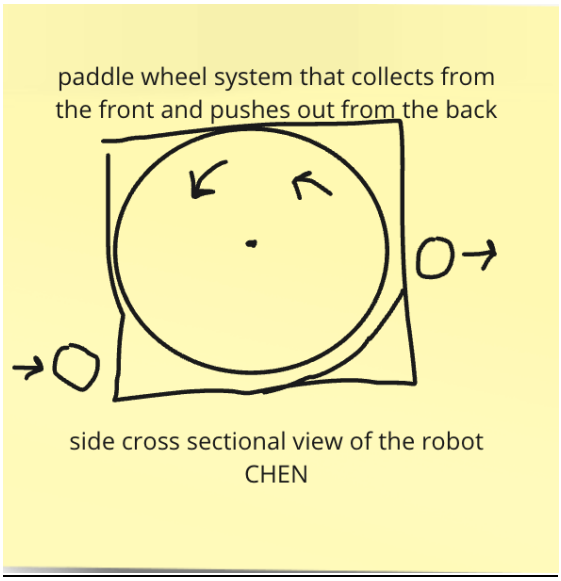
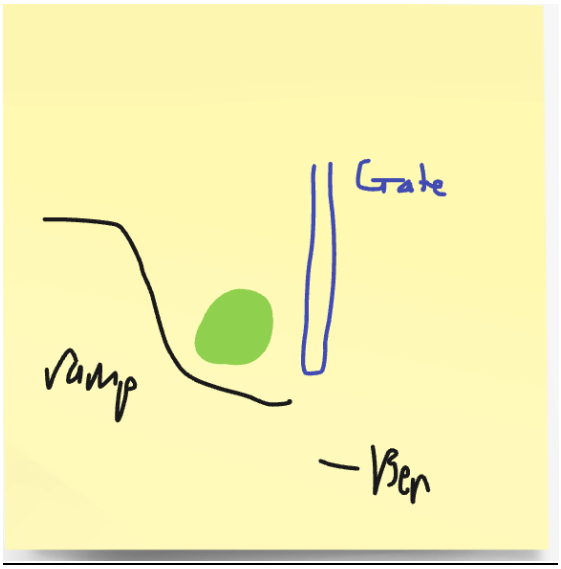
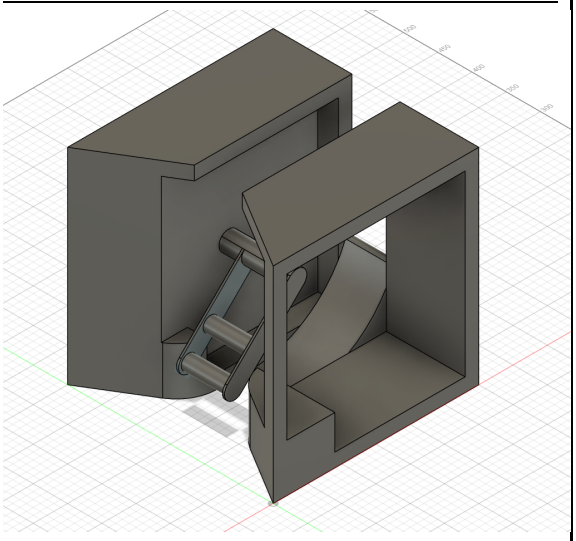
Figure 9 Flowchart of the Robot processes and priority function

Sub-Functions	**Explanation**
Initialize	Resets all parameters to zero and moves robot to the center of the arena.
Search for Ball	Scans the arena to detect the ball, ensures that it is a ball and not the opponent or wall before executing the next task.
Go to Ball	Aligns the robot directly in front of the ball, ready for collection.
Collect Ball	Captures ball and stores it in the chamber.
Deliver Ball	Ball is released upon triggering of back limit switches.
Edge Avoidance	To re-orientate robot in the event the IR edge sensors detect a line to prevent robot from exiting the playing area.
Opponent Avoidance	To detach itself or move away when opponent is detected before returning to the Search task.

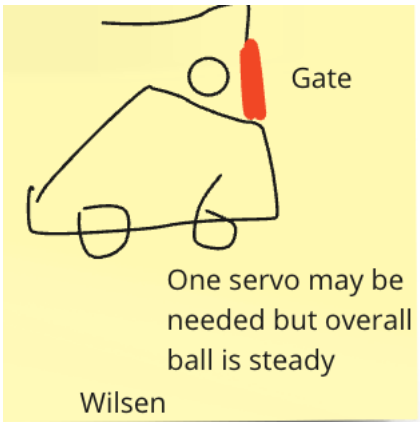
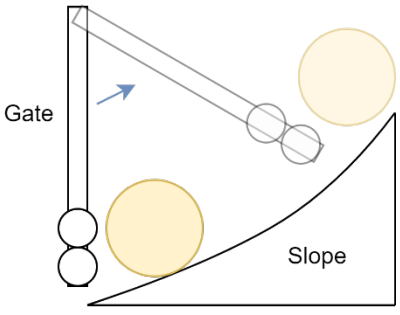
3.2 Morphological Chart

3.2.1 Ball Collector Design

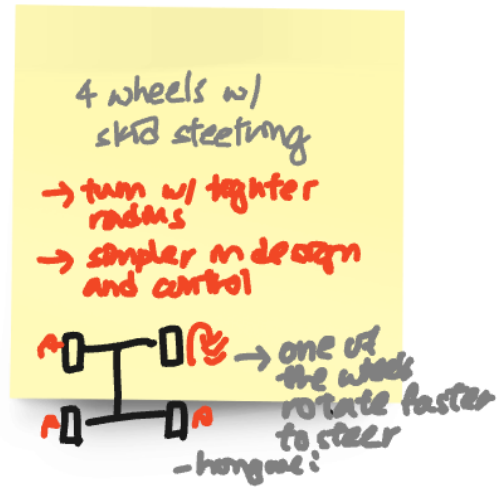
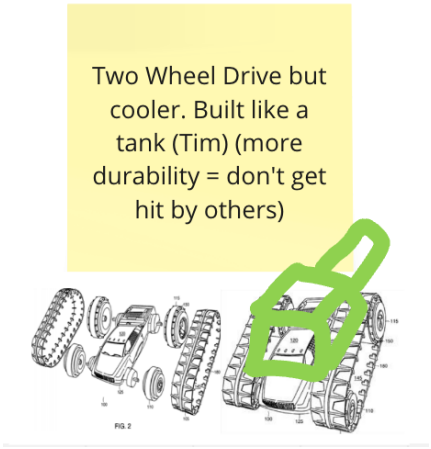
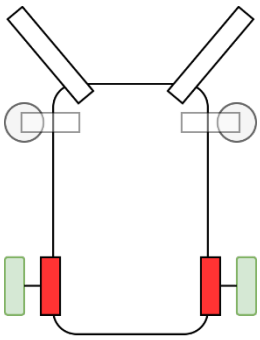
Ball Collector Design	**Description**
  Design 1: 360º Rotating platform	Rotating platform that acts as a lifting and scooping mechanism. Cons: • Rotating platform could push the ball away instead
 Design 2:	Hook to scrape and lift the ball. Cons: • Control mechanism is hard to design and might be inaccurate

	Roller wheel that collects from the front and pushes out from the back Cons: • Requires a large space
 	Rotating gate, like the rotating platform to design 1, but does not need to rotate 360º

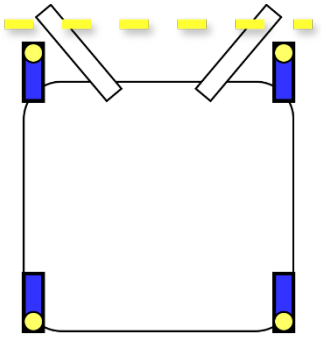
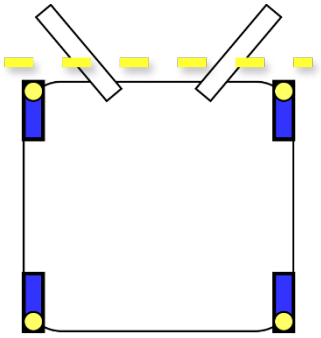
3.2.2 Ball Delivery Design

Ball Delivery Design	Description
	Need a servo motor for the ramp to raise one side of the platform so the ball can roll down. The torque needed may need to be higher compared to the other design as it needs to hold the weight of the platform and the ball.
	One servo motor needed to turn the gate so the ball can leave the ramp. Uses less torque compared to the dumpster truck design.
	One servo motor is needed to turn the gate from downward position to pushing up the ball leaving from the robot. The gate is equipped with rolling cylindrical tube to enable smooth delivery

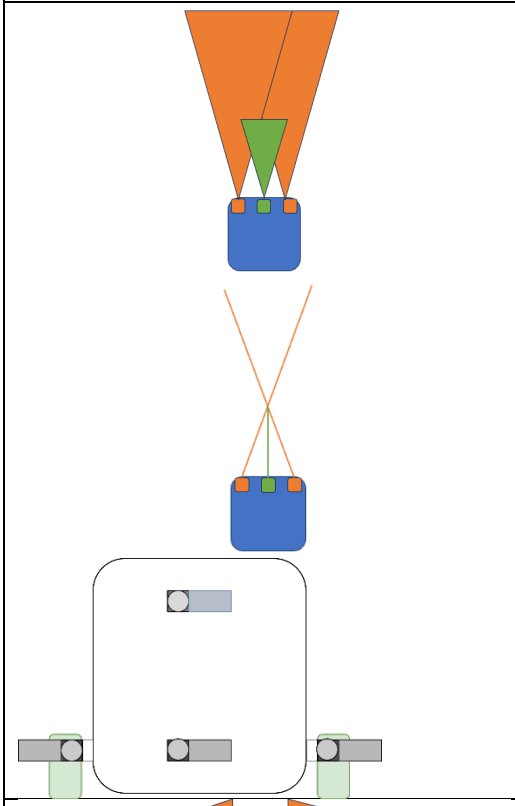
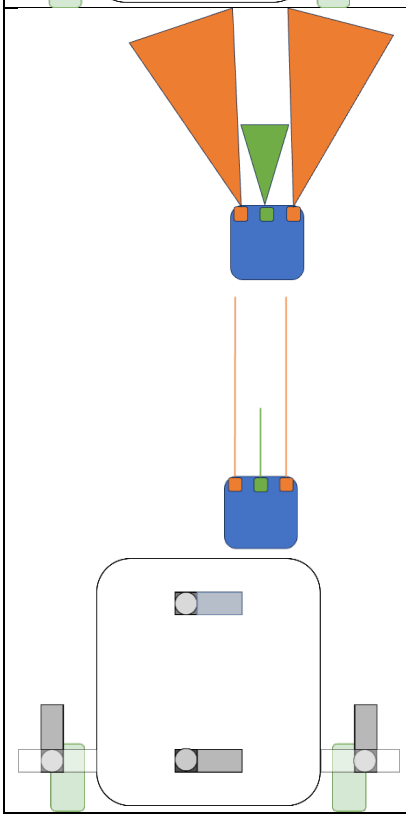
3.2.3 Drive System

Drive System Design	Description
	This drive system design revolves the full utilization of four-wheel system. The turning of the robot is done by controlling the rotational speed of one side of the robot to create speed differential resulting in a turning like motion. The advantage of this design is simpler control and higher stability.
	This drive system utilized four wheels with each side connected to a continues track. This enables the wheels to be hidden under the track and less prone of being hit by other robot or the ball. The disadvantage of this design is its higher friction and less accessibility for maintenance.
	This drive system design uses 2-wheel direct drive system with encoder coupled directly to the shaft of the wheel. This design was built with speed in mind and accuracy of movement. To achieve pivotal rotation, stability, and rotation, the wheel is placed near the back of the robot where much of the weight of the robot is. Additionally, two ball bearing wheel is added near the front of the robot

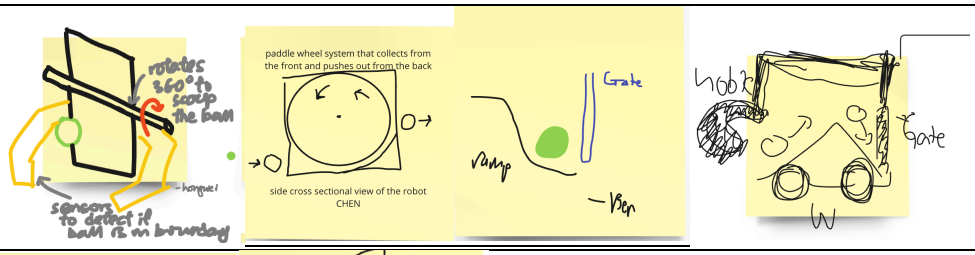
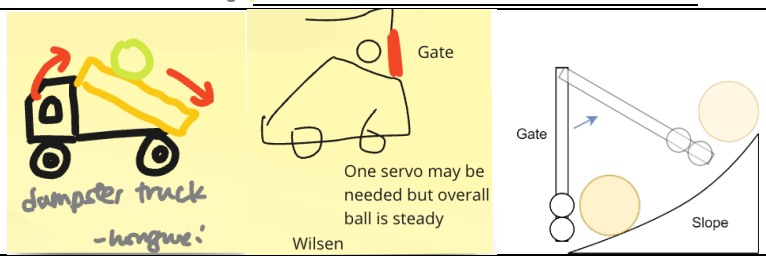
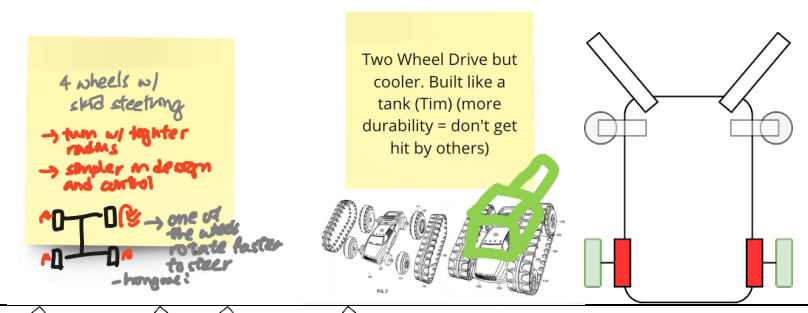
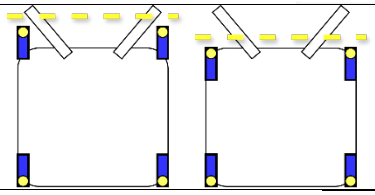
3.2.4 IR Sensor Placement

Line Tracking Sensor Placement	**Description**
	The placement of the IR line tracking sensor is placed on each corner of the robot profile to ensure the boundary can be detected before encroaching the side curb of the arena. Since the Line tracking sensor is rectangular in shape and the detection sensor is positioned on one end of the rectangle, the placement of the sensor was oriented in such that the detecting side is the most outward location of the sensor housing.
	The revised placement of the IR line tracking sensors was made to provide the robot space to collect the ball at the side curb and not detecting the line in precedence. Another design idea for the IR line tracking sensor was made to shift the front sensors. This was made to provide the robot space to collect the ball at the side curb and not detecting the line in precedence.

3.2.5 Sharp Sensor Placement

Sharp Distance Sensor	Description
	Two long distance sensor (10 cm to 80 cm) will be place on the left and right side of the robot to detect the ball and move accordingly. Another short distance sensor (4 cm to 30 cm) will be placed in the middle, when the ball is already right ahead, the robot can move straight to collect the ball. Two different IR path are shown (top and bottom) as in the initial iteration, the way the sensor detect objects are still unknown, in fact, the real path of the IR is a combination of straight line and parabolic line. Assuming that the IR path is conical as shown in the left, there will be parts in which they overlap, this way, less chance for blind spot in front of the robot.
	The placement of the sensor for the second design is like the first. The difference is in the path assumed taken by IR to reach the object. In this case, there will be blind spot if the ball is located straight in front of the robot but far away, the green sensor can only detect up to 25 cm. If the ball is beyond 25 cm in front, then the sensor will not be able to detect it.

3.2.6 Combined Morphological Chart

Component	
Ball Collection	
Ball Delivery	
Drive System	
Line IR Sensor	
Sharp Distance Sensor	

3.2.6.1 Design Idea 1

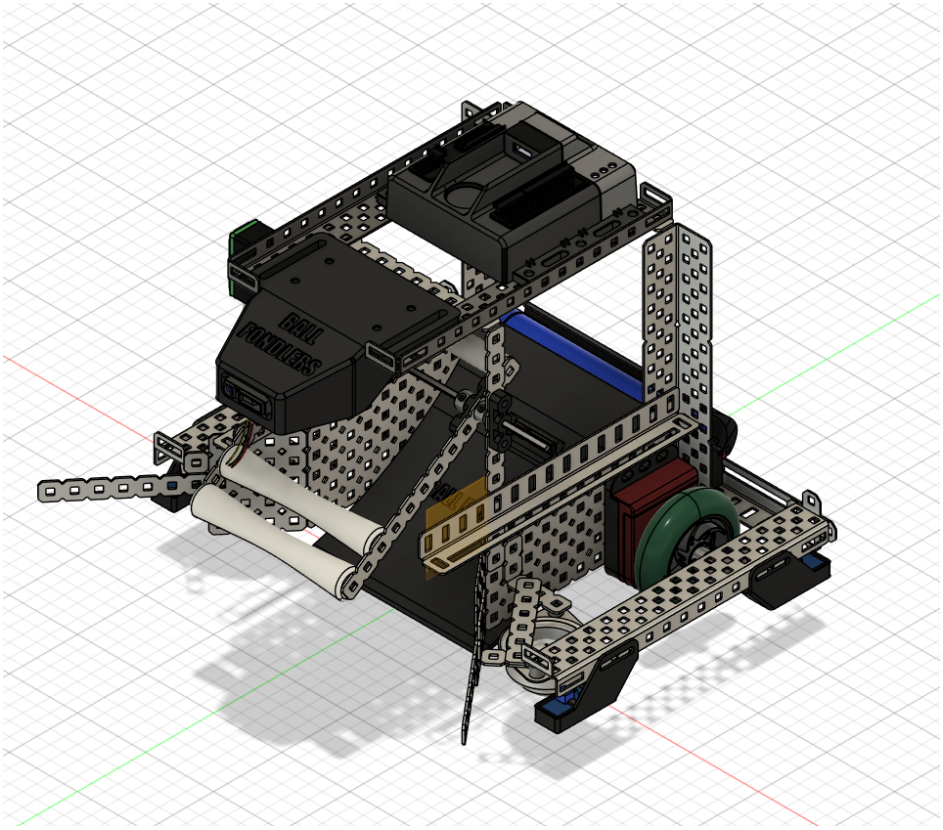


Figure 10: Design Idea 1

Component	Chosen Selection	Description
Ball Collector	Rotating 90° Gate system	The arm of the collector acts as a gate and swings from an open position to a closed position which are 90 degrees from one another.
Ball Delivery	Rotating Gate system	Rotating arm that acts as a lifting and scooping mechanism, movement is limited to 180 degrees.
Drive System	2 Wheel Direct Drive	This two wheeled drive system is a differential drive system. This configuration enabled most of the weight of the robot to be close to the wheels. This is to increase traction.
IR Sensor	4 IR within robot frame, with front allowance.	This placement of the line tracking module allows us to detect edges, while giving a front allowance to give some room to collect a ball at the

		edge, or near to the edge. This is at the cost of having delayed detection of a line in front of the robot.
Sharp Sensor	2 Sharp sensors facing straight	This configuration was selected with the consideration to maximize the search area and minimize blind spots. It is also chosen to keep the robot updated on the whereabouts of the robot relative to its orientation as it moves to approach the ball for a reactive-type algorithm. For example, when the ball detected on the right, slowly turn right while moving forward. Then, if the ball is now detected on the left due to too much turning, it will slowly turn left while moving forward, resulting in the robot getting closer to the ball. This logic removes the need for the robot to pinpoint exactly where the ball is but might require slightly more time for the robot to get to the ball even after adjusting the speed of the turning vs the speed of moving forward.

3.2.6.2 Design Idea 2

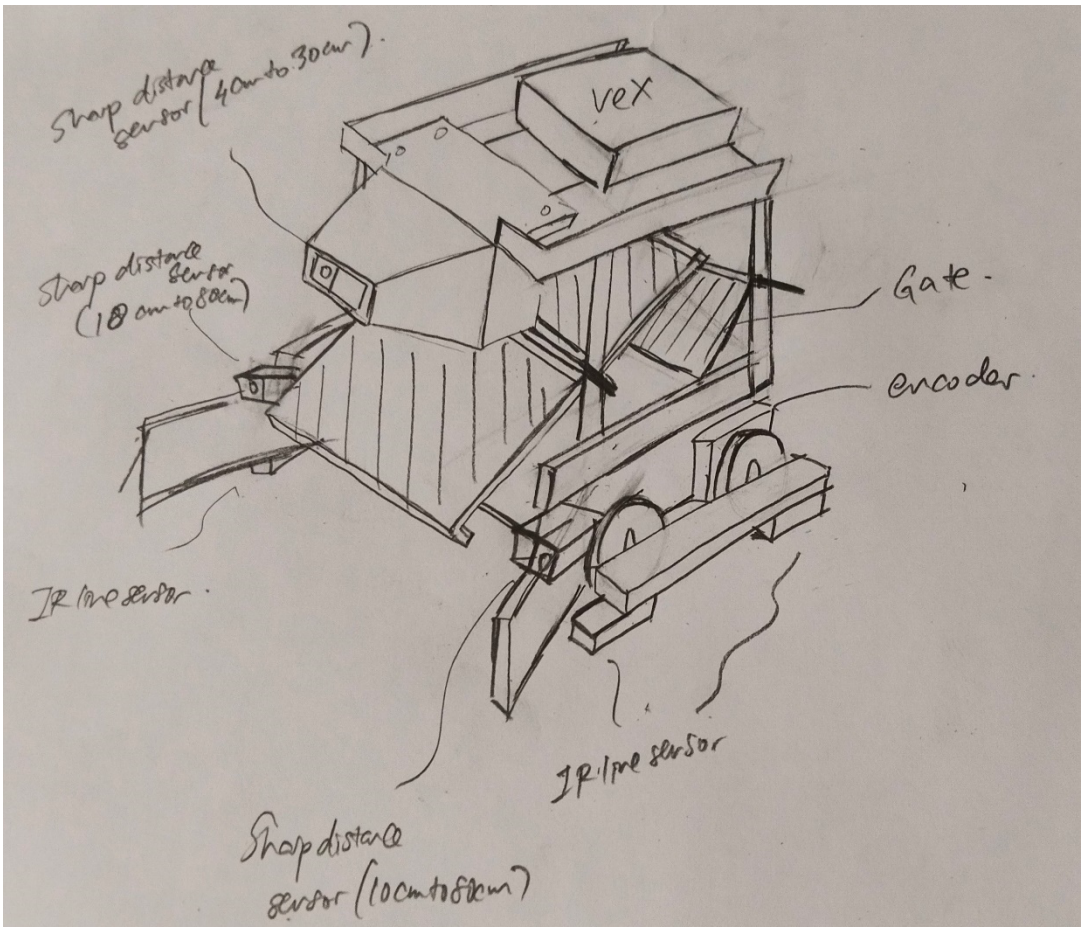


Figure 11: Design Idea 2

Component	Chosen Selection	Description
Ball Collector	Rotating Paddle System	There are two paddles attached to a central hub at 180 degrees from each other. This configuration would make this component self-balancing. This paddle would rotate in single direction to both collect and deliver the ball, as well as reset the collection.
Ball Delivery	Rotating Gate system	This mechanism was chosen due to its simplicity of the design by rotating the gate to release the ball when tasked to do so. A servo motor would be used to control the rotation of the gate where a limit switch is attached to the gate to check if the ball is in the chamber.

Drive System	4-wheel drive system	The 4-wheel drive system would be a skid steering system where both wheels on each side would rotate with the same velocities. This may also contribute to greater traction and load distribution of the robot.
IR Sensor	4 IR Line sensor placed at the outmost exterior of the robot	This placement of the line tracking module allows us to avoid edge at the earliest fashion. This placement of the line tracking however will prevent the collection of balls at the corner without a comprehensive software system.
Sharp Sensor	2 Sharp sensors facing diagonal.	The two sharp sensors are angled such that their line of detection intersects to allow for precise pinpoint of the tennis ball's location. As the robot rotates to search for the ball, the ball is detected when it intersects the point where the two sensors meet. However, the sensors are not very accurate which may result in inefficient detection of the ball.

3.3 Concept Evaluation

3.3.1 Evaluation Criteria Weight

Evaluation Criteria	**Details**	**Weightage (%)**
Ball collection capability	Ability to detect and collect the ball from arena.	25
Ball delivery reliability	Ability to detect the release point and release the ball from the robot	20
Search reliability	Ability to detect and differentiate ball with wall and move towards the ball	20

Line and wall detection	Ability to detect the line located near the boundary of the arena and also the arena's wall reliably and quickly	20
Complexity	Ease of building, parts replacement, and programming.	5
Opponent detection	Ability to detect opponent and avoid	5
Push resistance	Ability to withstand collisions without being pushed back by opponent	5

3.3.2 Evaluation Table

Evaluation Criterion		**Weightage (%)**	**Criteria Rating for Concept**	**Idea 1**		**Idea 2**	
				Value	**Weighted Value**	**Value**	**Weighted Value**
1	Ball collection capability	25	Scale of 1 to 5 1 - worst 5 - best	5	125	3	75
2	Ball delivery reliability	20		5	100	3	60
3	Line and wall detection	20		5	100	4	80
4	Search reliability	20		4	80	3	60
5	Complexity	5		2	10	3	15
6	Opponent detection	5		4	20	3	15
7	Push resistance	5		3	15	4	20
Total				28	350	23	325

3.3.3 Rationale Behind Results

3.3.3.1 Ball collection capability

The ball collection capability of our robot was optimized through a comparative analysis of various collection strategies. Our goal was to identify a method that could perform the dual functions of not only collecting the tennis balls but also securing them steadfastly once collected. The chosen strategy involved a roller system that effectively draws the ball into the robot's grasp and then, utilizing a combination of mechanical grip and strategic sensor placement, ensures that the ball is held securely. This method proved to be highly efficient as it consolidated multiple tasks into a single, streamlined device, enhancing the robot's operational effectiveness, and minimizing the need for additional components.

3.3.3.2 Ball release reliability

For the ball release mechanism, our design ethos centered on simplicity and compactness. We leveraged the natural force of gravity, designing the robot so that once the roller contracts, the collected tennis balls can roll off effortlessly from the top of the slope. This simple yet reliable method negates the need for complex release mechanisms. To maintain a compact and modular design, we innovatively utilized the battery's positioning, crafting it into a slope that guides the balls out of the robot. This not only ensures that the ball exits the robot without hindrance but also contributes to the robot's streamlined profile, allowing for efficient use of space and resources.

3.3.3.3 Search reliability

The sensor placements were selected to maximize range and accuracy of detection and to reduce complexity of processing the sensor information. With 3 sensors placed on the left, right, and the middle of the robot, facing to the front of the robot in a straight line, the robot can reliably detect any object in front of it, removing scenarios whereby the robot might not detect a ball in front of it and knocks it away instead. Additionally, since our distance sensor is not very accurate in telling us about the exact distance between the robot and the detected object, using an algorithm that pinpoints the exact location of the ball and goes to it is prone to errors. Therefore, the accuracy of the detection is also better with this sensor configuration as it enables the robot to adaptively react to the present situation rather than relying on past knowledge. Finally, an adaptive approach to searching the ball also reduces the complexity of the logic, making it into a simple "if detect on the right, go a little to the right, if detect on the left, go a little to the left, if detect in the middle, just go straight".

3.3.3.4 Line and wall detection

The placement of the IR sensors was important to ensure quick detection and reaction to move away from the line before the robot drives over the wall. Hence, the IR sensors were placed at the back two corners of the robot to ensure that the sensors would be the first part of the robot crossing over the line. This helped to allow for a quick reaction of the robot to re-orientate itself away from the wall. The front two sensor are placed further inwards to enable the collection mechanism to collect balls from the edges. In terms of safety, placement of the sensors further inwards prevents it from damage during collisions. Therefore, the sensor layout 1 was selected.

3.3.3.5 Complexity

For the complexity, the robot is in essence not very hard to build and assemble. Part replacement is not impossible, but it is not easy and will require time for some of the parts. The choice of 2-wheeled drive instead of 4-wheels is to reduce the complexity of the drive system so that power loss is kept to a minimum, and performing repairs on the robot would be faster.

3.3.3.6. Opponent detection

For our opponent detection, we relied on a short-range sharp distance sensor mounted atop our robot. This allows us to detect object which are located higher than the ball height, resulting in an identifiable opponent detection. We built this in mind believing that most of the opponent robot is structured to a certain height.

3.3.3.7 Push resistance

Push resistance of the robot is contributed largely by the drive system but also the collection and delivery systems. The drive system determines the amount of traction the robot would have where a 4-wheeled system would more likely have more traction than a two-wheeler. However, with the weight distribution of the collection–delivery system, and the electronics, the two wheeled configuration enables the mass to be concentrated on the driving wheels.

3.3.4 Concept Finalization

As seen in the Evaluation Table, Idea 1, accrues the most points overall and thus is the selected design for this project. Further discussions about this design will be explored in the embodiment design section.

4. Embodiment Design

Refining the selected layout from Chapter 3 into a stronger and improved design is crucial in embodiment design. This will be accomplished by adhering to the three rules and four principles of embodiment design.

4.1. Rules of Embodiment Design

4.1.1. Clarity

4.1.1.1. Function

The function of each component must be clearly defined and unambiguous to prevent failure.

Components	**Function**
Wheels	Provide motion for the robot
Collector Mechanism	Collect the ball and transfer it to the storage area

Ramp	Establish a connection from the collection area to storage
Storage Area	Store the ball before release
Sharp Sensor	Detect the ball and potential opponents
Infrared Sensor	Identify the yellow line boundary
Limit Switch	Connected with cam, acts as a positional encoder. Determine if the delivery area is reached
Compass	Offers exteroceptive sensory of orientation for the robot
Battery	Supply power to the robot
External Frame	Provide structural support to the robot

4.1.1.2. Layout

The design outlined in subsequent sections must clearly delineate the arrangement of the robot's components and the interfaces among them.

4.1.2 Simplicity

4.1.2.1. Mechanical Design

In the mechanical design of our robot, simplicity was embraced as a guiding principle to streamline production, reduce complexity, and enhance functionality. We meticulously refined the robot's frame, minimizing extraneous elements to accommodate a robust bumper system. This system fortifies the robot against impacts and collisions, fitting snugly within the competitive confines of 30x30x30 cm, and protects all vital components including sensors, wheels, and motors. Simultaneously, we maximized the internal volume dedicated to the ball collection area. This strategic expansion ensures that once within the robot's reach, the ball is efficiently guided and secured for transport, capitalizing on every available cubic centimeter to enhance the robot's ball-handling capacity within the regulations. This approach to simplification resulted in a robot that is both resilient in structure and proficient in its primary task, demonstrating that simplicity in design can yield a highly effective competitive machine.

4.1.2.2. Electrical Design

Implementing color-coded wiring for each component minimizes ambiguity in electrical connections. This facilitates easy understanding and adjustments during rewiring and troubleshooting. Electrical tapes are also added in between wires to make sure the connection between wires is not easily disconnected.

4.1.2.3. Software Functions

Software functions should be intuitive and easily achievable. Breaking down main functions into sub-functions enhances modularity and comprehension.

4.1.3. Safety

Safety stands as the building block of design considerations, ensuring the robot's reliability. Design elements such as robustness, reliability, ergonomic considerations, and accident prevention measures are essential. Employing Failure Modes and Effect Analysis (FMEA) aids in identifying and mitigating potential failure modes and their effects.

Furthermore, specific safety measures are taken:

- The overall robot components are shielded with additional metal parts.
- The external frame, constructed from metal parts, provides structural integrity to withstand collisions.
- Electrical components are strategically positioned within metal frames to shield them from collision damage.
- Neat wiring arrangements prevent tangling and potential damage during encounters with opponents.
- 3D printed mount to elevate the compass to mitigate magnetic interference from metal plates.

4.2. Principles of Embodiment Design

4.2.1. Force Transmission

4.2.1.1. Force Flowlines

The corner line sensors were identified as vulnerable in our initial robot design, particularly susceptible to environmental stresses and competitive interactions. To enhance their resilience, we iterated the design to include custom housings with rounded, filleted edges. These housings effectively encapsulate the sensors, substantially reducing stress concentration at the corners and providing a buffer against accidental impacts. By integrating these housings into the robot's framework, we significantly increased the durability and reliability of the sensors, ensuring they remain operational even in the most challenging conditions.

For the top-mounted distance sensor and the battery mount, our design iterations introduced dedicated solutions to increase the robot's functionality and survivability. The distance sensor was equipped with a protective housing that allows it to perform its sensing function without being knocked askew by other robots. The battery mount was re-engineered to be secure yet

detachable, enabling easy maintenance and replacement while also serving as a structural element guiding tennis balls into the collection area. This multi-functional approach not only secured the battery against the rigors of the match but also smoothed the path for ball collection, enhancing the robot's operational efficiency.

Finally, to protect the core structural and operational components like the encoders, wheels, and frame, we designed the original robot frame to be smaller than the overall dimensions of the robot, incorporating a bumper system that envelops the entire robot. This strategic design ensures that the surrounding bumper absorbs and dissipates the force in the event of a collision, shielding the robot's body and internal mechanisms from damage. The bumper acts as a sacrificial layer, taking the brunt of any impact and preserving the integrity of the robot's critical components, thereby maintaining its performance, and extending its service life in the competitive arena.

4.2.1.2. Direct and Shortest Force Transmission Path

In optimizing the robot for efficient force transmission, we adhered to the principle of creating the most direct and concise paths for force flow throughout its structure. By meticulously aligning components, such as the encoders and motors, along the central axis and close to the drive system, forces are transmitted with minimal deviation, circumventing the possibility of mechanical inefficiency, and reducing potential failure points. This streamlined force pathway, evident in the compact design of the robot's frame and the strategic placement of its core components, not only reinforces the structural integrity of the robot but also minimizes energy dissipation. Consequently, this leads to a robot that utilizes its power more effectively, ensuring both heightened performance and greater overall system reliability.

4.2.2. Division of Tasks

4.2.2.1. Division of Tasks for Different Functions

In our robot's embodiment design, we meticulously allocated specific functions to each component to avoid redundancy and ensure operational efficiency. The top-mounted distance sensor strategically scans for the presence of other robots, while the quartet of line sensors, positioned at each corner, continuously monitors the boundaries of the competition arena. This placement allows for precise navigational adjustments in real-time. Enhancing object detection capabilities, two additional distance sensors are mounted on the front corners, tuned specifically for tennis ball detection, ensuring that our robot can approach and position itself optimally for ball collection.

Within the ball collection mechanism, a distance sensor is embedded in the slope, designed to confirm the ball's presence and secure retention as it ascends within the robot's frame. Complementing this, limit switches are integrated with the rollers to provide real-time feedback on their position, indicating whether the collection gate is open or closed. This

sensor-switch synergy is vital in ascertaining that once a tennis ball is collected, it is firmly grasped by the rollers and ready for transport to the delivery point. This division of sensory tasks, with each element performing a distinct and vital role, harmonizes the robot's interactive functions and ensures a smooth, coordinated operation during the heat of competition.

4.2.2.2. Division of Tasks for Similar Functions

The gate's operation in securing and releasing the tennis ball is governed by a strategically placed servo motor, which is integral to the robot's ball handling capability. Once the ball is collected by the gate, the servo motor actuates a holding mechanism that clamps down on the ball, ensuring it is held securely in place. This firm grip is meticulously controlled to maintain possession of the ball as the robot navigates the field. The servo motor's precision allows for the ball to be retained without risk of slippage or accidental release, right up until the point of intentional delivery. When the robot arrives at the designated collection area, the servo motor is signaled to release, allowing for the ball to be smoothly transitioned onto the slope and into the collection zone. This servo motor is thus a critical component, enabling the robot to maintain a secure hold on the ball throughout various movements and only releasing it at the precise moment required.

4.2.3. Stability and Instability

4.2.3.1. Robot Structure

Stability is a cornerstone of our robot's structural design. By distributing heavier components such as batteries and motors towards the base, the robot maintains a low center of gravity. The symmetrical arrangement of these elements ensures a balanced weight distribution, which is essential for stable maneuvering. Additionally, the positioning of the ball collector and release mechanisms across diagonal ends of the robot acts to counterbalance the system, providing further stability during dynamic movements.

4.2.4. Maintenance Accessibility & Compliance with Regulations

In adherence to maintenance and regulatory compliance, our robot's design thoughtfully integrates a bumper system that conforms to the specified dimensions of 30x30x30 cm, optimizing the space for protection without exceeding competition size limitations. Furthermore, we've implemented a modular approach in our design, particularly with 3D printed parts such as the battery mount. This mount is detachable, enabling quick and effortless battery changes, which is crucial during competition and routine maintenance. Such detachability streamlines the process of replacing power sources and simplifies accessing internal components for inspection or repair. It ensures that all components are readily accessible, allowing for efficient checks and the tightening of screws to maintain structural integrity. These design choices ensure that our robot remains within competition standards while also prioritizing ease of maintenance for longevity and consistent performance.

4.3. Human-Centered Design Guidelines

Human-cantered design is integral to embodiment design, acknowledging that technical systems operate within a broader context. Throughout construction and operation, multiple stakeholders interact with the robot, with the operator playing a central role. Hence, ensuring the robot's design prioritizes ease of operation is crucial.

Safety during handling takes precedence. The robot's construction should eliminate any exposed sharp parts that could pose a risk of injury to the operator. It is imperative that all wires are adequately insulated, and any sharp edges should be smoothed to prevent cuts.

Furthermore, facilitating ease of operation is paramount. The power switch for the VEX controller should be readily accessible, enabling the operator to turn the robot on or off without the need for additional tools. Additionally, incorporating a conveniently located start button allows the operator to initiate robot functions effortlessly.

Last but not least, accessibility to maintenance tasks is essential for seamless operation. Routine maintenance activities, such as charging, should be simple for the operator to perform without requiring external tools. Thus, positioning the power connection to the battery in an easily accessible location is recommended.

5. Detailed Design

5.1 Mechanical Design

Utilizing Fusion360 for the design and development of a robot significantly enhanced the iterative improvement process through detailed visualization, which aids in identifying both aesthetic and functional areas for enhancement. Team-based workflows also ensure streamlined collaboration while also ensuring comprehensive documentation that captures design evolutions.

The finalized design of the robot in Fusion360 can be viewed below.

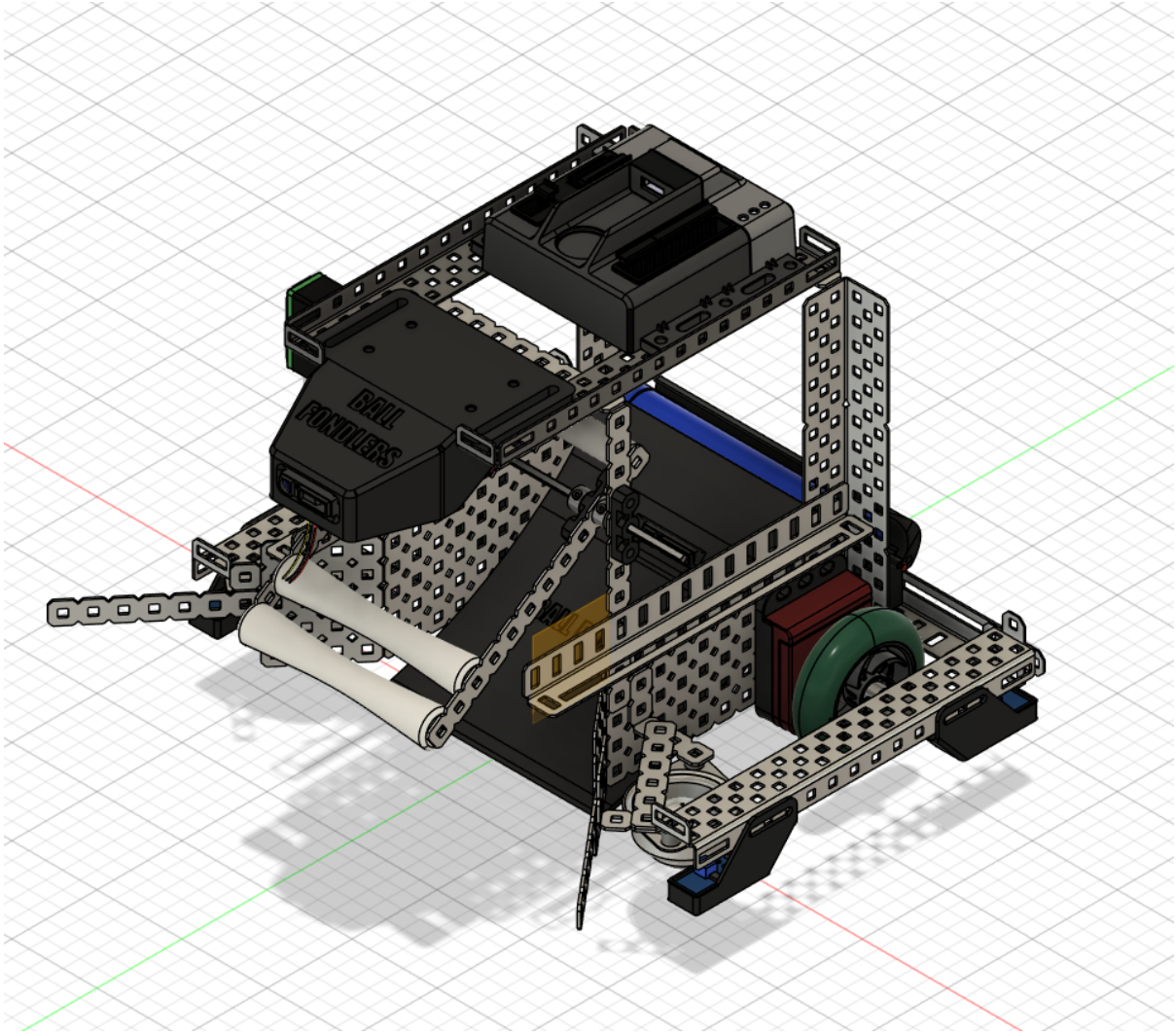


Figure 12: Robot Isometric View

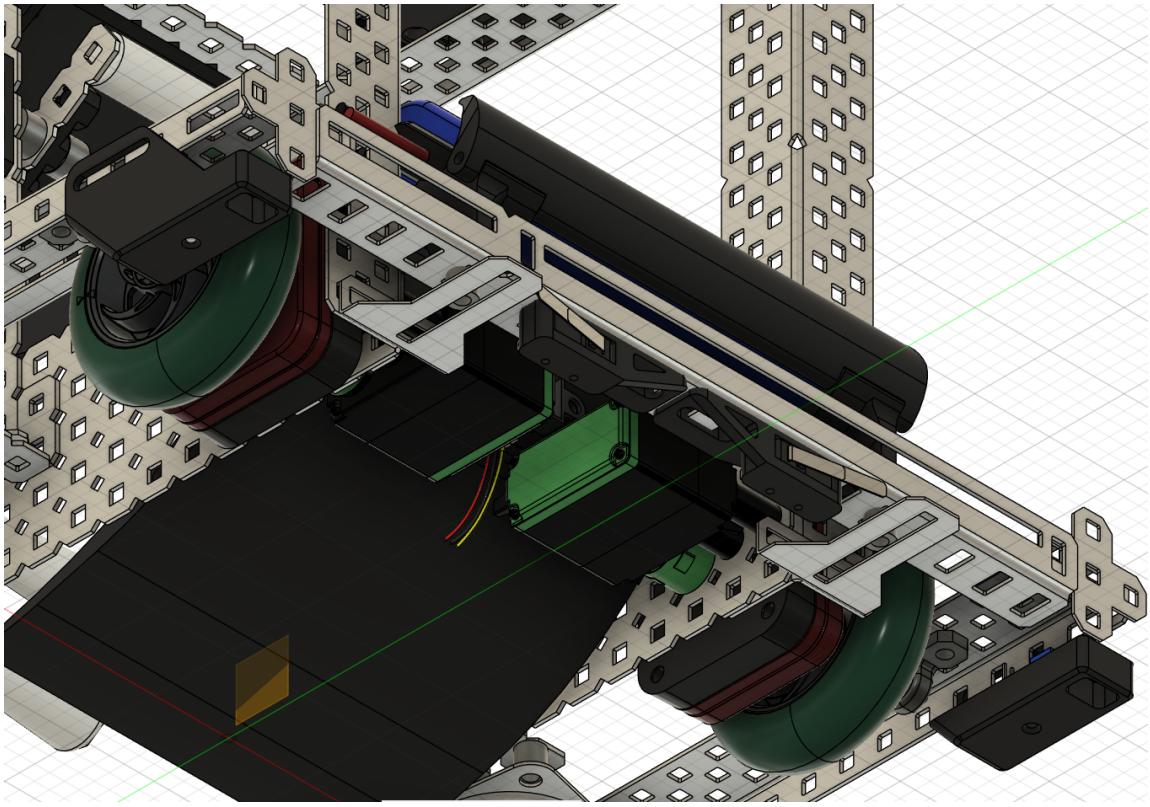


Figure 13: Robot bottom view

The robot's design incorporates critical components such as the encoder, motors, and the main unit of the short distance sensor in a strategically protected configuration. These elements are concealed behind the slope and encapsulated within the robot's robust frame, shielding them from potential collisions and damage from other robots in the competitive environment. This ensures continuous, undisturbed functionality of these sensitive parts. Complementing this safeguard, the robot is equipped with a bumper system that encases its entirety, wheels included. The bumpers act as a defensive perimeter, absorbing impacts and preventing direct hits to the robot's chassis and internal mechanisms. Such a design consideration is important in preserving the robot's structural and operational integrity during matches, where contact with other robots is likely.

5.1.1 Limit Switch – Bumper Design

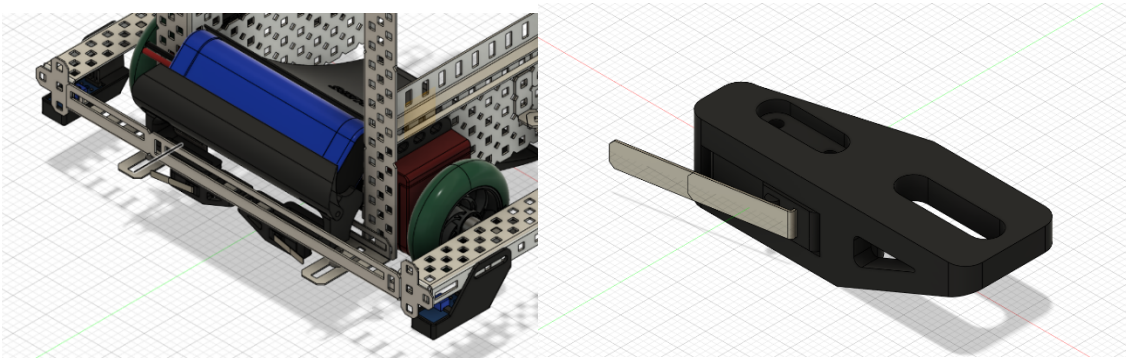


Figure 14: Bumper design

To enable the robot to detect when it has reached its starting area, a custom housing was designed for limit switches that were placed at the back of the robot's bumper. This housing was specifically engineered to hold two limit switches in separate enclosures, ensuring they are securely positioned while allowing for direct contact with the starting area's boundary. The design of the housing utilized a press-fit method, which not only streamlined the assembly process by eliminating the need for additional fasteners but also ensured a tight, secure fit that prevents the switches from dislodging during operation. This approach allowed for precise placement and reliable operation of the limit switches, enabling the robot to accurately sense when it has returned to its designated starting position through the physical actuation of the switches by contact with the starting area's edge.

5.1.2 Top Housing – Distance Sensor

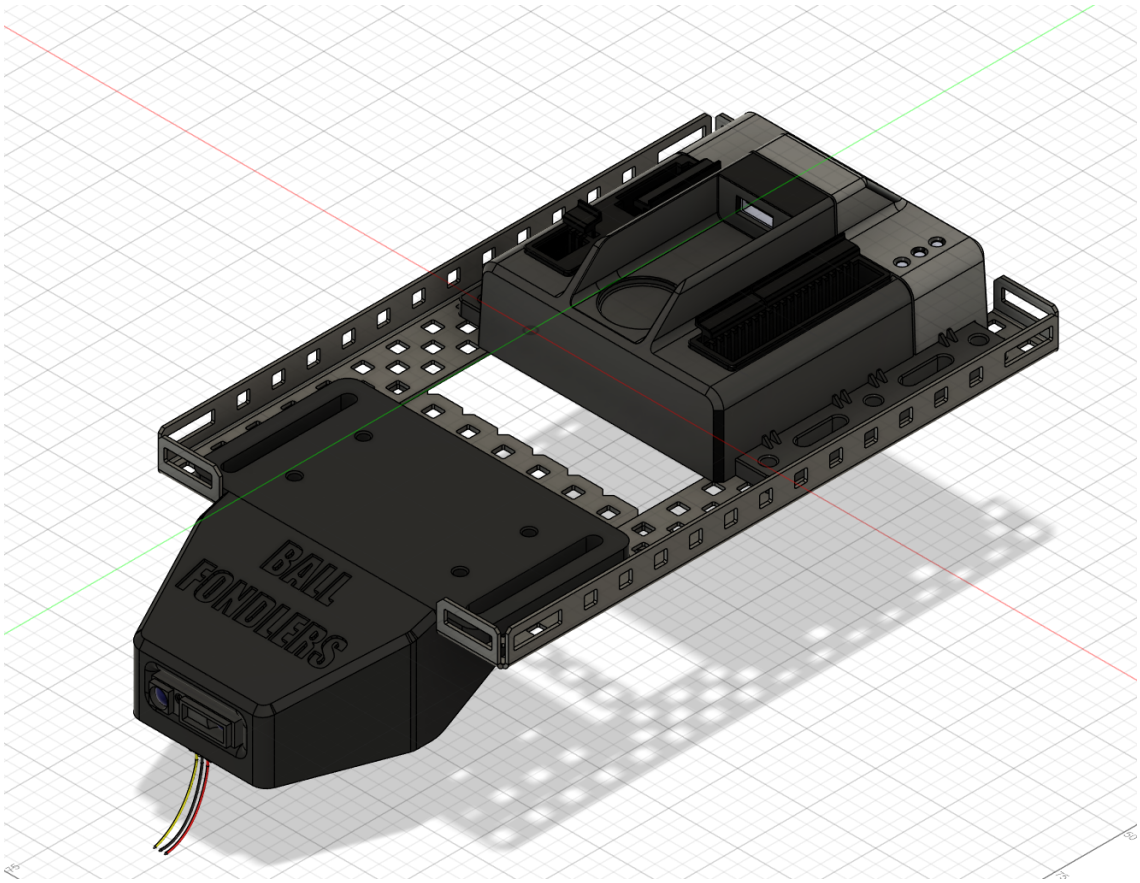


Figure 15: Top Housing

A housing was specially designed and mounted on the top side of the robot to accommodate the far distance sensor, an integral component for the quick detection of other robots and incoming objects. This housing was engineered with a focus on both protection and functionality. By encasing the sensor, the housing safeguards it from external impacts and vibrations, significantly reducing the risk of the sensor being dislodged or knocked off during the robot's operation. The design of the housing is such that it only exposes the front part of the sensor—the area necessary for its detection capabilities—thereby ensuring that the sensor's field of view is unobstructed while maintaining the integrity of the sensor's placement. This strategic exposure allows the sensor to effectively perform its function, detecting objects at a distance without being hindered by the housing itself. Moreover, the housing secures the sensor firmly in place, ensuring consistent performance and reliability in detecting other robots and objects, a crucial feature for navigating and interacting within its environment.

5.1.3 Collecting Mechanism – Roller System

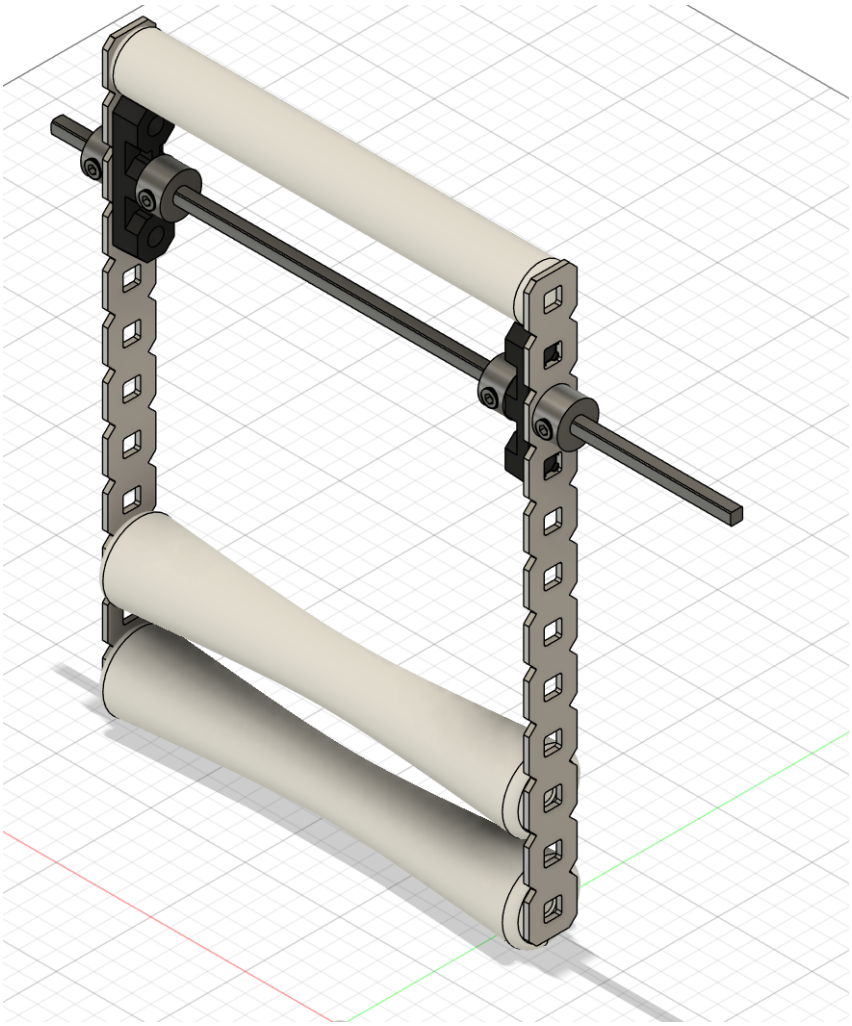


Figure 16: Collecting Mechanism

The roller system utilizes a pair of inward-rotating, convex cylindrical rollers to collect tennis balls, by leveraging the convex shape to funnel the balls towards the center and prevent them from veering off to the sides during collection. Opposite the fulcrum or pivot point, a counterweight is installed to ensure stability and balance of the system; it compensates for the weight of the rollers and the additional load of the tennis balls, mitigating the strain on the driving mechanism. This design choice ensures that the roller assembly remains level and consistently positioned, facilitating a steady and reliable ball collection process without necessitating frequent recalibration or risking misalignment that could disrupt the robot's operation.

5.1.4 Line Sensor Housing

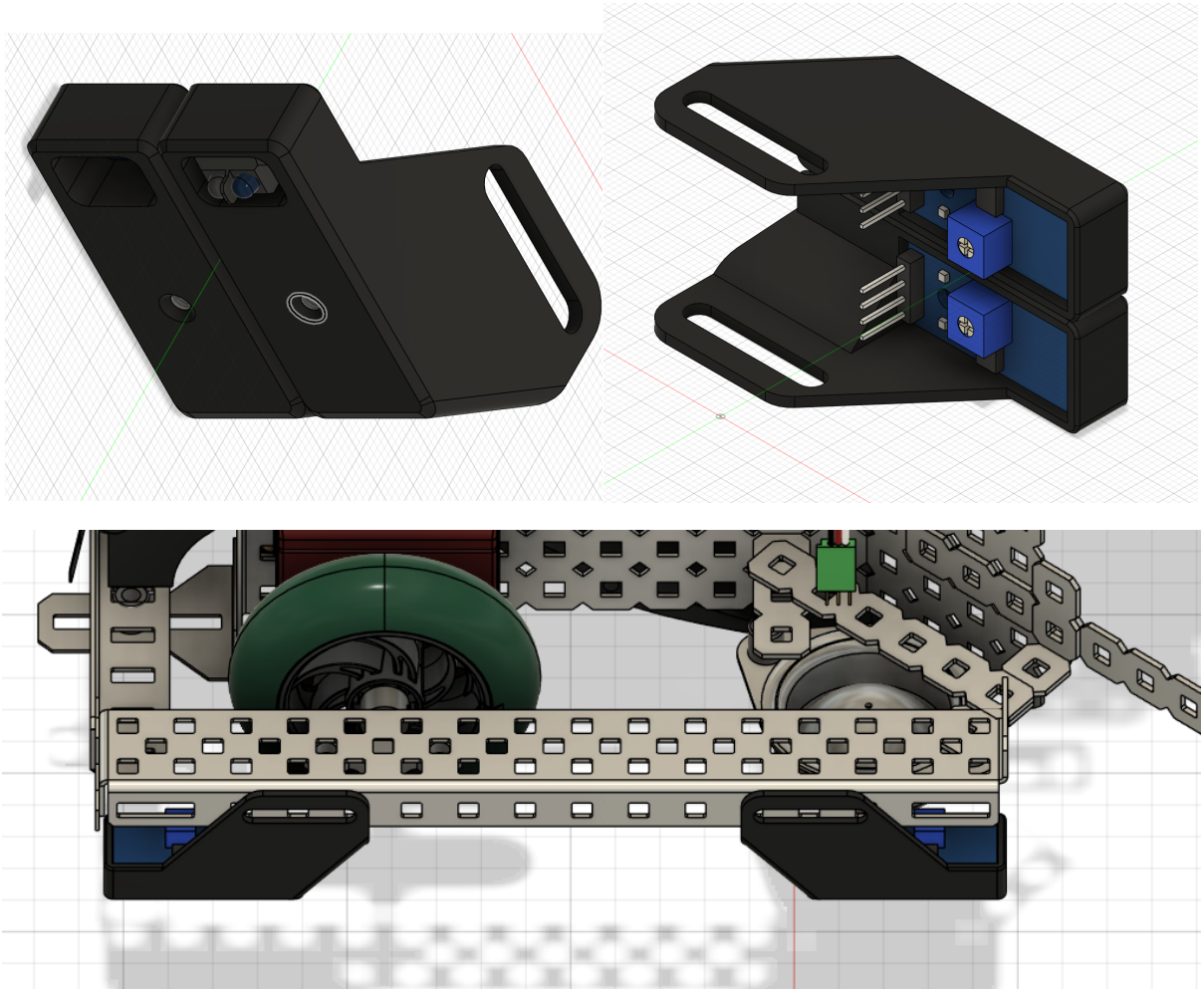


Figure 17: Limit switch housing

The robot incorporates strategically positioned line sensors within tightly engineered press-fit housings at each of its four corners, enabling it to detect arena boundaries accurately. These housings are designed to hold the sensors securely without additional fastening mechanisms, thereby minimizing the risk of sensor displacement due to operational vibrations or impacts. The bottom of each housing is purposefully exposed, allowing the line sensors to maintain proximity to the ground for optimal detection capability while being shielded elsewhere to prevent damage from environmental factors. This configuration ensures that the robot can accurately perceive and respond to the boundary lines, a critical function for maintaining the robot's course and strategy within the arena.

5.1.5 Battery Housing

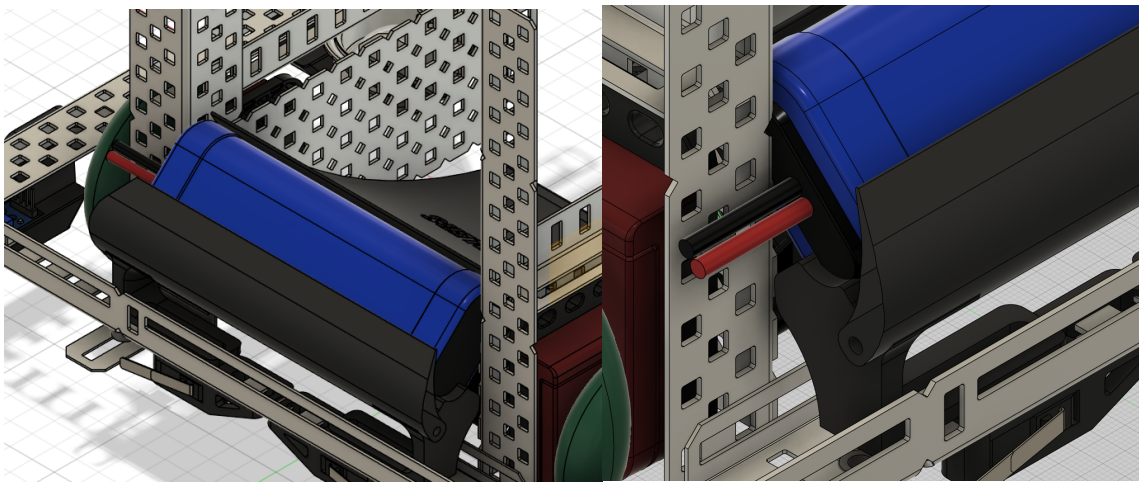


Figure 18: Battery Housing

The design of the robot incorporates a custom battery housing, which is press-fitted into the robot's frame, serving dual purposes. First, it enables tennis balls to glide off the battery's sloped surface directly into the collection zone. Second, an additional mount is secured at the back of the battery, anchoring it firmly in its downward-tilted position; this not only ensures that the battery remains steadfast amidst the rigors of the robot's movements and potential collisions but also optimizes the trajectory of the tennis balls towards the collection zone. The design of the housing ensures a smooth transition for the balls while simultaneously safeguarding the battery from external impacts, obstacles, or interference from other robots, which is critical for both the functionality and longevity of the robot.

5.1.6 Encoder Limit Switch Holder Design

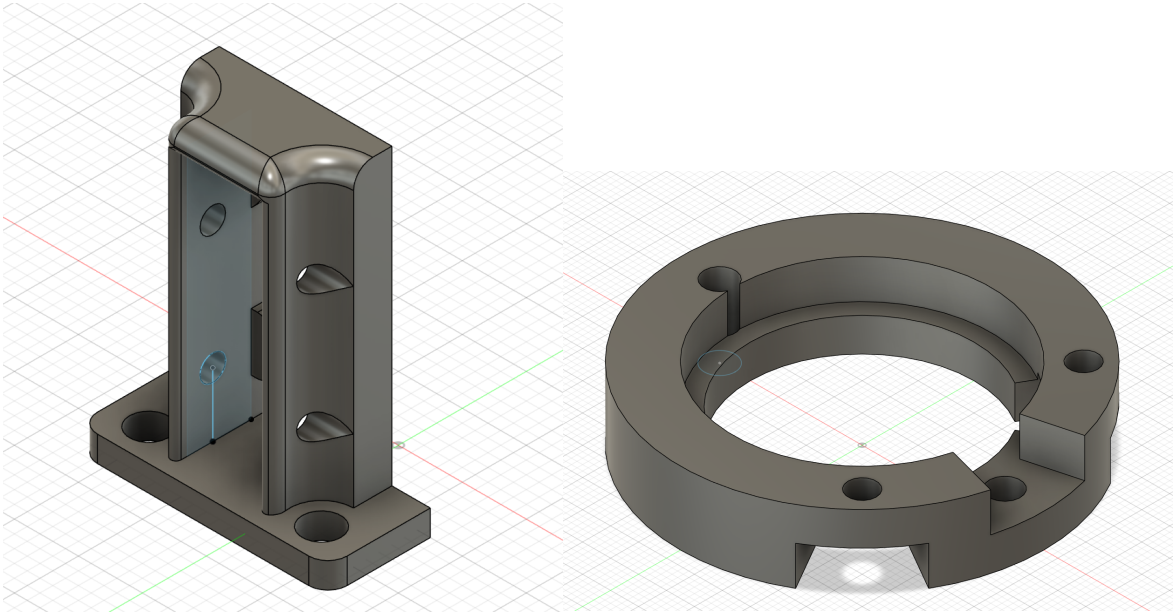


Figure 19: Encoder limit switch holder

In our robot's ball collection mechanism, a limit switch encoder is mounted onto the shaft of the roller that is responsible for gathering tennis balls. This encoder plays a critical role in tracking the roller's position as it descends to collect a ball and retracts to secure it in place. By continuously monitoring the roller's movement, the encoder provides real-time positional feedback to the control algorithm, which can precisely determine when the roller is in the down position—indicative of a successful ball collection—and when it is in the up position. This feedback loop is important for the algorithm to control the roller's operations, enabling the system to optimize the timing and coordination of the roller's movements, ensure efficient ball collection, and prevent unnecessary wear on the mechanism, thereby enhancing the robot's overall performance and reliability in the ball collection task. This component was designed to be highly adjustable to fine-tune the specific angles of the swing arm as required. By using a simple cam mechanism along with 3 different limit switches, it can independently detect the open, closed, and ejected positions of the swing arm. Initially tested to operate with only two limit switches, it was originally meant to locate position based on a binary input. Open position would result in 01, closed would result in 11, and eject would result in 10. This was later found to be inaccurate and prone to false positives. An encoder was vital for the swing arm as positional value was extremely important. Due to the limited electronic parts given, both encoders were utilized in the drive wheels. To solve this limitation, limit switches were used instead.

5.1.7 2-Wheel Drive with 2-Ball Bearings

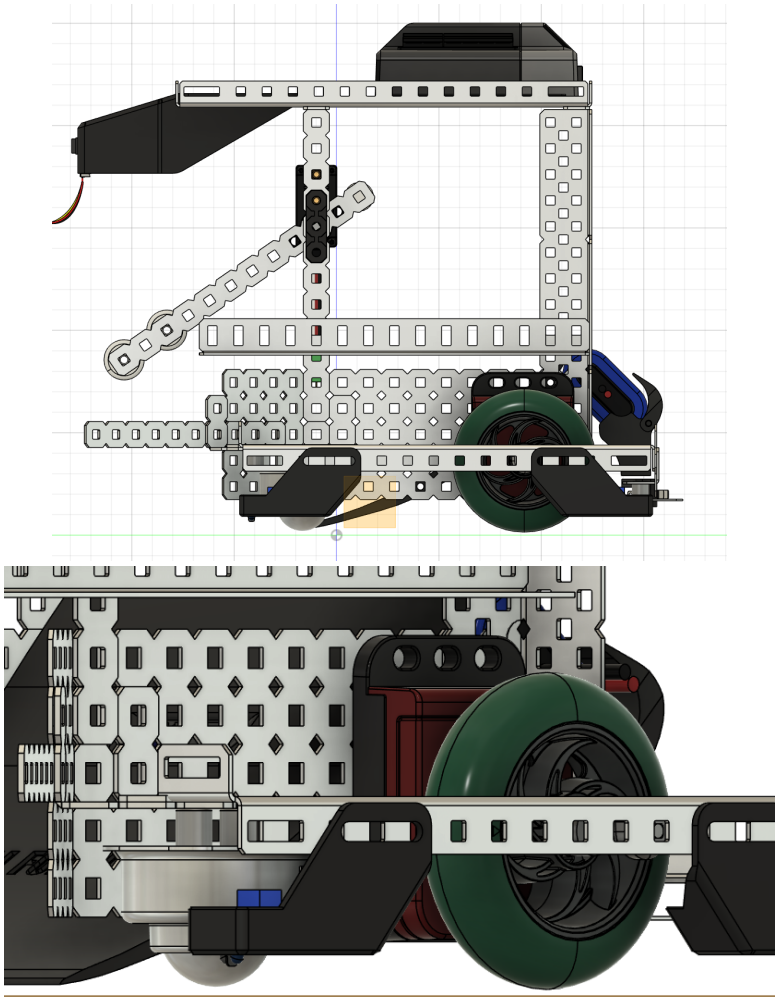


Figure 20: Driving mechanism

The robot's mobility is engineered around a two-wheel drive system, which provides precise control over its movement and maneuverability. Each drive wheel is connected to a motor that dictates the speed and direction, allowing for agile and responsive navigation within the operational environment. Complementing the drive system, two ball bearings are strategically positioned to bolster the structural integrity of the device. These bearings facilitate the smooth rotation of the wheels and minimize frictional forces, which not only contributes to the longevity of the moving parts but also ensures stability and balance. The combination of the two-wheel drive for movement control and the ball bearings for mechanical support culminates in a robot that is both robust in its construction and adept in its handling.

5.1.8 Slope Design

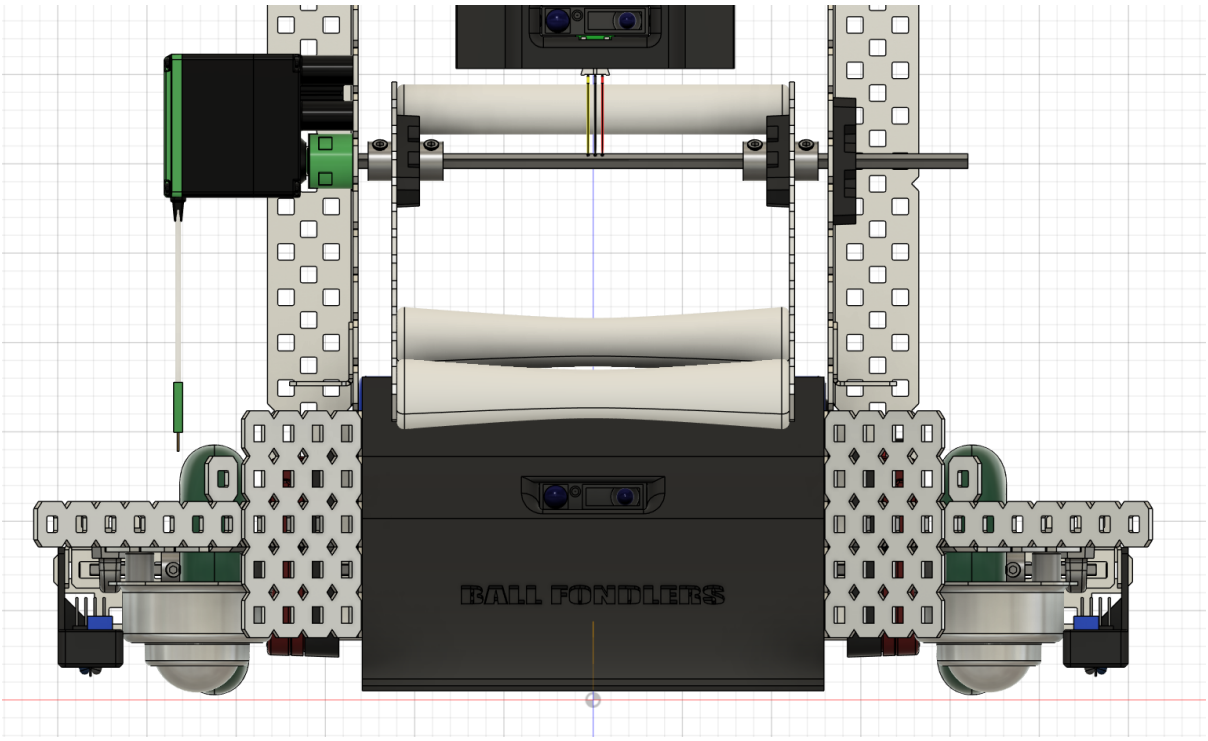


Figure 21: Slope Design

The robot features a strategically designed slope with a custom cutout to house a short distance sensor, essential for detecting the successful capture of tennis balls by the roller mechanism. This cutout is precisely engineered to expose only the vital sensing elements, thereby minimizing the sensor's vulnerability to any potential damage from incoming tennis balls. The sensor's placement within the slope is meticulously calibrated to ensure that it can accurately register the presence of a ball without being obstructed or impacted by the ball's motion. This design choice underscores a harmonious balance between functionality and durability—by safeguarding the sensor while enabling it to perform its critical role in the ball collection process, ensuring seamless operation and longevity of the robot's collection system.

6. Detailed Calculations

6.1. Dynamic Load Analysis

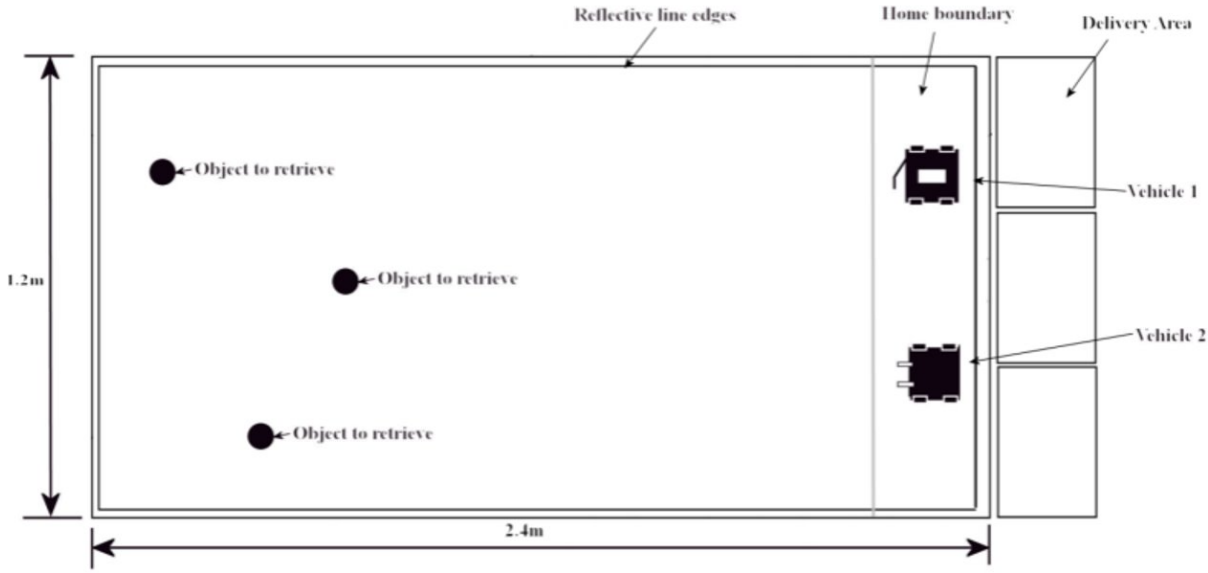


Figure 22: Competition Arena

For a total of 3 minutes for each round and 3 balls in total placed in the playing field as shown in the Figure above, we would need to find out the minimum distance needed for the robot to travel and collect all the balls.

$$Min.\ distance\ to\ collect\ all\ 3\ balls\ =\ (2.4-0.4)\ \times\ 6\ =\ 12m$$

Considering collision between opponent robots and accidental movement of the balls, an additional 2m distance is added to total the minimum distance travelled by the robot to 14m.

$$Circumference\ of\ wheel\ =\ 0.06296m$$

$$Dist.\ per\ revolution\ =\ \pi\ \times\ 0.06296\ =\ 0.198\ \frac{m}{rev}$$

$$Number\ of\ revolutions\ =\ \frac{14}{0.198}\ =\ 70\ rev$$

$$Motor\ Speed\ =\ \frac{Number\ of\ Revolutions}{Time}\ =\ \frac{70}{60\times 3}\ \times\ 60\ =\ 23.3\ RPM$$

Hence, the minimum speed of the motor required for the robot to travel 14m in 3 minutes is 23.3 RPM.

6.2 Static Load Analysis

Gravitational Force on the Robot:

- Mass of the robot (m): 0.75 kg
- Acceleration due to gravity (g): 9.81 m/s²

$$F_{gravity} = m \times g = 0.75 \times 9.81 = 7.36N$$

This calculation shows the force exerted by the robot on the ground due to gravity, influencing its stability and traction.

6.3 Collision Force Estimation

Impact Force During Collisions:

Given sudden stops and interactions with obstacles or other robots, we consider the robot coming to a halt from its travel speed in a short time (), estimated conservatively as 0.1 seconds.

$$a_{collision} = \frac{\Delta v}{\Delta t} = \frac{0.0778 m/s}{0.1 s} = 0.778\ m/s^2$$

$$F_{impact} = m \times a_{collision} = 0.75 \times 0.778 = 0.5835N$$

6.4 Energy Consumption Calculations

Estimated Power Usage:

Assuming an average power consumption of 10 Watts and a battery capacity of 21.6 Wh:

$$Operating\ time\ = \frac{Energy}{Power\ Consumption} = \frac{21.6\ Wh}{10W} = 2.16\ hours$$

This duration comfortably exceeds the requirement of 3 minutes per round, confirming the adequacy of the battery capacity for competition demands.

7. Electrical and Electronics System

7.1. Motors

Two motors are provided for this project: Vex 2-wire 393 motor and Vex continuous rotation motor.

The characteristics of the motors are provided below:

Motor Type	Free Speed (RPM)	Stall Torque (Nm)	Stall Current (A)	Free Current (A)
Vex 2-wire 393 motor	100 @ 7.2V	1.68	3.6	0.15
Vex continuous rotation motor	100 @ 7.5V	0.73	1.6	0.14

To aid in control of the motors, the no-load relationship between the signal to the motor and the wheel velocity is investigated and modelled.

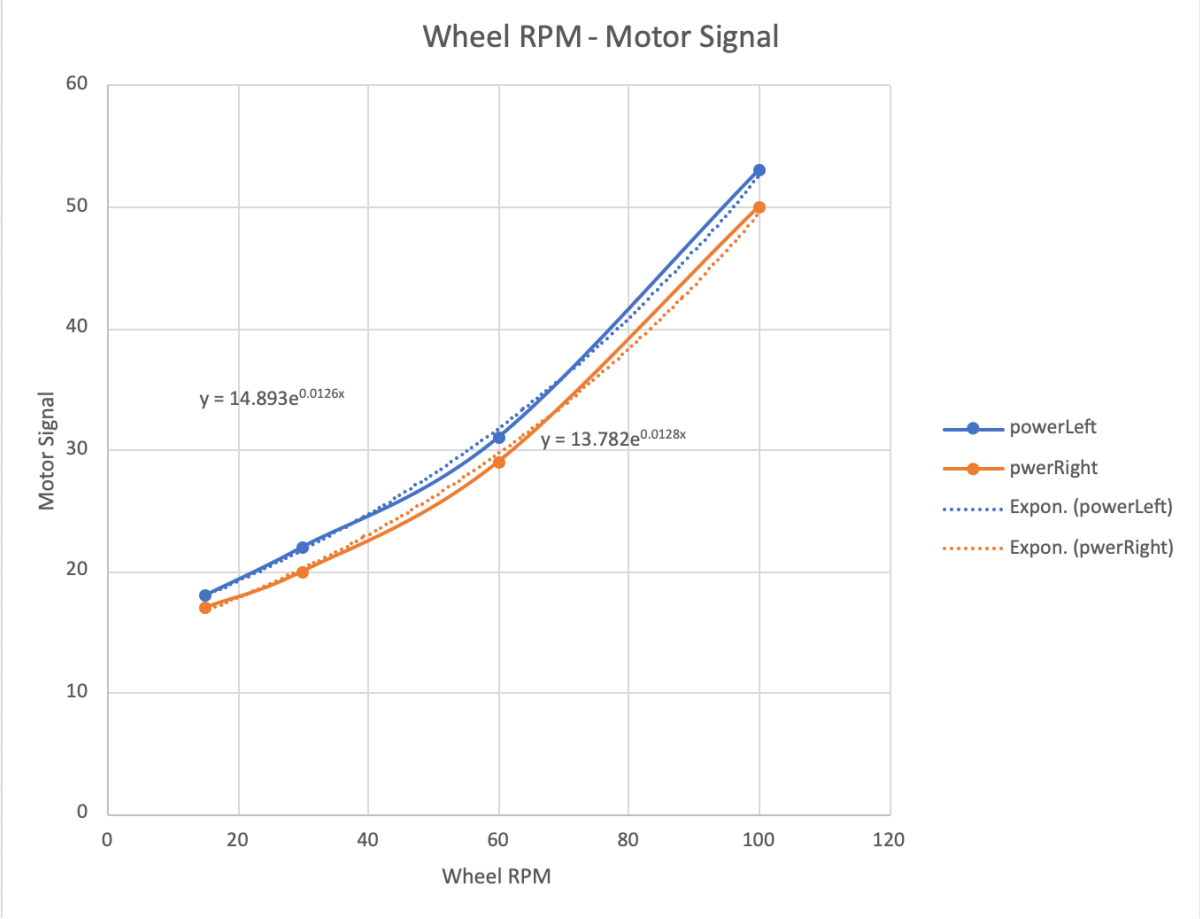


Figure 23: Motor Modelling Chart

7.2. Sharp Sensor

The robot is equipped with three **Long Range Infrared (IR) sensors** (GP2Y0A21SK0F) along with one **Short Range IR sensor** (GP2Y0A41SK0F) to detect objects and maneuver through the arena. These IR sensors gauge the distance between the robot and nearby objects, facilitating precise navigation towards the tennis ball. Moreover, they serve to identify obstacles, enabling the robot to steer clear of collisions.

Integration of these sensors into the robot equips it with essential sensing and detection capabilities, streamlining the ball retrieval process. As a result, the robot operates autonomously during the competition, efficiently collecting tennis balls without requiring human intervention.

The characteristics of the given sharp sensors are summarized in below:

Sensor Type	**Optimal Measuring Distance (cm)**
Long range IR sensor (GP2Y0A21SK0F)	10 - 80
Short range IR sensor (GP2Y0A41SK0F)	4 - 30

During testing however, it was noted that long range IR sensor (GP2Y0A21SK0F) was only able to accurately measure up to 25cm.

7.2.1. Sharp Sensor Calibration

Output for both sensors are as follows:

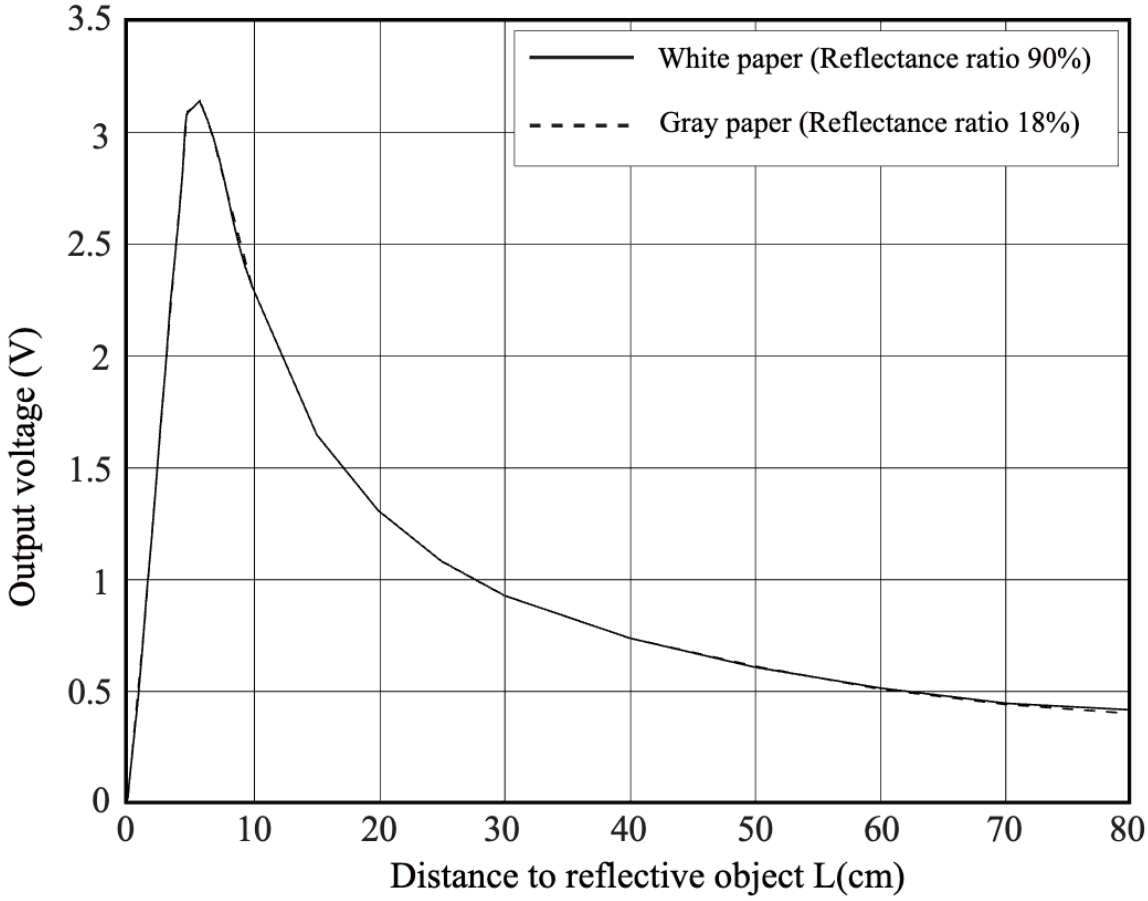


Figure 24:Long range GP2Y0A21YK0F Output [2]

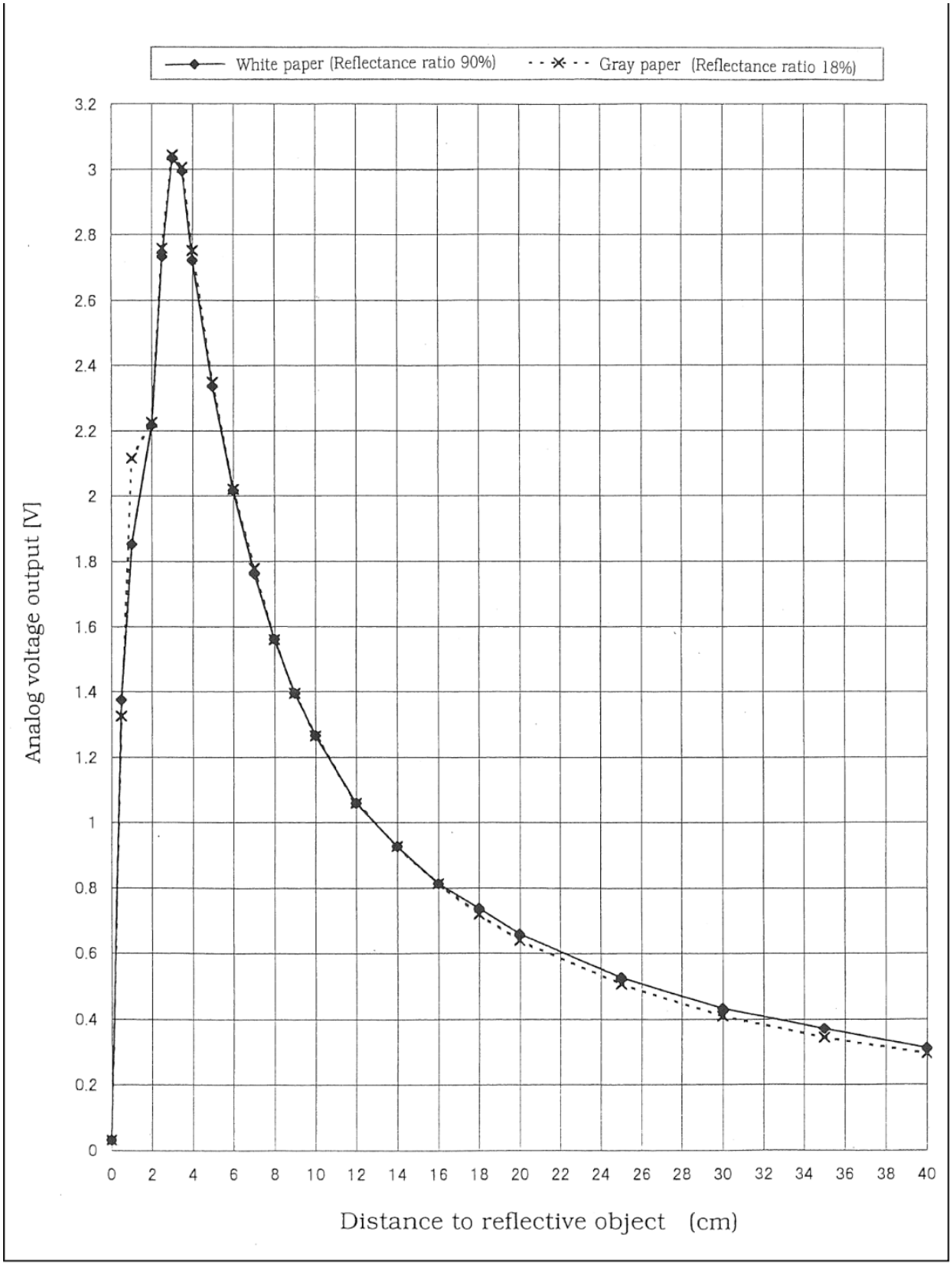
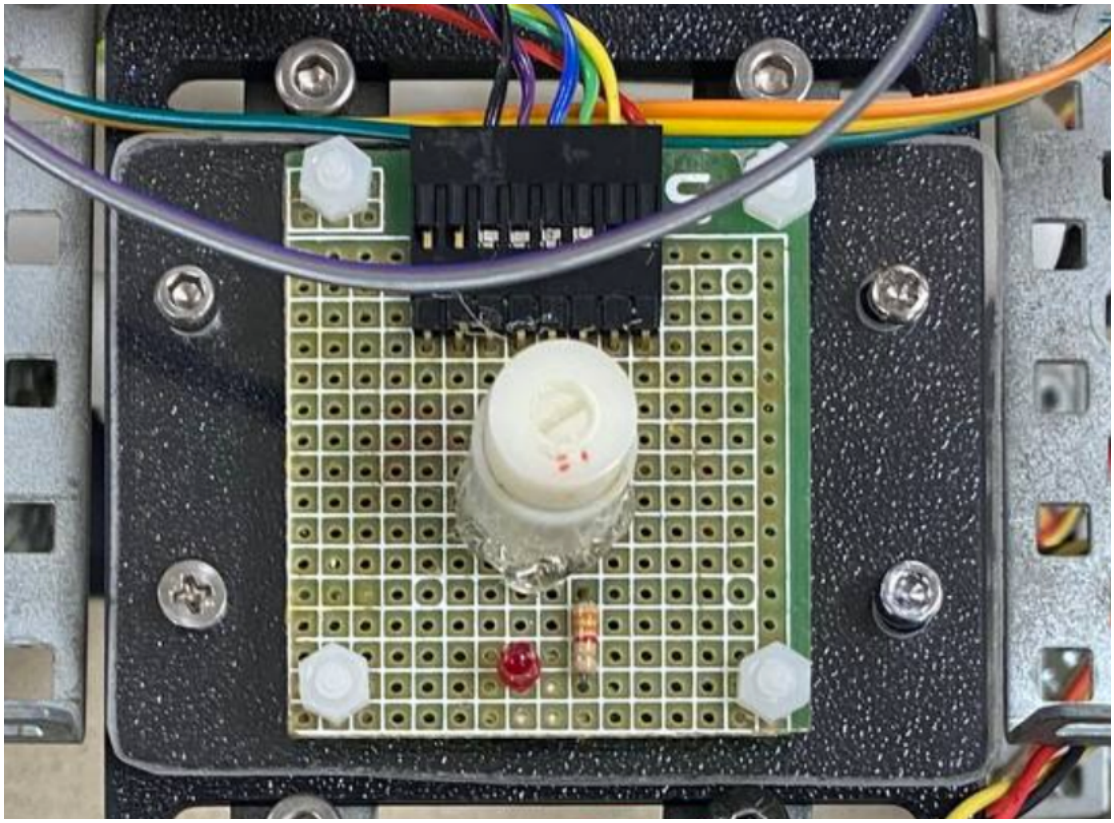


Figure 25: Short range GP2Y0A41SK0F [3]

7.3. Magnetometer



The magnetometer is an exteroceptive sensor that detects the Earth's magnetic field. This enables us to not rely solely on wheel encoders which are prone to drift. This model of magnetometer can only sense the four main compass headings: North, South, East and West using its four digital output pins. The additional four headings (NE, NW, SE, and SW) are inferred from two triggered readings of the magnetometer. Therefore it is assumed that this magnetometer has a resolution of 45 degrees.

7.4. IR Sensor

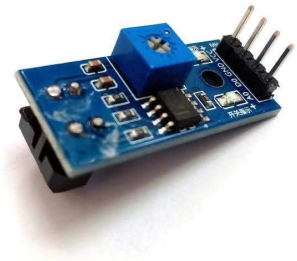


Figure 26: IR Sensor

The IR Line tracking sensor used in this project is IR Line Tracking Module. The maximum quantity given was 4. During our project, we used IR Line Tracking Module to detect the boundary line of the arena. This is done by reading the input of the IR, which are higher at the boundary due to the reflective tape placed around the arena. The number we tested are as shown for the analog input. We then provide the average of the light and dark number as a threshold to be utilized in our software.

Light (boundary tape):

FL: 190

FR: 234

BL: 150

BR: 137

Dark (arena platform):

FL: 2467

FR: 1880

BL: 2376

BR: 1502

The threshold for detection of the line is the mean value between light and dark readings.

7.5. Wiring Diagram

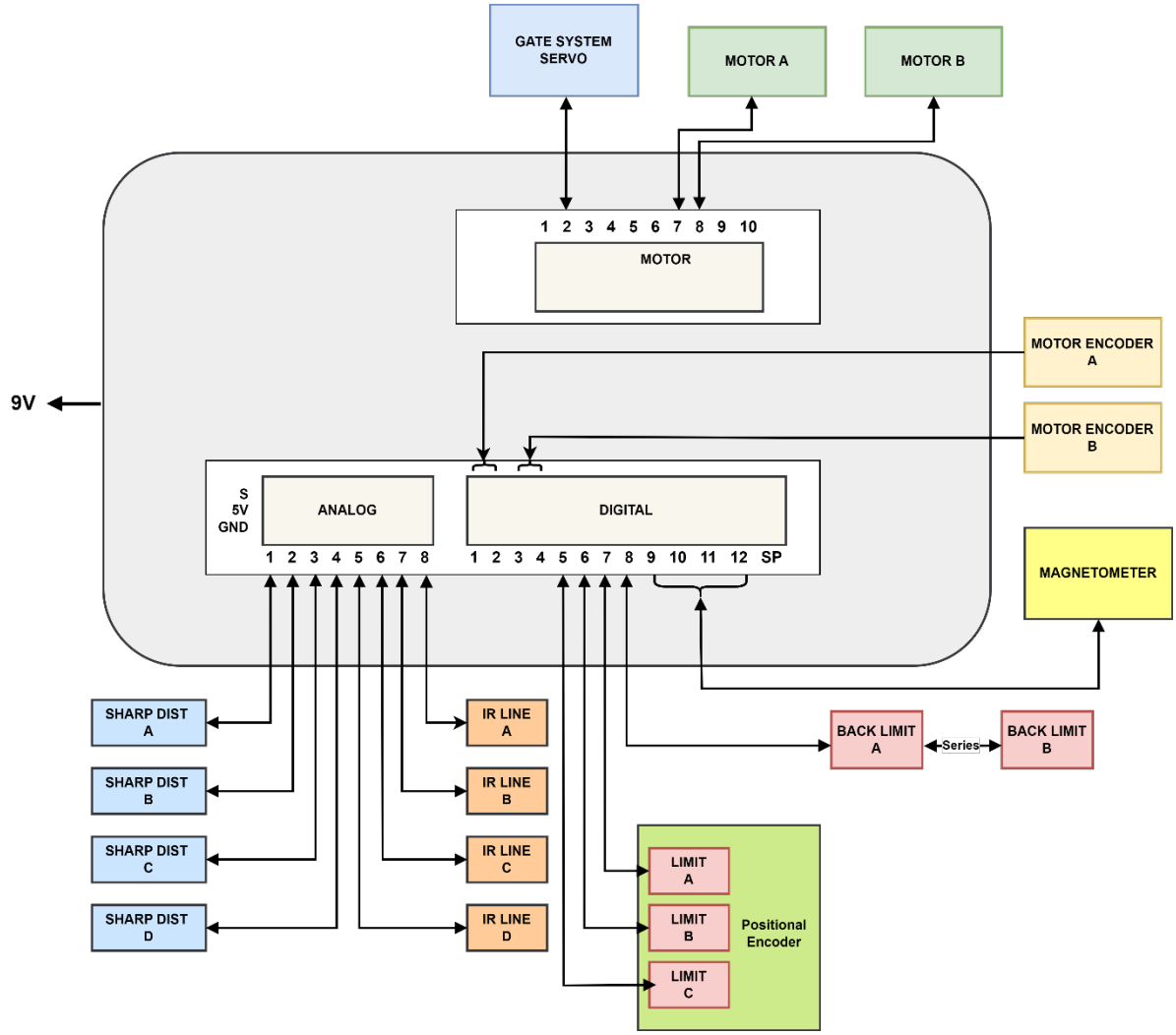


Figure 27. Circuit Diagram

8. Software Design

8.1 System Architecture

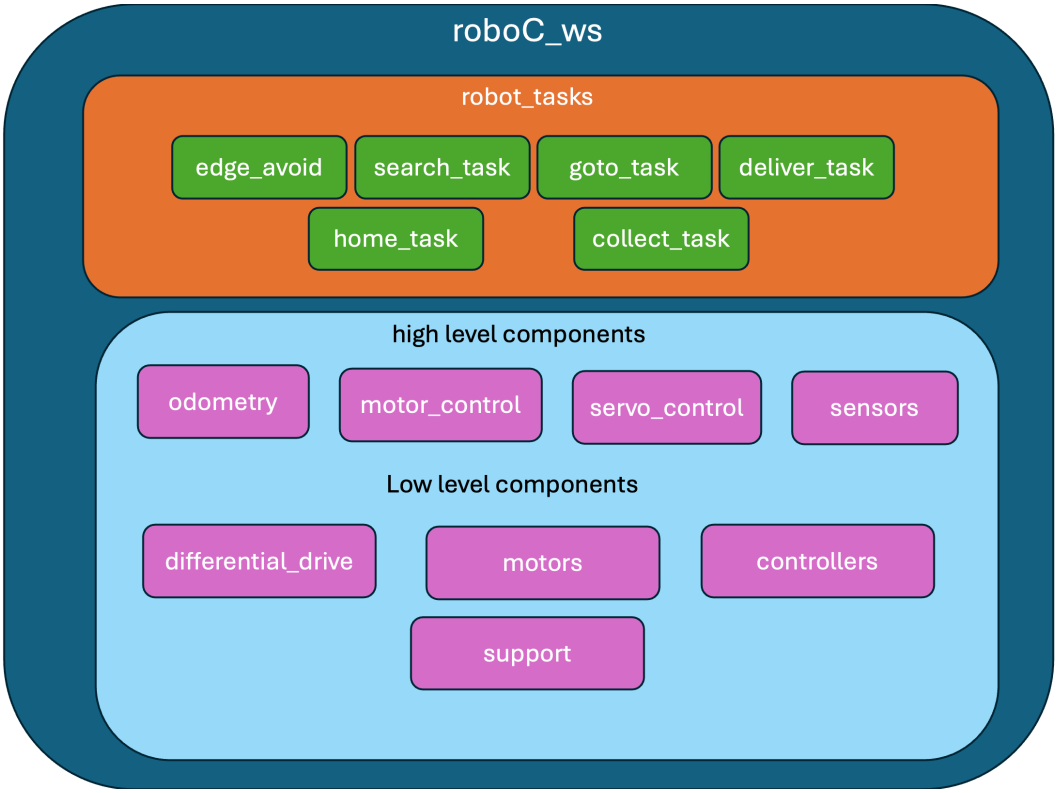


Figure 28 Overall System Architecture

The code is organized in a hierarchical manner. On the high level, are the robot tasks which is the decision-making portion of the robot. It takes in sensor data as inputs and outputs commands for the actuators of the robot and the task to execute next. The components to the robot are for sensing, actuation, and odometry. The code base was uploaded and developed on GitHub (Appendix A).

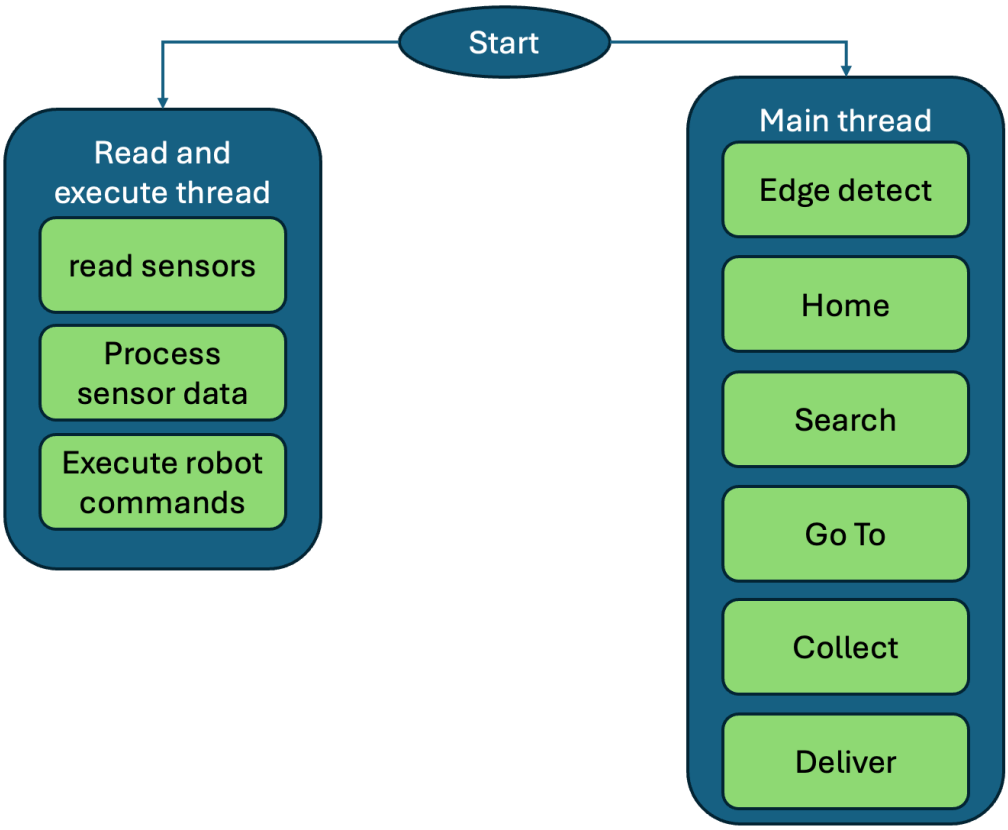


Figure 29 Robot task execution structure.

The robot has two threads running loops in discrete time where the entire loop is run through at each time step without any blocking code in time steps of 50 [ms]. This allows the robot to be aware of the latest information of the robot states whenever it is being queried by a robot task. Additionally, this is done to ensure that robot commands are executed for a fixed amount of time and to ensure that the calculations to obtain speed has accurate timing.

The first is the read and execute thread. This thread contains the functions to read and process the information from the sensors. This processing includes filtering, fitting to the curve sensor calibration curves, and passing through thresholds. It also executes the commands of the robot for a fixed period so that the state model estimates for odometry are as accurate as possible.

The main thread contains the robot tasks which are the decision-making component of the robot. A task state is used to keep track of which task is to be run at each time step. This can also be used to create exceptions to performing the avoidance tasks.

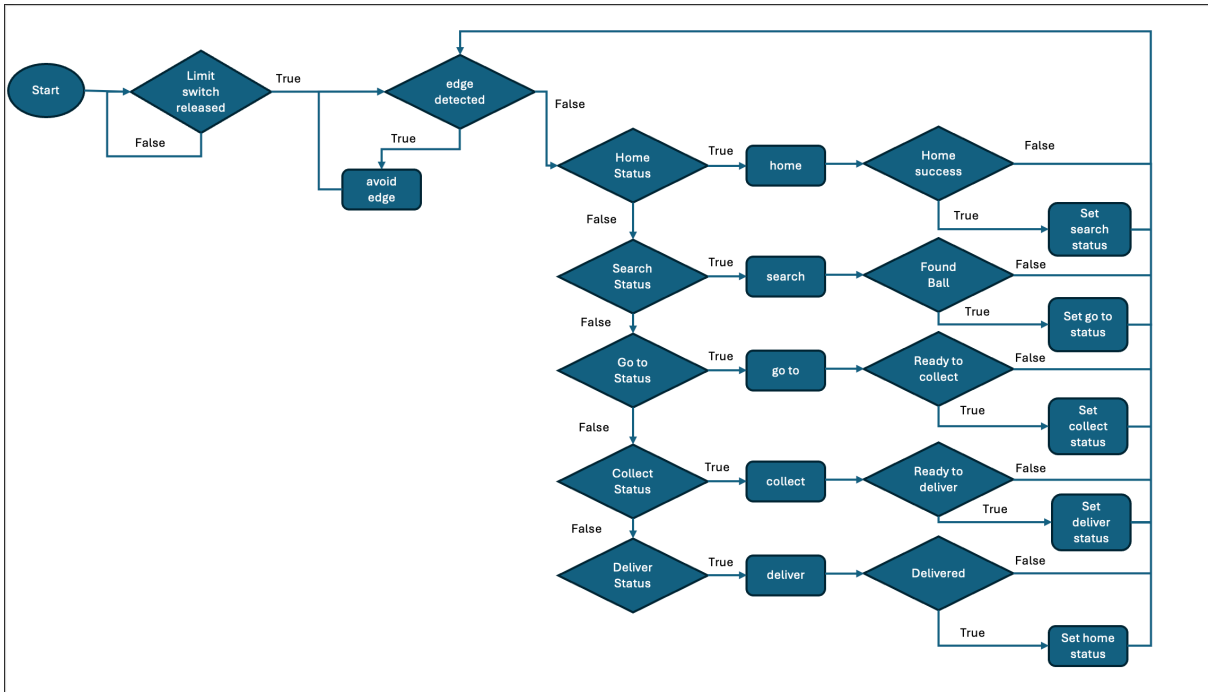


Figure 30 Block Diagram of main loop of robot program.

At startup, the robot initializes its odometry coordinates (x, y, yaw, linear velocity, and angular velocity) while waiting for the limit switch to be released.

8.2 Robot Description and Model

8.2.1 Robot Description

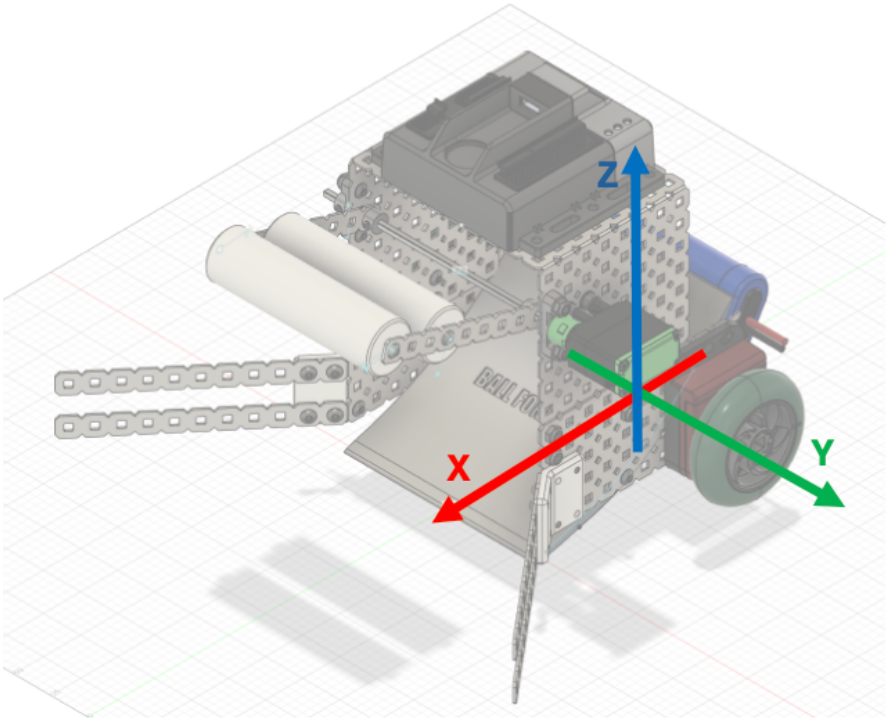


Figure 31 Coordinate frame of the robot.

The y-axis of the robot is colinear with the differential drive axle. The x-axis runs perpendicular to the y-axis, through the center of the robot, and parallel to the ground. Finally, the z-axis is determined with the right-hand rule. The yaw of the robot is described from -180 to 180 degrees from the x-axis of the arena coordinate system to the robot's x-axis.

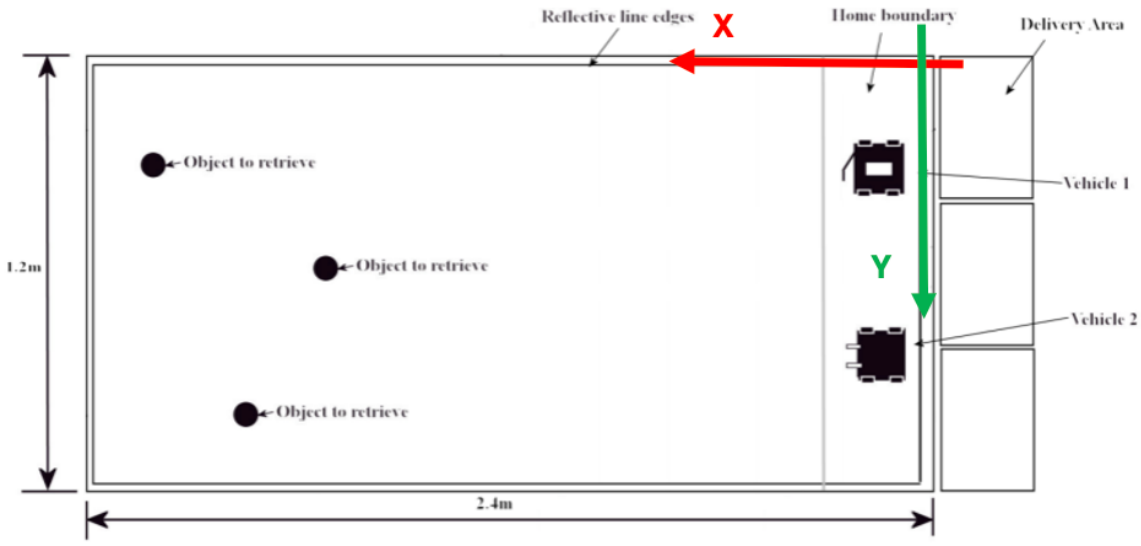


Figure 32 Arena coordinate system.

8.2.2 Robot Model

The robot is modelled as a differential drive model. This model is used to formulate the odometry and the wheel velocities.

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} \cos\varphi & 0 \\ \sin\varphi & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_x \\ \omega_Z \end{bmatrix}$$

Equation 1 Differential Drive Model

Where:

$v_x : robot\ linear\ velocity\ in\ the\ x-axis, [rad\ s^{-1}]$

$\omega_Z : robot\ angular\ velocity\ about\ the\ z-axis, [m\ s^{-1}]$

$\dot{x} : linear\ velocity\ in\ the\ global\ x-axis$

$\dot{y} : linear\ velocity\ in\ the\ global\ y-axis$

$\dot{\varphi} : angular\ velocity\ about\ the\ global\ z-axis$

$\varphi : yaw\ of\ the\ robot\ wrt\ to\ the\ global\ x-axis$

8.3 Robot Actuation

8.3.1 Drive – Velocity Control

For information input to our robot, we have sharp distance sensors, line sensors, a compass, limit switches, and motor encoders. Their readings, however, must be processed before it is able to provide useful information for our robot. Hence, functions are created to do processing on the readings and incorporating them to the basic logics of the robot.

The drive system of robot is controlled through velocity control. These velocities for a non-holonomic differential drive robot are linear velocities in the x-axis and angular velocities about the z-axis. To actuate these velocities, they must be translated into wheel rotations.

$$\omega_R = \frac{v_x}{R} + \frac{T\omega_Z}{2R}$$

Equation 2 Right wheel angular velocity given robot velocities.

$$\omega_L = \frac{v_x}{R} - \frac{T\omega_Z}{2R}$$

Equation 3 Left wheel angular rotation given robot velocities.

Where:

$v_x : robot\ linear\ velocity\ in\ the\ x-axis, [rad\ s^{-1}]$

$\omega_Z : robot\ angular\ velocity\ about\ the\ z-axis, [m\ s^{-1}]$

$\omega_L : left\ wheel\ angular\ velocity, [rad\ s^{-1}]$

$\omega_R : right\ wheel\ angular\ velocity, [rad\ s^{-1}]$

$R : wheel\ radius, [m]$

$T : robot\ track\ width, [m]$

Since the motors are torque controlled, a feedback controller is used to match the output speed to the desired speed by controlling the power to the motors. The selected closed loop control was the Proportional-Integral-Derivative (PID) controller.

$$u(t) = K_p\, e(t) + K_i \int e(t)\ dt + K_d \frac{de}{dt} + b$$

Equation 4 Proportional-Integral-Derivative Controller

Where:

$u(t): control\ signal\ at\ time\ step\ t$
$e(t): error\ at\ time\ step\ t$
$K : Controller\ gain$
$b : output\ bias$

The controller was designed with a bias where the bias was equal to the open loop motor power for a commanded wheel angular velocity. This is to provide an initial guess for the controller output so that the PID algorithm would have a smaller error to correct for. Additionally, the output of the controller is then clipped by the maximum motor control signal of 127 as good practice.

8.3.2 Arm – Position Control

```
/**
 * @brief Sets the desired position of the servo and actuates if it is not there
 * @param set_position desired position of the servo in DEG
 * @return direction of the arm
*/
float robot_arm_move(float set_position, float arm_position, int current_task){
    static float prev_direction;
    float err_arm = set_position - arm_position;
    float set_servo_power = 0;
    // position of arm is -1 at most open, and 180 at most closed
    // when motor is set to positive, arm swings inwards
    if (fabs(err_arm) > SERVO_TOLERANCE){
        // set_servo_power = sgn(err_arm) * MAX_POWER * SERVO_GAIN;
        set_servo_power = err_arm * SERVO_KP;
        if (current_task == DELIVER){
            set_servo_power = 127.0 * SERVO_KP * sgn(err_arm);
        }
        else{
            set_servo_power = limit_byte(set_servo_power);
        }
    }
    motor[servo] = -set_servo_power;
    if (sgn(err_arm) == 0){
        return prev_direction;
    }
    prev_direction = sgn(err_arm);
    return sgn(err_arm);
}
```

Figure 33 Code snippet for arm positional control.

The arm of the robot is controlled positionally. Utilizing a Proportional feedback control loop, the control signal to the arm is manipulated to obtain a desired arm position. The rationale for only applying a P controller is because the resolution of the arm position is very large (45 degrees).

8.4 Robot Sensing

The robot is equipped with sharp distance sensors, line sensors, a compass, limit switches, and motor encoders. Their readings, however, must be processed before it is able to provide useful information for our robot. Hence, functions are created to do processing on the readings and incorporating them to the basic logics of the robot.

Aside from the compass and limit switches, the inputs are analog. For all the analogue sensor readings, we first needed to apply a low pass filter to try and filter out high frequency fluctuations in the sensor readings. The filter functions are thus created and tuned for each of the sensors.

After filtering, each sensors' readings are then modified for easier troubleshooting and logic creation. This part is only for the sharp distance sensors and the compass. For the sharp distance sensors, a graph is plotted through Python to find a general equation to convert the sensor reading from voltage (in volts) to distance (in cm). Compass, on the other hand, has its readings processed to return the direction it is facing to. For example, north, northeast, east, etc. With this function implemented, the logic creation and hyperparameter tuning becomes easier as the numbers output from the reading makes sense.

Finally, functions for basic logics are also created. For example, a logic to detect and differentiate balls from opponents are created based on the sharp distance sensor readings. A logic to detect the back wall is also created by incorporating readings from the compass and odometry from encoder readings with the sharp distance sensor readings. Logic is also implemented for line sensors to determine which edge the robot is hitting based on the line sensor readings.

8.4.1 Analog Sensor Filtering

The analog sensor readings which include the sharp distance sensors and line sensors of the robot undergo filtering to remove noise from the readings. One such filter is a first order low pass filter [2]. This low pass filter is deployed for all filters to sensor's output.

$$u_{t,out} = \alpha\, u_{in} + (1 - \alpha)u$$

Additionally, a median filter is used for the sharp distance sensors due to their noisy readings. This median filter has a windows size of 3 so to not take too much memory and to keep the readings closer to the new reading.

```
float median_filtering(float current, float prev1, float prev2){
    float max_ = max(x1: current, x2: max(x1: prev1, x2: prev2));
    float min_ = min(x1: current, x2: min(x1: prev1, x2: prev2));
    if (prev2 > min_ && prev2 < max_){
        return prev2;
    }
    else if (prev1 > min_ && prev1 < max_){
        return prev1;
    }
    else{
        return current;
    }
}
```

Figure 34 Code snippet of median filter with window size 3

8.4.2 Robot Pose Filtering

The odometry can be obtained from multiple sources which may either be proprioceptive or exteroceptive [3]. This includes the:

1. Commanded Velocities as these are accurate due to the feedback loop controller. Then are put through the robot model equations to obtain the robot's pose. This is used to make a state model prediction.
2. Wheel encoders as these keep track of the wheel rotations. Then are put through the robot model equations to obtain the robot's pose.
3. Magnetometer is limited to providing information about the yaw of the robot. It is assumed that the magnetometer has a resolution of 45 degrees.

These sources of odometry of the robot is combined to obtain a more accurate reading of the robot's pose than by any single means of tracking. The selected filter for this purpose is the Alpha-Beta-Gamma filter [4]. Generally,

$$prediction = state\ model\ prediction + sensor\ gain * sensor\ innovation$$

Each source is assigned gains which act as weights to how confident the outputs of the source are. The commanded velocities and the wheel encoders being proprioceptive sensors have equal weightage and thus the gain is 50%.

Meanwhile, the magnetometer being an exteroceptive sensor and a resolution of 45 degrees is given 100%. However, this 100% confidence is placed at the change in sensor readings rather than the readings themselves. This involves discretizing the current state model prediction into 45 degrees bins and comparing that to the current reading of the magnetometer.

```
/**
 * @brief Updates the yaw angle of the robot's odometry based on various sensor inputs.
 * @param yaw The current yaw angle of the robot.
 * @param angZ The angular velocity of the robot.
 * @param rpmL The left wheel's rotational speed in RPM (Revolutions Per Minute).
 * @param rpmR The right wheel's rotational speed in RPM (Revolutions Per Minute).
 * @param magneto_yaw The yaw angle measured by the magnetometer.
 * @param dt The time step for the update.
 * @return The updated yaw angle of the robot's odometry.
 */
float update_odometry_yaw(float yaw, float angZ, float rpmL, float rpmR, float magneto_yaw, float dt) {
    float output_yaw;
    // Predict the state of the robot
    float state_yaw = yaw + angZ * dt;

    // Predict the state of the robot using encoders
    float encoder_yaw = yaw + calculate_angular_z(rpmL, rpmR) * dt;

    float encoder_yaw_innovation = encoder_yaw - state_yaw;
    float magnetometer_yaw_innovation = magneto_yaw - discretise_yaw(yaw: state_yaw);
    if (abs(magnetometer_yaw_innovation) > 0.0){
        output_yaw = state_yaw + MAGNETO_FILTER * magnetometer_yaw_innovation;
    }
    else{
        output_yaw = state_yaw + ENCODER_FILTER * encoder_yaw_innovation;
    }

    output_yaw = wrap_to_pi(angle: output_yaw);

    return output_yaw;
}
```

Figure 35 Code Snippet of sensor fusion of robot yaw.

8.5 Robot Task

The robot tasks consist of all the actions the robot needs to take to carry out the process to search for the ball, go to the ball, collect, and deliver the ball then go back to searching for a ball. Each robot task takes sensor readings and robot states as inputs. It makes decisions based on this input and returns robot commands as well as the task to run in the next cycle depending on success criteria or failure criteria of the task completion.

8.5.1 Release to Start Button

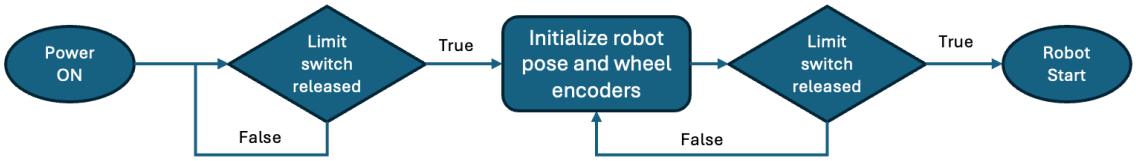


Figure 36 Release to start block diagram.

The robot uses a release to start function on startup where the limit switch at the arm delivery position is used. The objective of this is to enable the robot to run instantly at the start of the clock as to have a competitive edge over opponents with similar drive systems.

8.5.2 Avoid Edge Status

This Avoid Edge task's main aim is to ensure the robot remains within the boundary marked out by the yellow lines. To detect these yellow lines, IR sensors have been placed at all 4 corners of our robot. When one or more of these IR sensors have been triggered, together with the orientation the robot is facing, the robot is able to find out which wall it has hit. From there, the robot would re-orientate itself to ensure that it moves away from the line.

If the robot is facing perpendicular to the wall, the robot would rotate 90 degrees and continue with the previous task it was doing before the edge is detected. The robot's location is also considered where if it is located at the top half of the arena, it would rotate 90 degrees counterclockwise and vice versa. Furthermore, if the robot is facing the wall at an angle, we will orientate the robot as shown by the equation below:

If the robot is facing perpendicular to the wall, the robot would rotate at 90 degrees and continue with the Search task. The robot's location is also considered where if it is located at the top half of the arena, it would rotate 90 degrees counterclockwise and vice versa. Furthermore, if the robot is facing the wall at an angle, we will orientate the robot as shown by the equation below:

$$\alpha = 90 - yaw$$

$$rotate\ angle = 180 - 2 \times \alpha$$

Where the values are taken as shown in the Table below.

The direction is rotation follows as per the robot facing the wall perpendicularly, where if it's in the top half of the arena, it rotates counterclockwise and vice versa.

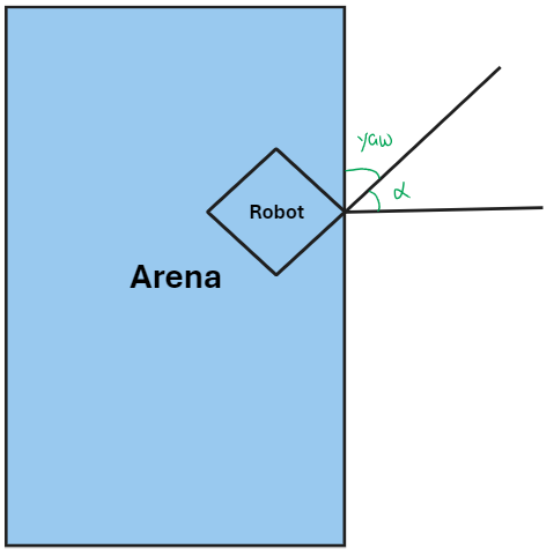


Figure 37 Schematic showing the angles needed to be used to calculate the rotation angle to move the robot away from the wall.

Yaw value	FR	FL	FR or FL	α	distance with starting point(cm)	rotate angle	rotate direction
-175<Yaw<=-85	1	0	/	180-yaw	/	180-2* α	ccw
	0	1	/	yaw-90	/	180-2* α	cw
-85<Yaw<=-75	/	/	1	/	>120	90	cw
	/	/	1	/	<120	90	ccw
-75<Yaw<=-5	1	0	/	90-yaw	/	180-2* α	ccw
	0	1	/	yaw	/	180-2* α	cw
-5<Yaw<=5	/	/	1	/	/	180	cw
5<Yaw<=85	1	0	/	yaw	/	180-2* α	ccw
	0	1	/	90-yaw	/	180-2* α	cw
85<Yaw<=95	/	/	1	/	>120	90	ccw
	/	/	1	/	<120	90	cw
95<Yaw<=175	1	0	/	yaw-90	/	180-2* α	ccw
	0	1	/	180-yaw	/	180-2* α	cw
175<Yaw or Yaw<-175	/	/	1	/	/	180	cw

Table 3 Direction and angle of rotation of the robot based on its orientation and sensors triggered

8.5.3 Home Task

The objective of the home task is to bring the robot to the center of the arena where the likelihood of detecting a ball is the highest. It does this by first rotating the robot to face the front wall then to move towards the center of the arena while making small adjustments to the angular velocity so that the yaw is kept at zero. While moving, it checks if any balls are detected. The success criteria of this task are if the distance travelled by the robot is half the arena length or if a ball is detected while enroute.

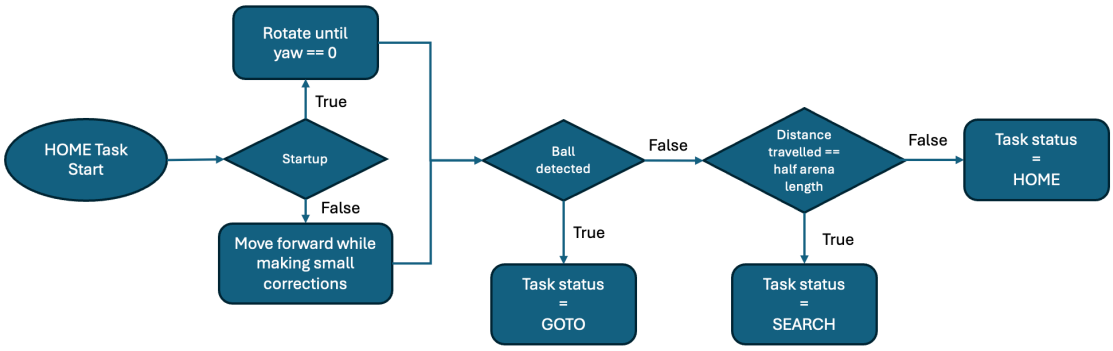


Figure 38 Block Diagram of Home Task

8.5.4 Search Task

The search algorithm is simple. It tells the robot to rotate counterclockwise while looking for the presence of a ball. If it has done a full 360-degree rotation without detecting any ball, it will change its search position by moving forward around 40 centimeters before continuing to rotate and scan for ball again. If anything is detected (except for the opponent robot if it is detected with the short distance sensor on the top of the robot), it will then record the yaw position at which it detects the ball and passes the task over to Goto.

Unfortunately, due to the limitations of the hardware, doing an exact 360-degree rotation is impossible. There is always a slight error in the final yaw position, which in most cases is undershooting. Therefore, the program is created to do the rotation for the scanning in an alternating direction to try and compensate for the inaccuracy.

After testing, we also spotted another problem with the interaction between Search and Edge Avoidance task. If the robot detects an edge while rotating to scan for ball, edge avoidance task would be triggered to move away a little from the edge. However, it is not far enough away such that, as the robot goes to Search task again, it will rotate again and hit the same edge again multiple times before breaking away. Therefore, we program it such that the robot will move further away from the edge if the edge is detected during its search for ball.

8.5.5 Go To Task

The Goto algorithm is the tricky part. It is responsible in steering the robot towards the ball when the Search task detects something. Had the detections from the Search task is accurate and it always detects a ball, it would be easy to set the robot to move towards it depending on which sensor is triggered. However, the actuality is not as pretty as hoped. Therefore, software protocols are implemented to combat these limitations.

Firstly, since the robot cannot immediately stop its motion when it detects the ball, an algorithm is set to orientate the robot to where it last detected the ball based on recorded yaw readings. However, the yaw position is only accurate to 45 degrees as our compass only has 8 different, equally -spaced states. There is only so much we can do with odometry to try and improve this accuracy. Because of this, if after going to the last detected ball's yaw the robot still has yet to detect anything, it will do a sweeping motion up to 45 degrees away from the last detected ball's yaw to hopefully find the ball. If it does not, then it will run Search task again.

```
//turn to ball_yaw when first started
if(goto_startup_phase){
    back_to_search_counter = 0;
    go_to_target_yaw(ball_yaw, yaw, 1);

    //stop go_to_target_yaw() when the robot has either faced towards ball_yaw or the ball is already detected
    ball_location = get_ball_location(left_sensor_dist, right_sensor_dist, mid_sensor_dist, top_sensor_dist, opp_detected, yaw);
        if(fabs(yaw-ball_yaw) <= YAW_TOLERANCE || ball_location != 0){
            goto_startup_phase = 0;
                goto_linX = 0.0;
            goto_angZ = 0.0;

            //check if sweep is needed (needed when already reached ball_yaw, but no detections are ever found)
            if(ball_location == 0){
            //initialize sweeping movement
                    goto_sweep_counter = 0;
                    activate_goto_sweep = 1;
                    goto_linX = 0.0;
                    goto_angZ = -15.0;
            }
        }
    return GOTO;
}
```

Figure 39: Code snippet on how the robot reorientates itself towards the last detected ball

When something is detected, it will go towards it. If the object then enters the collection chamber, the middle sensor reading will be able to recognize that it is ready for collection, passing it onto Collect task. However, if during the approach the short distance sensor at the top detects something with a distance like the one detected by the middle sensor, it will recognize that it is an opponent and will instead back away from it and continue to search for a ball.

Another case is if the robot detects the back wall instead (the place to deliver the ball). In this case, the left, right, and middle sensors will all detect the wall. In this case, the robot is programmed to reorient itself away from the wall, and then go back to Home task.

```
//WALL
if(activate_wall_protocol){
    go_to_target_yaw(0, yaw, 0);

    if(fabs(yaw) < YAW_TOLERANCE){
        activate_wall_protocol = 0;
            return HOME;
    }
    return GOTO;
}
```

Figure 40: Code snippet on how the robot performs when detecting a wall

Finally, in cases where the balls are on the edges of the arena, we noticed that the robot's line sensors will detect the arena's edge and execute Edge Avoidance task before it can collect the ball. To combat this, a variable 'goto_ignore_edge' is created for the scheduling between Goto task and Edge Avoidance task. With it, we program the robot to try capturing the ball by closing the gate if it hits an edge during Goto task before passing over the control into Edge Avoidance task. After finishing edge avoidance, it will then check whether a ball is successfully captured inside the chamber. If it does, it will directly go to Deliver task. Otherwise, it will go back to Search task.

8.5.6 Collect Task

In this task, once the Goto ball has completed and the various sharp distance sensors confirm that it is indeed a ball and not a wall or opponent detected, the servo would then be activated to bring the collector arm down to a 90 degrees position to collect the ball. After the collector arm has captured the ball, the middle sharp distance sensor placed within the ramp would detect and confirm if a ball is indeed in the chamber. If the ball has been confirmed to be collected, it would move on to the Delivery stage, else it would return to the Search task to look for the ball again.

In this task, once the Goto ball has completed and the various sharp distance sensors confirm that it is indeed a ball and not a wall (mid distance sensor) or opponent detected (top distance sensor), the servo would then be activated to bring the collector arm down to a 90 degrees position, pushing the ball into the chamber in the process to collect the ball. After the collector arm has captured the ball, the middle sharp distance sensor placed within the ramp would detect and confirm if a ball is indeed in the chamber. If the ball has been confirmed to be collected, it would move on to the Delivery stage, else it would return to the Search task to look for the ball again.

```
//check if ball is ready to collect
if(distance_sensor_mid <= READY_TO_COLLECT_THRESHOLD){
    collect_linX = 0.1;
    collect_angZ = 0.0;
    collect_servo = SERVO_COLLECT_POSITION;
}
else if(detect_ball(99.9 ,99.9 , distance_sensor_mid, distance_sensor_top, opp_detected, 0.0) == 1 && distance_sensor_mid <= BALL_THRESHOLD_MID){
    collect_linX = 0.2;
    collect_angZ = 0.0;
    return COLLECT;
}
else{
    collect_reset_wait --;
    collect_servo = 0.0;
    if (collect_reset_wait <= 0){
        return SEARCH;
    }
}
```

Figure 41: Code to call for collection of ball or return to Search task if ball in chamber not triggered

8.5.7 Deliver Task

Once the ball has been collected and is in the chamber, the Deliver task is called. Here, the robot's goal is to reorientate itself with its back to the back wall (yaw = 0) then move towards it while making minor adjustments. This is accomplished by determining the yaw error, then applying an angular velocity in the negative direction of the yaw error. Once the robot has arrived at the back wall, two limit switches and the line sensors will be activated. The limit switches are to distinguish the back wall's edge from the other edges, and the line sensors are to prevent opponent robots from prematurely triggering the swing arm. The servo would then be activated to swing the arm backwards to push the ball from the chamber into the delivery box. Once the task has been completed, the robot would reset itself back to home status and reset the robot's position in the x-axis before going out to retrieve the next ball.

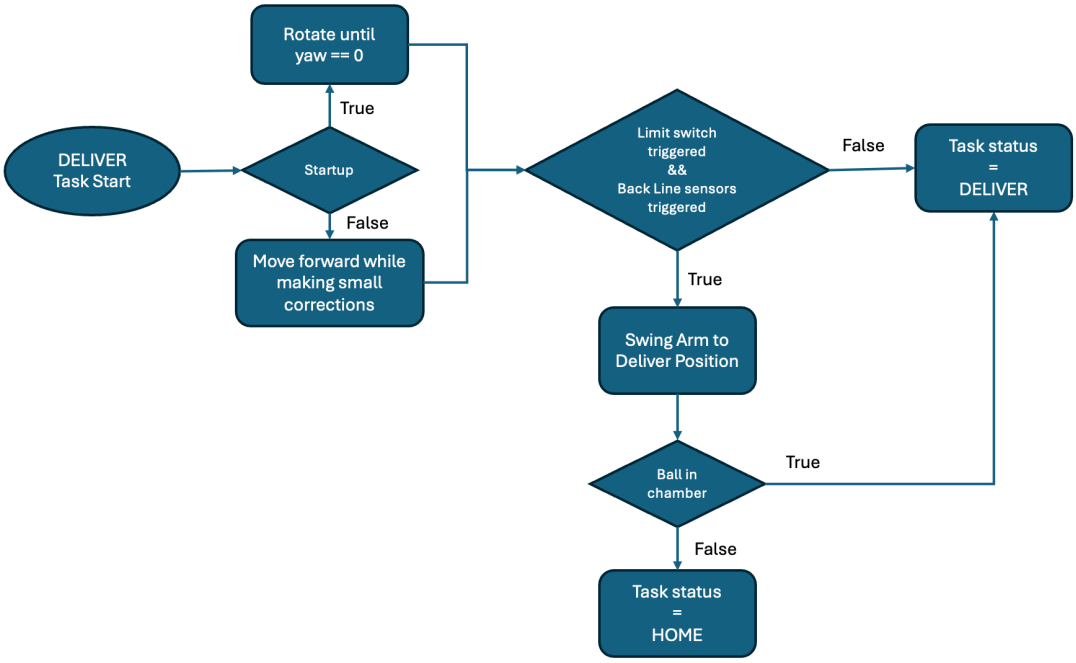


Figure 42 Block Diagram of Delivery Task

9. Improvements and Limitations

9.1 Robot Checklist

As mentioned in the Competition Outcome section, the robot had hardware issues that led to the robot faults. A checklist of features of the robot to test would help identify these issues prior to competing in the arena.

9.2 Drive System Gearing

During the competition, it was observed that the robots geared up were faster which allowed them to collect balls faster. Since all teams are given the same equipment, the drive system of the robot could have been geared up to improve robot speed.

9.3 Servo Arm

The arm position of the robot was inferred from the timing of limit switch triggers in the swing arm encoder. An issue that arose was when the robot collided with other robots, the swing arm position would go off as the arm is pushed inwards from the closed position.

10. Reflection

No	Name	Reflection
1	Timothy	Reflecting on this project, I began as a novice in the realm of CAD design, with Fusion 360 being a territory largely uncharted for me, my only experience being in MA3071 Lab. My journey started with simple clicks and tentative sketches, but as the project advanced and together with Hong Wei's guidance, so did my proficiency. With each passing day, the complexity of shapes and the subtlety of designs I could craft grew exponentially. All the way from week 1, our team's spirit was a truly melting pot of ideas, where collective wisdom was tempered into innovative solutions, where we started creating different ball collecting designs from week 3. As the weeks went by, although we were confronted with a myriad of challenges that tested our resolve, we found unity in our relentless pursuit of solutions (And for that I'm thankful we managed to persevere and push through). The line sensor's suboptimal height, the robot's initially oversized frame, and the front ball bearings' failure to contact the ground were but a few of the puzzles we embraced. Our approach was rooted in collaboration; we gathered, debated, and iteratively refined our designs, each iteration adding a layer of improvement and insight. The week leading up to the competition was a crucible of intensity. We met nearly every day, and our focus was singularly trained on tweaking, rebuilding, and enhancing our robot, showing our dedication and testaments to our commitment to excellence. Among the countless memories of triumphs and defeats, one stands out distinctly: in week 10, with the robot still veering off course, the hardware team made a bold decision to disassemble the entire construct. We tested various motors, meticulously searching for the perfect combination that would grant our robot the precision to navigate straight. This moment, emblematic of our collective resolve, was not just a technical triumph but also a defining experience in collaboration and perseverance. Although our final position was 5th out of 8 teams, this did not dim the sense of achievement that radiated from us as champions of the lower bracket. We took immense pride in our shared journey - the intense camaraderie, the late nights punctuated by the hum of the 3D printer, and even the last-minute print failures on the day of the competition. Each setback was met with a steadfast spirit and a determination to succeed. The robot, with all its sensors snug in their housings and its sturdy frame, was a testament to our dedication and growth. It was more than a machine; it was a chapter in our lives where we all turned unfamiliarity into expertise, obstacles into steppingstones, and a group of individuals into a cohesive, problem-solving team. The competition may have ended, but the skills and memories forged will last a lifetime 😊.

2	Hong Wei	I always enjoyed doing group projects as this is a chance for me to collaborate with other talented individuals and see how our ideas can come together for the success of the project. As part of the mechanical team, I was privileged to be able to work alongside Timothy and Chen in designing the robot. Our project presented a unique challenge: working with a limited hardware kit and limitations in 3D printed components. This constraint pushed us to think creatively and find innovative solutions to achieve our goals. One of the most significant lessons that I learned was the difference between what I envision in my CAD design and real-life implementation. While our initial designs looked promising on paper, translating them into a functional robot revealed unforeseen obstacles. This experience has taught me the importance of practical testing and the need of swift iterations in the design process. Moreover, communication with the software and electrical teams highlighted the critical role of understanding design requirements. You could have a perfect mechanical design, but if it does not align with the specific needs, it still could not be used at the end of the day. Part of our group's success in delivering a working robot lies in our team's open communication and our willingness to take in feedback. Even though our team placed 5th out of 8 teams, I am still extremely proud of our efforts. Throughout the project, we worked together tirelessly, and almost every single member of the team was present in each of our discussions and meetings. In general, my experience in MA4012 has been transformative and I would say I have grown both professionally and personally. This project has well prepared me to be a better engineer and I am more confident in my ability to tackle future engineering challenges.
3	Kelvin Lee	Reflecting on my project journey, several key lessons emerge that warrant careful consideration for future learnings. Foremost among these is the imperative to enhance communication channels between teams. My project encountered delays and inefficiencies stemming from inadequate communication, resulting in repeated testing, component changes, and reiterations. Clearer lines of communication would have mitigated these setbacks and fostered smoother progress. Furthermore, I acknowledge the importance of swift and frequent testing. While theoretical groundwork is essential, my experience underscores the value of practical experimentation. Problems often manifest during testing that may not be evident in theoretical planning. By prioritizing early and continuous testing, I can promptly identify and address issues, preventing prolonged delays and costly revisions. Another critical insight gained from my project is the significance of testing components within the context of the entire system. The team's tendency to isolate components for testing led to unforeseen performance differentials upon integration into the overall robot. Moving forward, I recognize the need to allocate more time and

		resources to holistic system testing, ensuring that each component functions optimally within the larger framework. Moreover, I recognize the importance of leveraging existing solutions rather than reinventing the wheel. My project experienced unnecessary delays and uncertainties when attempting to develop novel solutions instead of building upon established practices. Embracing proven methodologies and technologies allows me to allocate resources more efficiently and focus on refining and enhancing existing systems, rather than starting from scratch. Additionally, an important lesson learned is the necessity of testing with opponents. I observed that robot behavior can vary significantly when tested in the presence of opponents compared to solitary testing conditions. Testing against opponents provides invaluable insights into real-world scenarios and enables me to fine-tune my strategies and algorithms accordingly. Lastly, I wanted to say that this mod taught me a lot of skills that will make me question more things in the future. To close, I would like to thank the other group members that have spent their last few weeks in university dripping in sweat to complete the project and achieve a great outcome!
4	Benjamin Teh	This tennis ball robot challenge provided a platform for me to practice, enhance and demonstrate my skills the learnings that I have picked up throughout my university. This included both soft skills such as communication, collaboration, and leadership, as well as hard skill such as code organization, system design, and algorithms. As the team leader I think I did somewhat well in organizing the team into functional groups in the initial stages of the project as well as establish communication (weekly meetings), project management (project management), and project execution (GitHub and Fusion360) means. However, I think I did not maintain this consistency throughout the project as the workload from assignments outside this project came full swing. I could've done more to keep an eye on all on going progress for all functional teams. This is to ensure that all parties were kept informed to avoid miscommunication for the interfaces between mechanical, electronic and software components, and to brainstorm solutions to edge cases. I am grateful to the team because despite this, each member managed themselves well and filled in for each other's shortcomings. Working in the software team, I realize that what may seem obvious to one, may not seem as obvious to another due to differences in coding styles, familiarity with software, and experience in coding. This comes in the form of how to use the software development tools, how the code is abstracted and structured, and the logical flow of the robot tasks. However, through discussions, many code comments, and trust, each member's ideas and work were successfully integrated into the robot's

		tasks. Here, the phrase “Keep it short and simple” is paramount in coding within a project team. Finally, in the days leading up to and during the competition. My team banded together to get all three components: mechanical, electrical, and software in harmony with each other. Improvements and changes to the design of the hardware were quickly implemented and tested. Software edge cases were found during testing at the lab’s arena and considered for. Yet on the day itself, there were new cases to be considered especially cases with interacting with the opponent robot. I am grateful to the team’s quick identification of problems and quick ideation to solutions. About the project and tournament, I think that teams could be slightly smaller so that there could be more teams during the competition, and so that more people may learn more from the experience as there would be more opportunities to fill in. Additionally, as an alternate means to determine the winner, I suggest a Swiss-System tournament like in chess so the very best teams are brought to the finals as well as to allow the course to grow in number of teams participating. Overall, this experience was invaluable to me. In making many attempts, I made a lot of mistakes, and in making many mistakes, I have learnt much. I am sure that the lessons and bonds that I’ve gathered here will be with me for life. I hope this module continues for years to come. “GO BALL FONDLERS!!!”
5	Bryan Timothy Galenius	This project provided me with an opportunity to learn from it and from my peers. Coming into this project, I thought we would be creating only 1 file of code, having lots of functions inside that one file, including the main function. I learned proper code structuring during this project from Benjamin as he is more experienced in this. In the end, I was able to structure my codes properly, properly contributing to the team. I also learnt for the first time how to properly use GitHub since this is my first, collaborative programming project. I also learnt that often, things that are supposed to work theoretically, performs far from expectation. I am used to things not working as the theories expected it to, but this project I think by far is the most far from the theories. Although in this project theory can help us guide to the general solution (in this case, the architecture of the code and how the flow of tasks should work along with their basic logics), ample time shall be made for actual, fully integrated tests as major changes can keep happening. Additionally, communication between sub-teams is paramount. There was a time where the software team had to rework the code as the mechanical team did not timely inform us that they no longer put any limit switch inside the ball chamber to detect whether the ball is inside. Instead,

		they moved the distance sensor behind, inside the chamber so that it could serve to search for the ball and detect when the ball is inside. There was also another time where the software team was super puzzled on why the sensor readings are not as expected. As we are experienced with Arduino, we thought the analogue inputs would be also in terms of bits, not in millivolts. Software team worked an entire weekend to try and solve this problem, before the whole team meet again on Monday during class and then I talked to electrical team about the problem, finding out that they knew that the VEX analogue input returns millivolts instead of bits. This could have saved the team quite a lot of time. In the future, to solve this problem, it is wise to have a routine meeting between the different sub-teams to ensure alignment between the teams and to update all the sub-teams should any sub-team encounter any issue. It could be just a brief summary of the problem that they are facing and might perhaps have nothing to do with the other sub-teams. But for the rare cases where the solution is already held by other sub-teams, just like our bits vs millivolts case, it could have saved a lot of time. In the end, when it was almost time for the competition, all the sub-teams came together to work on the robot. The amount of progress we achieved during this time was a lot. We keep testing on the spot, fixing and discussing problems together as a group, and covering as much edge cases as possible. This further shows the importance of collaboration with sub-teams. Overall, this project has provided me an unforgettable learning experience. I think universities should be filled with more of these kinds of courses rather than just theories.
6	Fan Zifu	In my reflections on this project, I want to highlight the challenges I faced and the important lessons I learnt during the software part. As a member of the software team, I was presented with a completely new challenge: working on programming a group project for the first time. Initially, I needed to learn programming skills quickly, as it was a programming language I had not used before. This was an exciting opportunity for me, but also a challenging journey. In the initial stages of the project, I spent time learning the programming language and how to share code using GitHub, as well as understanding how to communicate the software effectively with the robot hardware. Along the way, I found myself needing to constantly adapt and refine my programming skills to meet the needs of the project. As the project progressed, I gradually began to feel the joy and challenge of programming. It was especially fulfilling when we could control our robot to move through the code we wrote. As the competition approached, I faced a series of intense challenges with the rest of the team. We needed to make sure that our software could control the robot correctly, that the hardware was stable, that the sensor values were accurate. This required us to work closely together, solve problems promptly, and stay focused on the team's goals. Although we did end up having a mishap in the competition and only got a fifth-place finish. But the entire process was much more important than the ranking.

		Looking back on the whole project experience, I realize the importance of communication. Effective communication helps us better understand the project requirements and promotes cooperation among the team, which leads to the project's success. I also learnt how to stay calm under pressure and how to work together with the team to solve problems. These valuable experiences and lessons will have a profound impact on my future study and career development and make me more confident to face future challenges.
7	Eunice	Coming into this project, I was very comprehensive about the contributions I would make to this team as I wasn't very confident in my skills. However, with the constant support from my teammates who told me to just try my best and for Benjamin who was always ready to lend the helping hand whenever I was stuck or confused about how to go about the code, I was able to encourage myself to just do whatever i could to support my team. As part of the software team, I learned many new ways to go about writing codes where each one of had very different styles of coding such as how to organise the codes to make it easier for tuning and editing from Benjamin. This was one part where we had to communicate well adapt to one another styles to ensure that we will be able to integrate the code smoothly. Moreover, it was the first time was using GitHub to do my codes in a group project which was a little confusing at first but with the help of my software teammates, I managed to learn how to use it. Furthermore, I realised the importance of communication between sub-teams, where at one point, there were changes in design that some of us were not aware of leading to some confusion in the codes being done up. During the competition, our team definitely had the most heart-stopping day. It was as prof had prophesized since Week 1, "Whatever you thought won't go wrong will all go wrong on the day of the competition.":((. It felt as if all our 12 weeks of work had literally gone down the drain in an instant until we came clutch in the loser bracket, getting 5th place as redemption for ourselves. It definitely wasn't the smoothest ride, but it was definitely without a doubt one of the best experience I've had in any mod these 4 years. I cannot thank my team enough for tanking the entire project and putting so many sleepless nights into it. Thank you!!! I have once again realised the importance of teamwork and communication in order to get through the entire process, not just in building the robot, but also, taking care of one another and ensuring everyone's coping well as well.

8	Lee Chen Hui	This tennis ball competition has definitely been one of the highlights of university. I have always enjoyed the application of my knowledge, building robots and creating products just from my imagination ideas. This project pushed my collaboration skills, communication skills, along with time management. Our group, the Ball Fondlers. Was and is extremely coordinated and allowed for extremely precise work distribution. Our team all put in an amazing effort, working within our sub-groups of mechanical, electrical and software. The communications between the sub-teams were extremely proficient, allowing for troubleshooting and optimization of the robot. Members within subgroups assisted other subgroups when there were tough problems. Personally, I felt the challenge when attempting to design the robot swing arm along with the swing arm encoder. This proved difficult due to the tight integration between hardware, software, and electronic integration. Creating multiple CAD designs and attempting to find the limitations of my 3D printer, the designs challenged my ability to design products within a group. Usually working alone, it was simple for me to consider all the restrictions. Working together with a larger group required a lot more back and forth confirmations before iterations came to fruition. This issue was only exacerbated when all of us were forced to juggle the time management of our final year courses and FYP projects along with this project. Finding time for all the members of the teams to meet was nearly impossible. Many of the meetings had to settle for smaller groups that usually lacked members to progress fluently. An adjustment I would recommend to the competition is to reduce the restrictions. I believe that due to all the restrictions, only a few robot types could actually be built. A lot of strategies and ideas are not allowed. This limited the creativity that could be implemented in the competition. I believe that when loopholes are discovered, the rules don't need to be immediately changed to ban the strategy. I believe that teams should find workarounds that counter the strategy. This makes the competition much more interesting and challenging.
9	Wilson	Reflecting on the past 12 weeks + 1 week (recess week) working on this project has been quite an eye-opening experience for me. Throughout my university years, I have never been part of an electrical team in a group project before. In fact, whenever robotics projects came up, I instinctively gravitated towards the software side of things. However, delving into the electrical aspect of the project introduced me to a set of challenges I hadn’t previously considered, particularly dealing with sensors. One particular revelation that struck me was the deceptive nature of distance measurement by the sharp distance sensors provided. In our case, realizing that a range from 10 cm to 80 cm actually meant

		10 cm to 25 cm was quite astounding. This realization emphasized the critical importance of rigorously testing and verifying each sensor's performance to ensure we have an accurate information of their true practical capabilities. Even though I had prior knowledge of the importance of communication in group settings after taking Engineering Communication, this project reiterated its significance in a more subtle way. There was a week in which we were engaged in our respective tasks in the lab. While the software team was working to decipher the meaning and representation of the output values from the sensors, it dawned on us in the electrical team, including myself, that we had already known that the output was in millivolts. This truly highlight a crucial breakdown in communication between subgroups. Despite having this information that could have expedited software team progress, we failed to effectively communicate with them. Reflecting on this experience, I recognized the importance of proactive communication with other subgroups. While each team may focus on specific aspects of the project, it is essential to maintain open communication to ensure that important insights and information are shared promptly. Additionally, I also believe that testing and iteration is crucial in any robotics projects because theoretical planning can only take us so far. While meticulous planning and design are essential, it's during testing and iteration phases that we truly discover how our robots perform in real world. Overall, I would say that this project truly taught me important lessons about robotics projects, encompassing both technical and non-technical aspects. From technical standpoint, I gained a deeper understanding of the complexities involved in designing, building, and testing robotic system. Beyond technical aspect, this project underscored the significance of effective communication. Personally, I think it would be nice if more courses are made similar to this where we can learn more about robotics by making robots directly and work in groups.
10	Wilsen Lotan	Looking back to the very first session of this robotics module. I never thought a competition-based module would be one of my memorable experiences in my university life. Soon to be working in a field not so relevant to robotics, I found my passion rekindled in practical problem solving, group work, and ultimately competition. As soon as we matched and grouped, I realized I was surrounded by great engineers who were passionate about taking this project to the next level. Observing the team dynamics, and the interest of my teammates, I decided to take the opportunity to work on the electrical portion of the project. This was an eye-opening journey for me, as I and the electrical team tested various components, especially the sensors of our robot extensively.

		Early on some of the week, we allocated time outside of our class to test the components. However, tthe results were not what we hoped for, as the sensor reliability fell low compared to what we expected, we were even doubting the quality of our sensor. Over the course of this testing period, we would go on and ask for sensors and component replacement, having tested more than 10 sharp distance sensors of the same kind, and many more. This is due to the design requirement the software team asks us, in hopes of having accurate readings and theoretically accurate performance. We were even asked to develop an electrical filter, and instruments to solve the problem electronically. This is where I found that communication is really paramount in the group. As each component of the team has its wants and needs, clashes of interest and arguments are bound to happen. Fortunately, we were able to come to an amicable agreement. However, I realized that we had reached that stage too late. We shouldn't have wasted our resources on fixing problems that shouldn't even be solved. I knew that two MCU communications that were originally intended for our robot design were not realizable in such a short time span. Not being the leader, alongside a grouping of ten members gets the best of me. Moreover, the crunch for FYP and other responsibilities takes away the crucial time for the majority of the team members. Nevertheless, after realizing the gravity of our actions, in week 11, I strive to push and recover from our mistakes. Fortunately, I was blessed with teammates who were willing to dedicate themselves too for this project. Benjamin, the team leader, deserves the most recognition for handling the robot and goes as far as creating an arena in his own room and testing beyond the lab operational hours. While we ended up on 5th place out of eight teams, I was more than happy of our achievement. This robot is a labor of love of everyone involved. Despite mishaps on the early part of the competition, we never gave up and were able to win the rest of the matches, which shows our spirit and determination. Overall, I'm more than proud to be part of this amazing team

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Appendix

Appendix A GitHub

https://github.com/bentjh01/MA4012_Ball_Fondlers.git

Appendix B Datasheet

B1. Sharp Distance Sensor (4 to 30cm)

SHARP GP2Y0A41SK0F

GP2Y0A41SK0F

Distance Measuring Sensor Unit
Measuring distance : 4 to 30 cm
Analog output type

■Description

GP2Y0A41SK0F is a distance measuring sensor unit, composed of an integrated combination of PSD (position sensitive detector) , IR-LED (infrared emitting diode) and signal processing circuit.
The variety of the reflectivity of the object, the environmental temperature and the operating duration are not influenced easily to the distance detection because of adopting the triangulation method.
This device outputs the voltage corresponding to the detection distance. So this sensor can also be used as a proximity sensor.

■Agency approvals/Compliance

1. Compliant with RoHS directive (2011/65/EU)

■Applications

1. Cleaning robot
2. Personal robot
3. Sanitary

■Features

1. Distance measuring sensor is united with PSD, infrared LED and signal processing circuit
2. Short measuring cycle (16.5ms)
3. Distance measuring range : 4 to 30 cm
4. Package size (29.5 × 13.0 × 13.5mm)
5. Analog output type

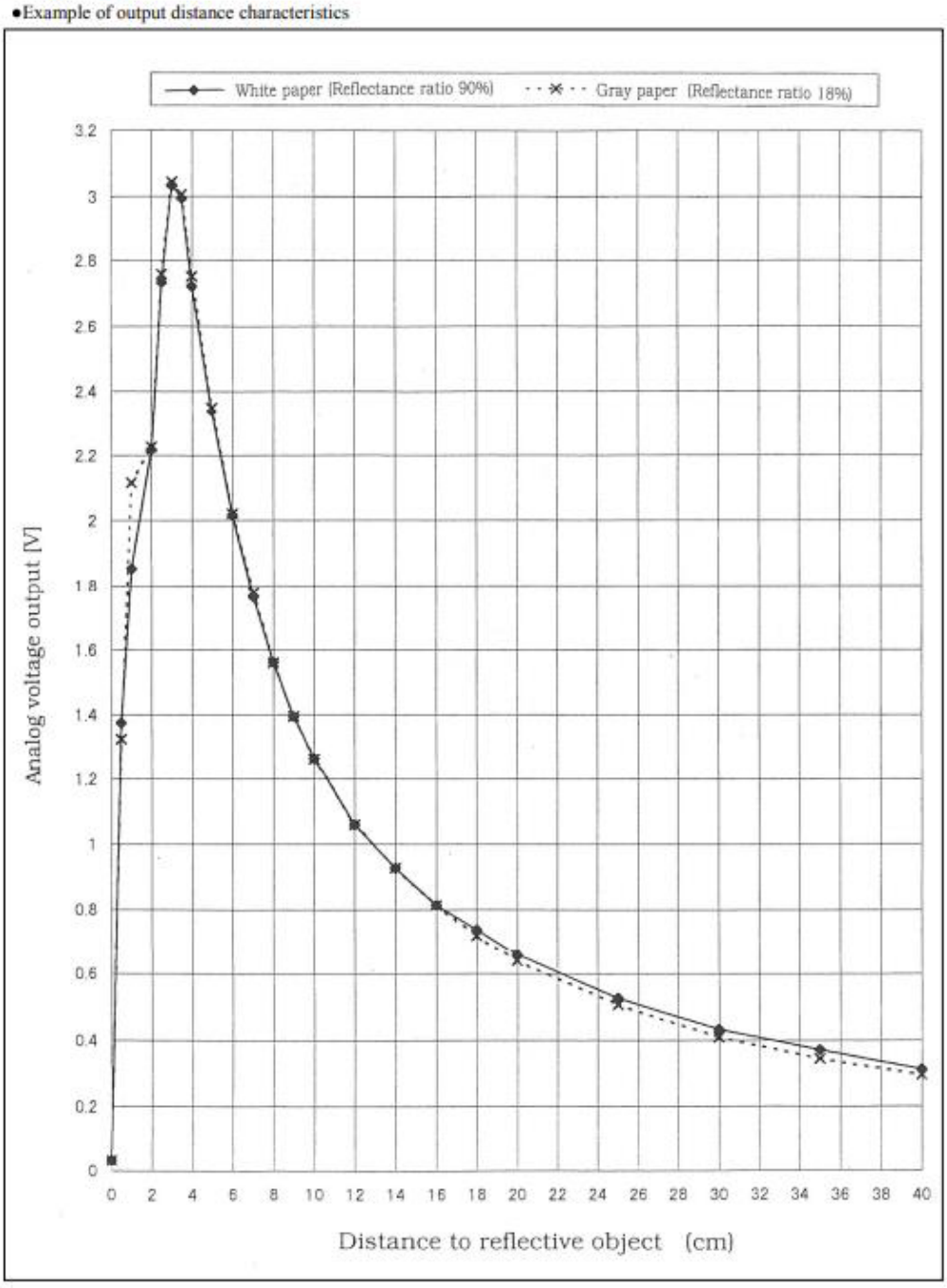
Notice The content of data sheet is subject to change without prior notice.
In the absence of confirmation by device specification sheets, SHARP takes no responsibility for any defects that may occur in equipment using any SHARP devices shown in catalogs, data books, etc. Contact SHARP in order to obtain the latest device specification sheets before using any SHARP device.

Sheet No.: OP13008EN

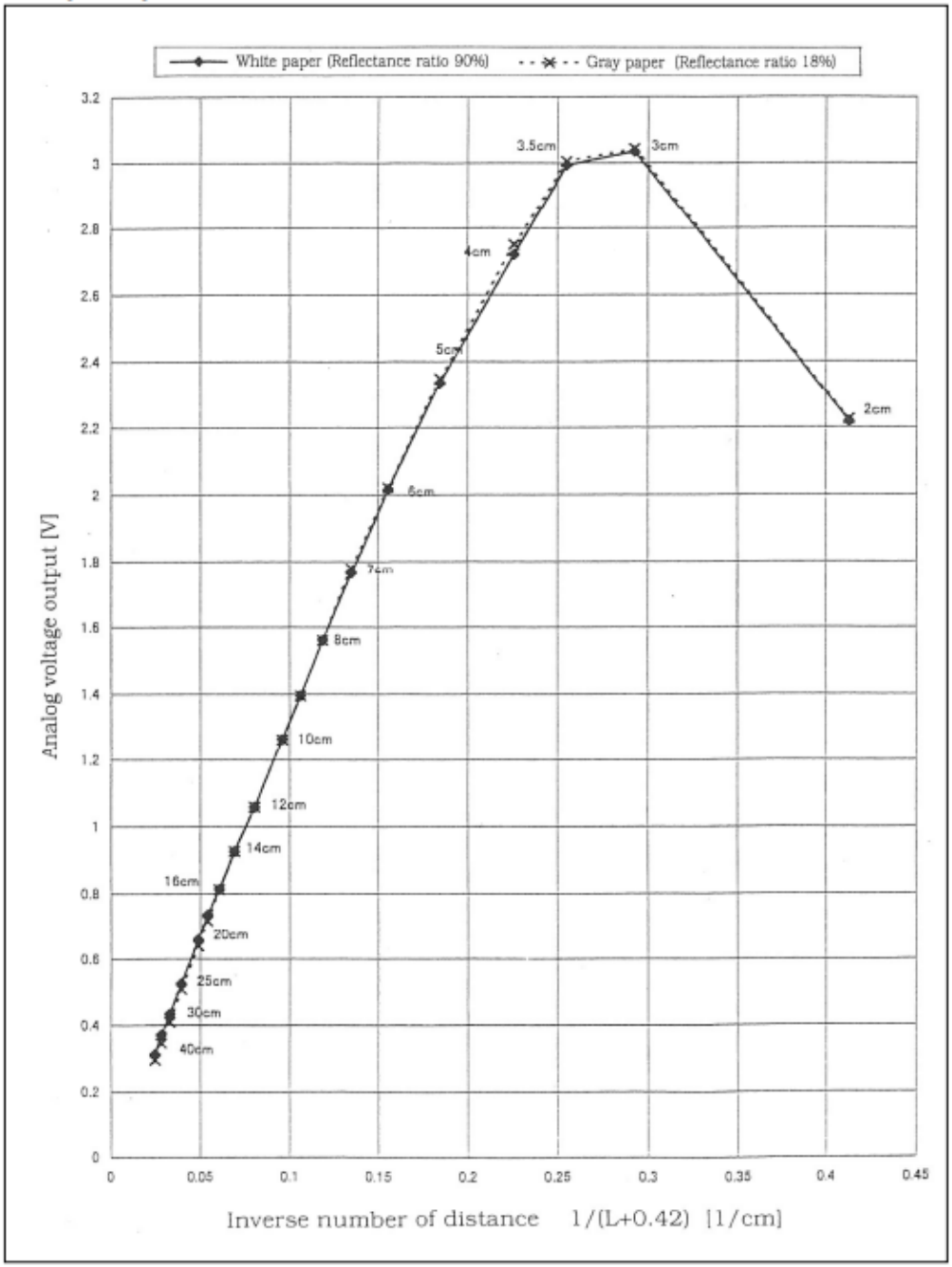
Figure 43: Sharp Distance Sensor 4-30cm

■Supplements

●Example of output distance characteristics



●Example of output distance characteristics with the inverse of distance



B2. Sharp Distance Sensor Long range (10 to 80 cm)

GP2Y0A21YK0F

Distance Measuring Sensor Unit
Measuring distance: 10 to 80 cm
Analog output type

■Description

GP2Y0A21YK0F is a distance measuring sensor unit, composed of an integrated combination of PSD (position sensitive detector) , IRED (infrared emitting diode) and signal processing circuit.
The variety of the reflectivity of the object, the environmental temperature and the operating duration are not influenced easily to the distance detection because of adopting the triangulation method.
This device outputs the voltage corresponding to the detection distance. So this sensor can also be used as a proximity sensor.

■Features

1. Distance measuring range : 10 to 80 cm
2. Analog output type
3. Package size : 29.5×13×13.5 mm
4. Consumption current : Typ. 30 mA
5. Supply voltage : 4.5 to 5.5 V

■Agency approvals/Compliance

1. Compliant with RoHS directive (2011/65/EU)

■Applications

1. Touch-less switch
   (Sanitary equipment, Control of illumination, etc.)
2. Robot cleaner
3. Sensor for energy saving
   (ATM, Copier, Vending machine)
4. Amusement equipment
   (Robot, Arcade game machine)

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Figure 44: Sharp Distance Sensor 10-80cm

GP2Y0A21YK0F

Distance Measuring Sensor Unit
Measuring distance: 10 to 80 cm
Analog output type

■Description

GP2Y0A21YK0F is a distance measuring sensor unit, composed of an integrated combination of PSD (position sensitive detector) , IRED (infrared emitting diode) and signal processing circuit.
The variety of the reflectivity of the object, the environmental temperature and the operating duration are not influenced easily to the distance detection because of adopting the triangulation method.
This device outputs the voltage corresponding to the detection distance. So this sensor can also be used as a proximity sensor.

■Features

1. Distance measuring range : 10 to 80 cm
2. Analog output type
3. Package size : 29.5×13×13.5 mm
4. Consumption current : Typ. 30 mA
5. Supply voltage : 4.5 to 5.5 V

■Agency approvals/Compliance

1. Compliant with RoHS directive (2011/65/EU)

■Applications

1. Touch-less switch
   (Sanitary equipment, Control of illumination, etc.)
2. Robot cleaner
3. Sensor for energy saving
   (ATM, Copier, Vending machine)
4. Amusement equipment
   (Robot, Arcade game machine)

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B3. 2-Wire Motor (393)

Motion Accessories

2-Wire Motor 393

The 2-Wire Motor 393 is the primary actuator used in the VEX Robotics Design System. Build rotational mechanisms using this motor. Drive bases, rotational joints, conveyor belts, or anything that spins can be built using the 2-Wire Motor 393.

Connection

The 2-Wire Motor 393 does not have an internal motor controller.
The motor connects directly to 2-Wire Motor Ports (Ports 1 & 10 on the Cortex Microcontroller). For 3-Wire Motor Ports (Ports 2-9 on the Cortex), a Motor Controller must be used in between the Microcontroller and the motor.

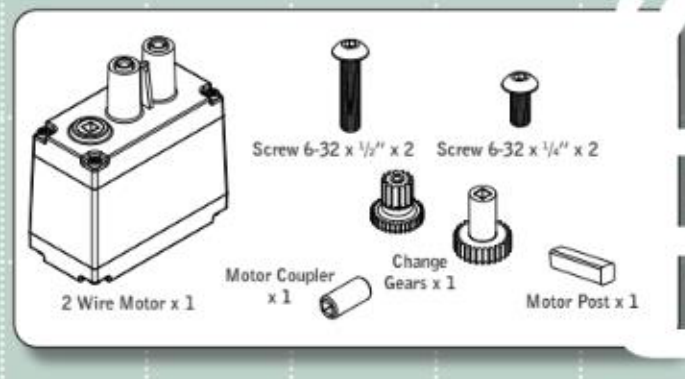
High Speed Option

Want to go faster than the standard motor but still have the same output torque as the standard motor? No problem! The 2 Wire Motor 393 kit can be configured into "high speed" or "turbo" versions. Simply follow the "Gear Change Procedure" step-by-step instructions to increase the output speed by up to 140%.

Motor Specifications

All motor specifications are at 7.2 volts. Actual motor specifications are within 20% of the values below.

Description	As Shipped	High Speed Option	Turbo Option (not included)
Stall Torque	1.67 N-m [14.76 in-lb]	1.04 N-m [9.2 in-lb]	0.70 N-m [6.2 in-lb]
Free Speed	100 RPM	160 RPM	240 RPM
Stall Current	4.8 Amps		
Free Current	0.37 Amps		



Limited 90-day Warranty
This product is warranted by VEX Robotics, Inc. against manufacturing defects in material and workmanship under normal use for ninety (90) days from the date of purchase from authorized Innovation First dealers. For complete warranty details and exclusions, check with your dealer.

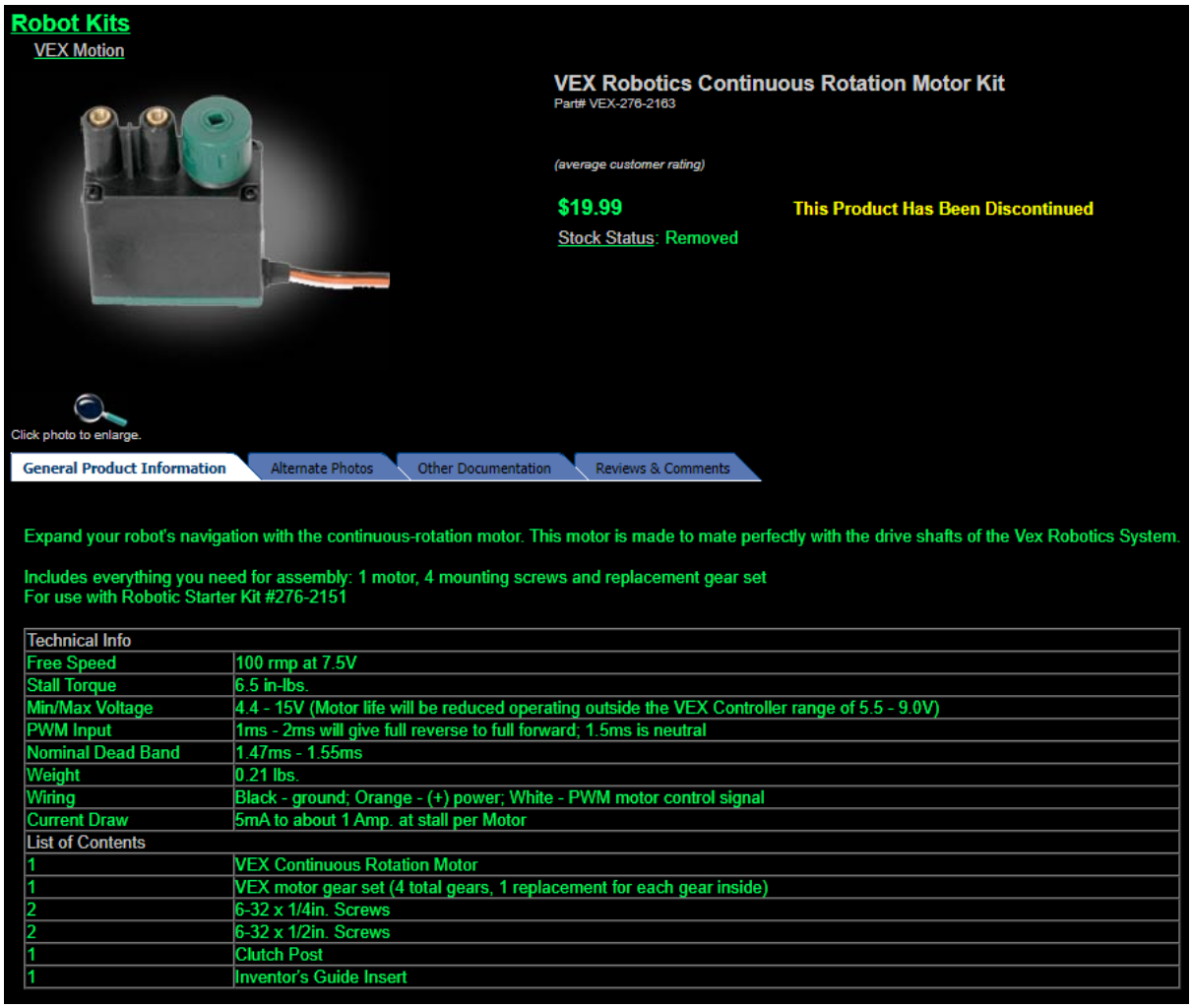
VEX Robotics, Inc.
1519 IH 30 W
Greenville, TX 75402

0414

For More Information, and additional Parts & Pieces refer to:
www.vexrobotics.com

Figure 45: 2-Wire Motor

B4. 3-Pin Motor



Technical Info	
Free Speed	100 rmp at 7.5V
Stall Torque	6.5 in-lbs.
Min/Max Voltage	4.4 - 15V (Motor life will be reduced operating outside the VEX Controller range of 5.5 - 9.0V)
PWM Input	1ms - 2ms will give full reverse to full forward; 1.5ms is neutral
Nominal Dead Band	1.47ms - 1.55ms
Weight	0.21 lbs.
Wiring	Black - ground; Orange - (+) power; White - PWM motor control signal
Current Draw	5mA to about 1 Amp. at stall per Motor
List of Contents	
1	VEX Continuous Rotation Motor
1	VEX motor gear set (4 total gears, 1 replacement for each gear inside)
2	6-32 x 1/4in. Screws
2	6-32 x 1/2in. Screws
1	Clutch Post
1	Inventor's Guide Insert

Figure 46: 3-pin motor

Appendix C Engineering Drawing

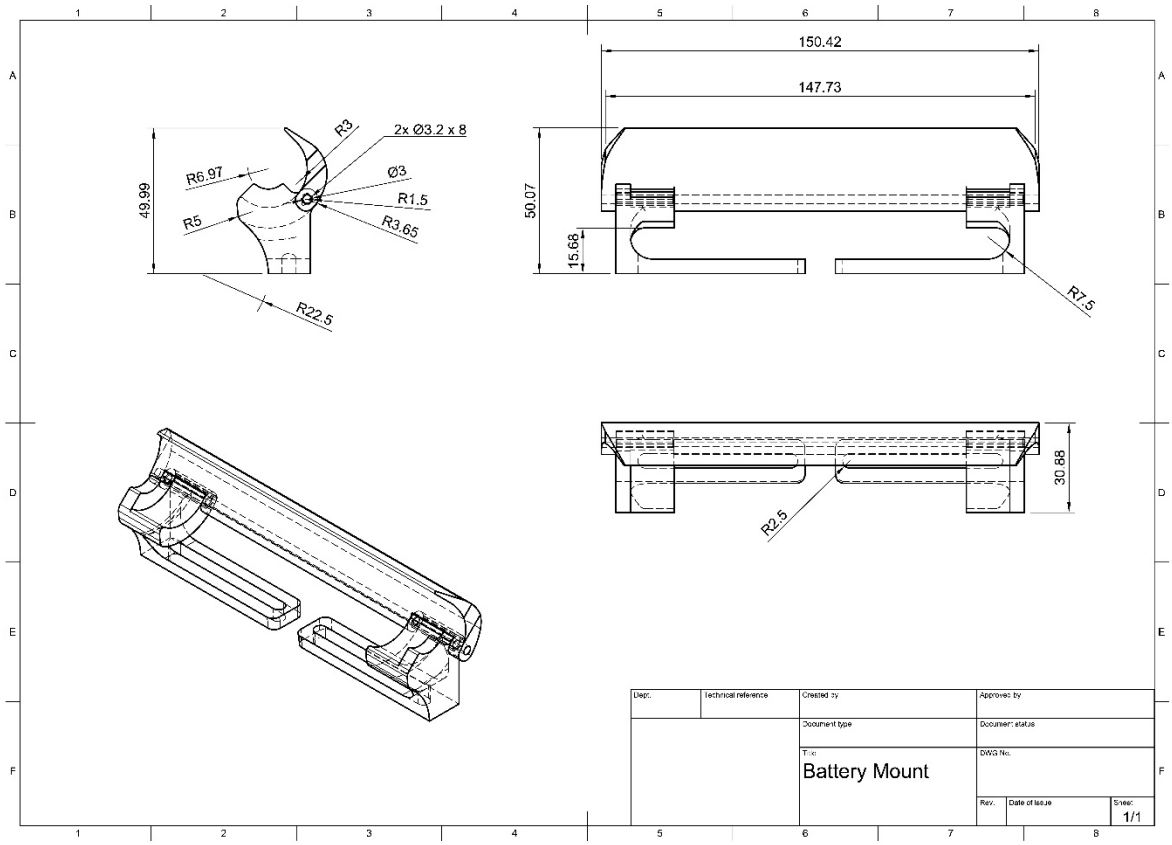


Figure 47: Battery mount

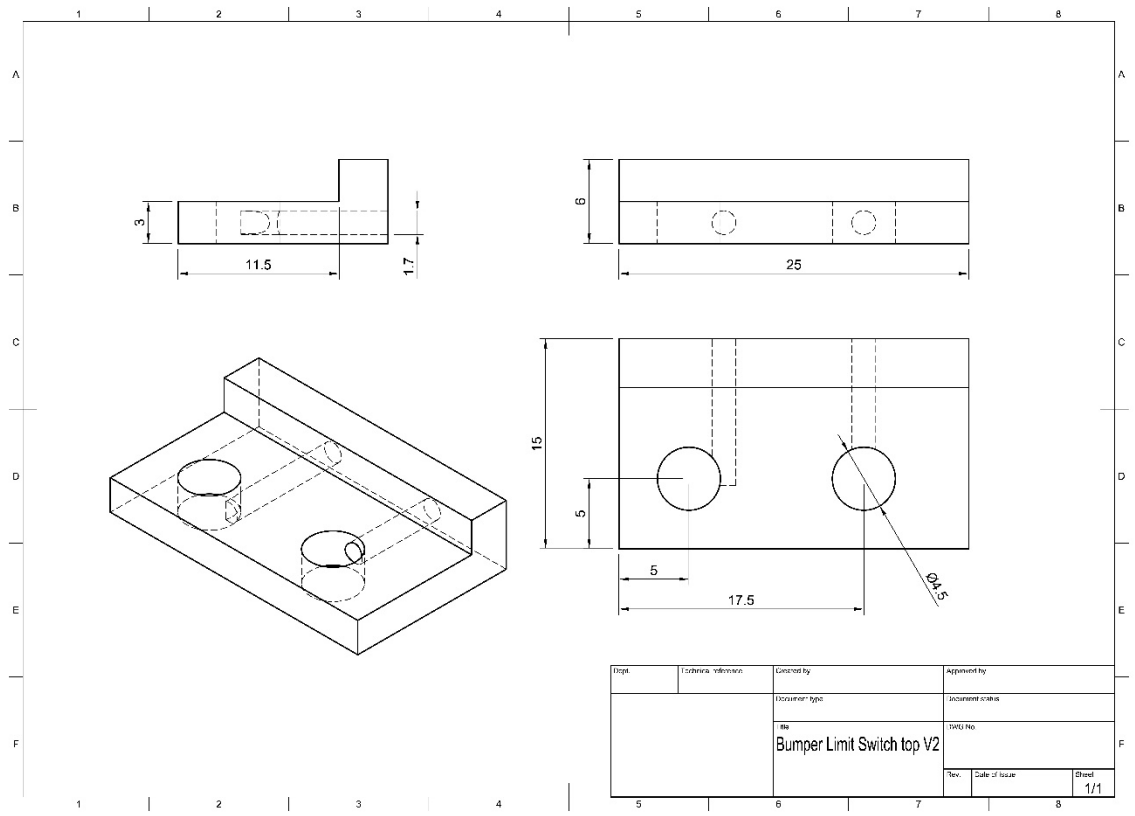


Figure 48: Bumper Limit Switch top V2

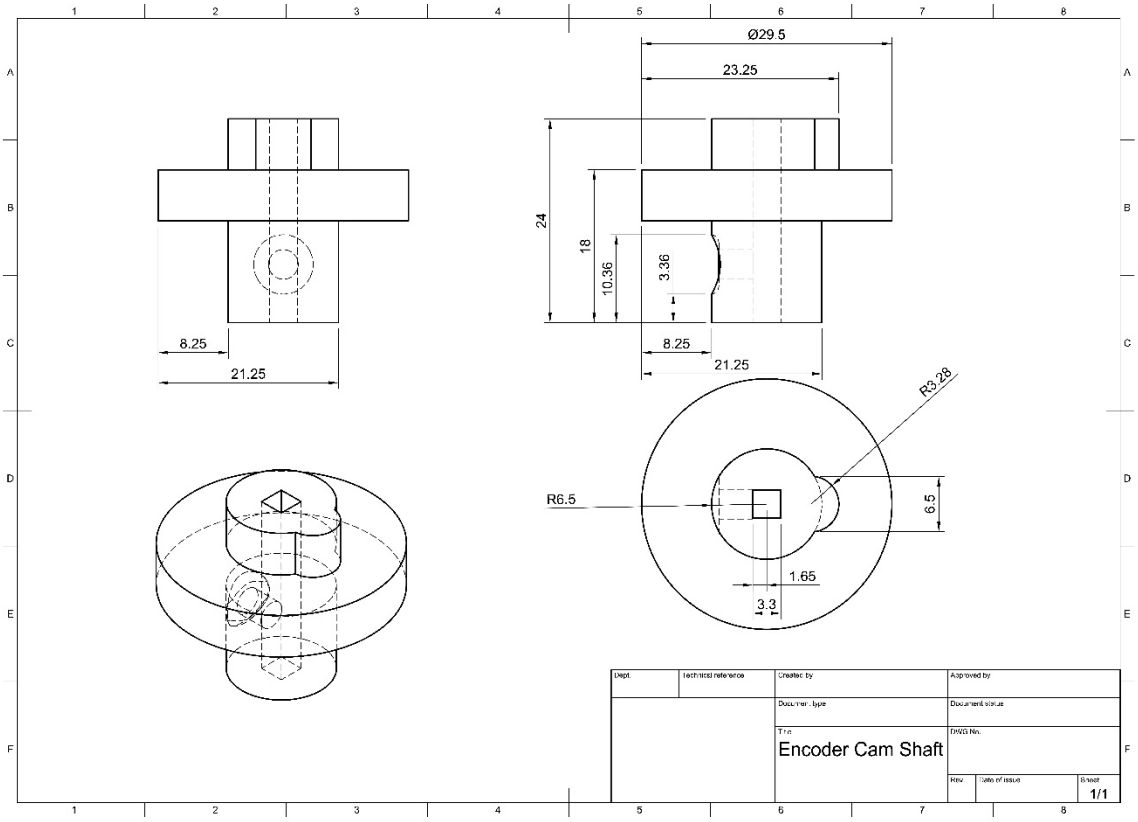


Figure 49: Encoder Cam Shaft

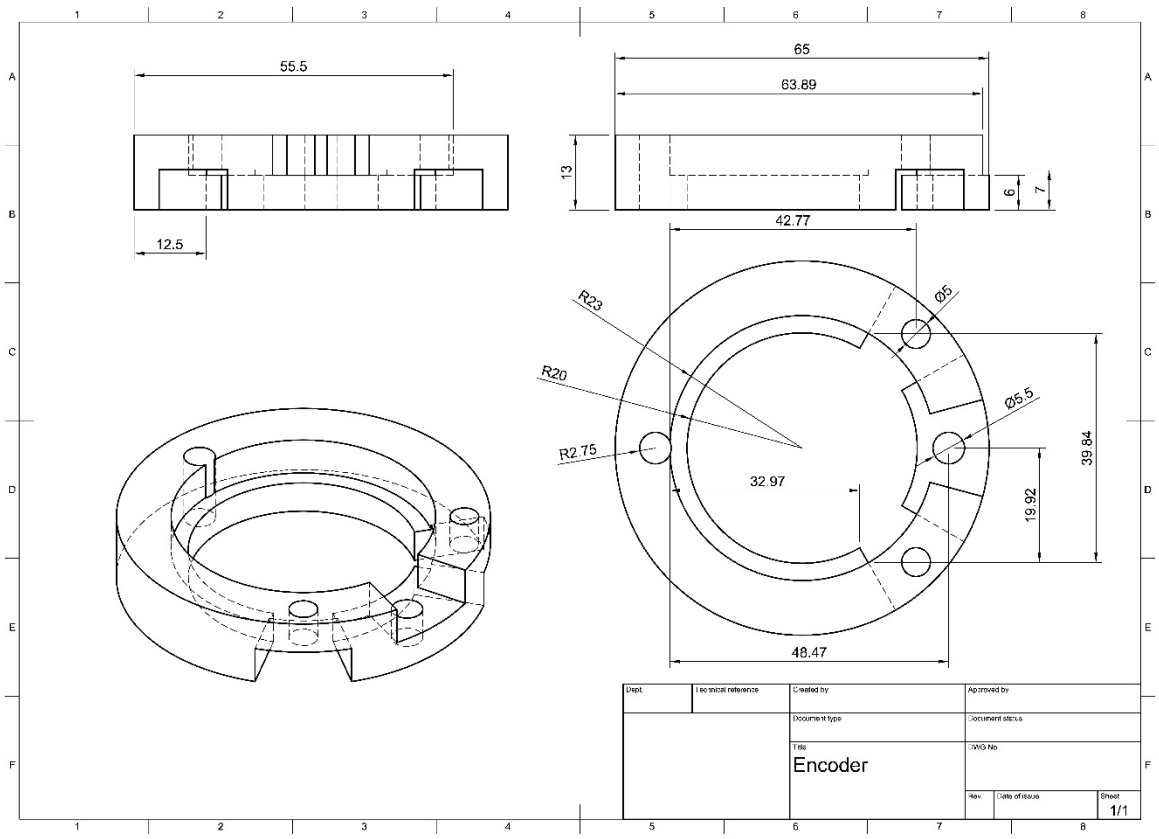


Figure 50: Encoder

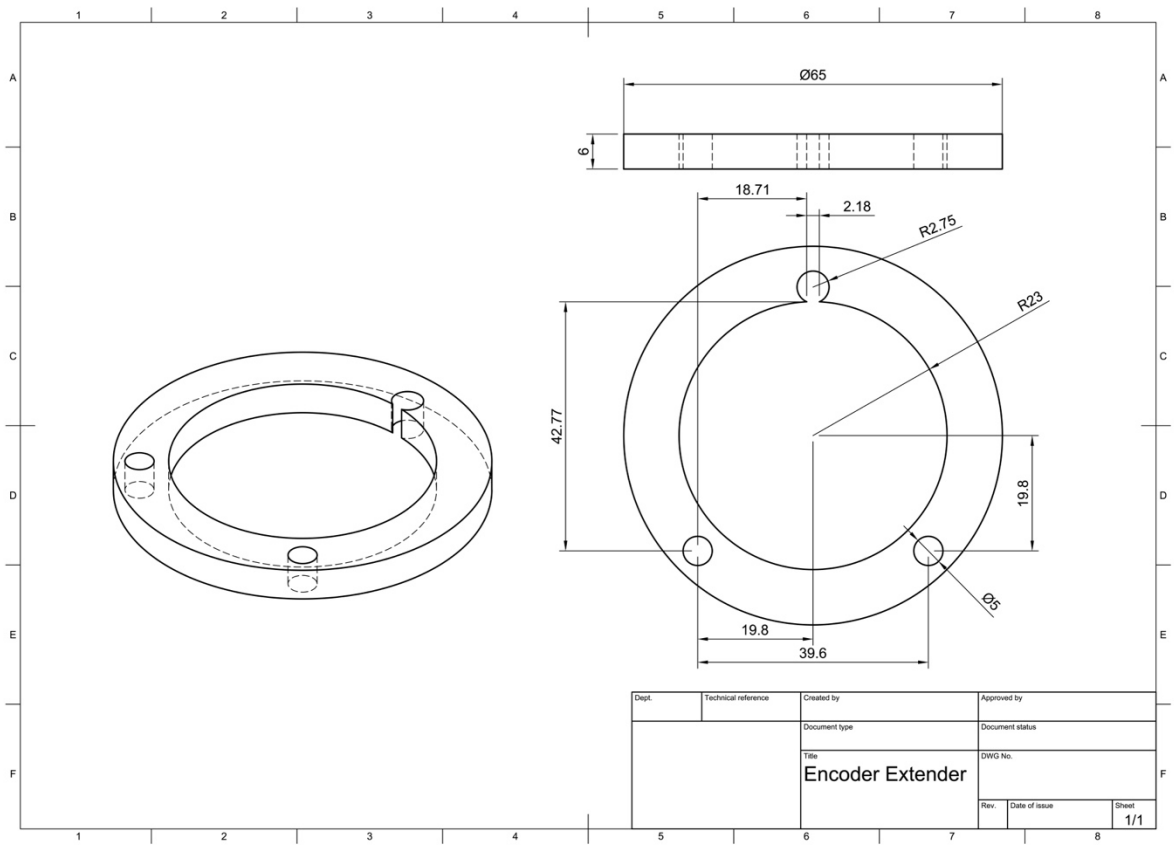


Figure 51: Encoder Extender

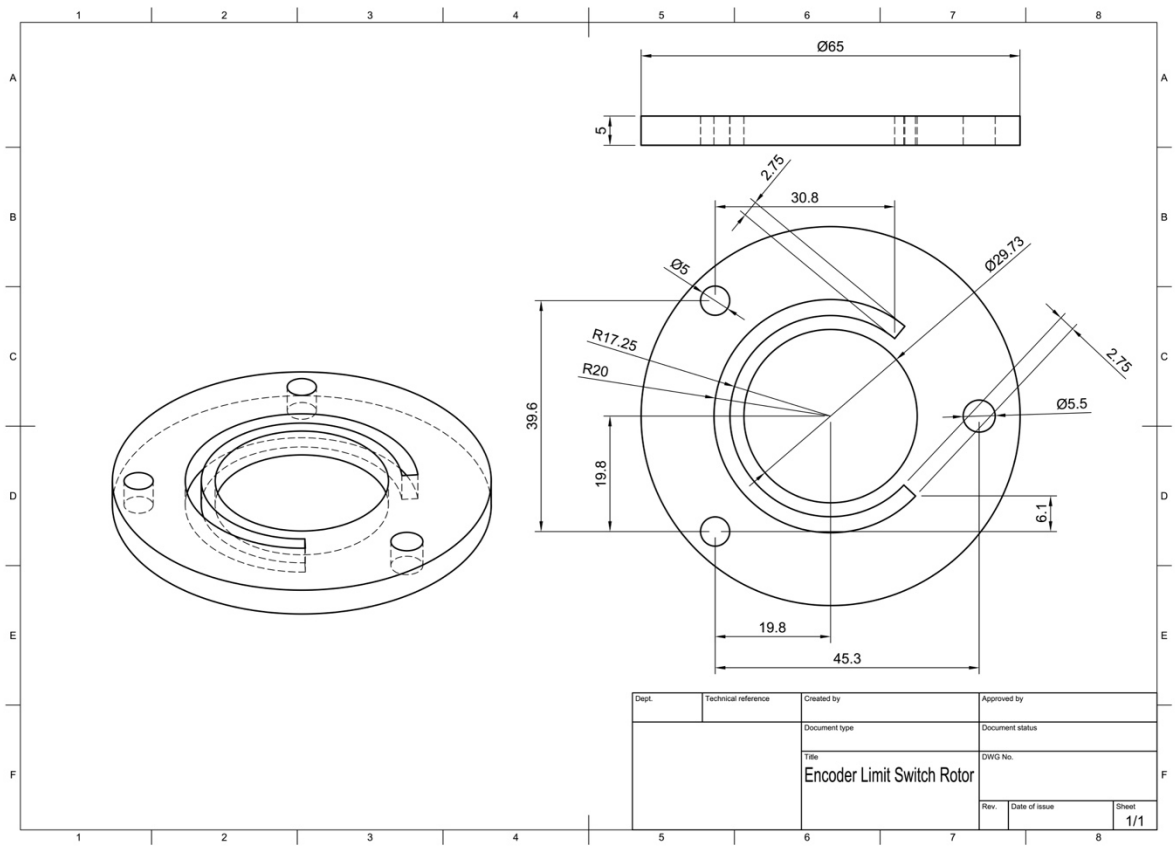


Figure 52: Encoder Limit Switch Rotor

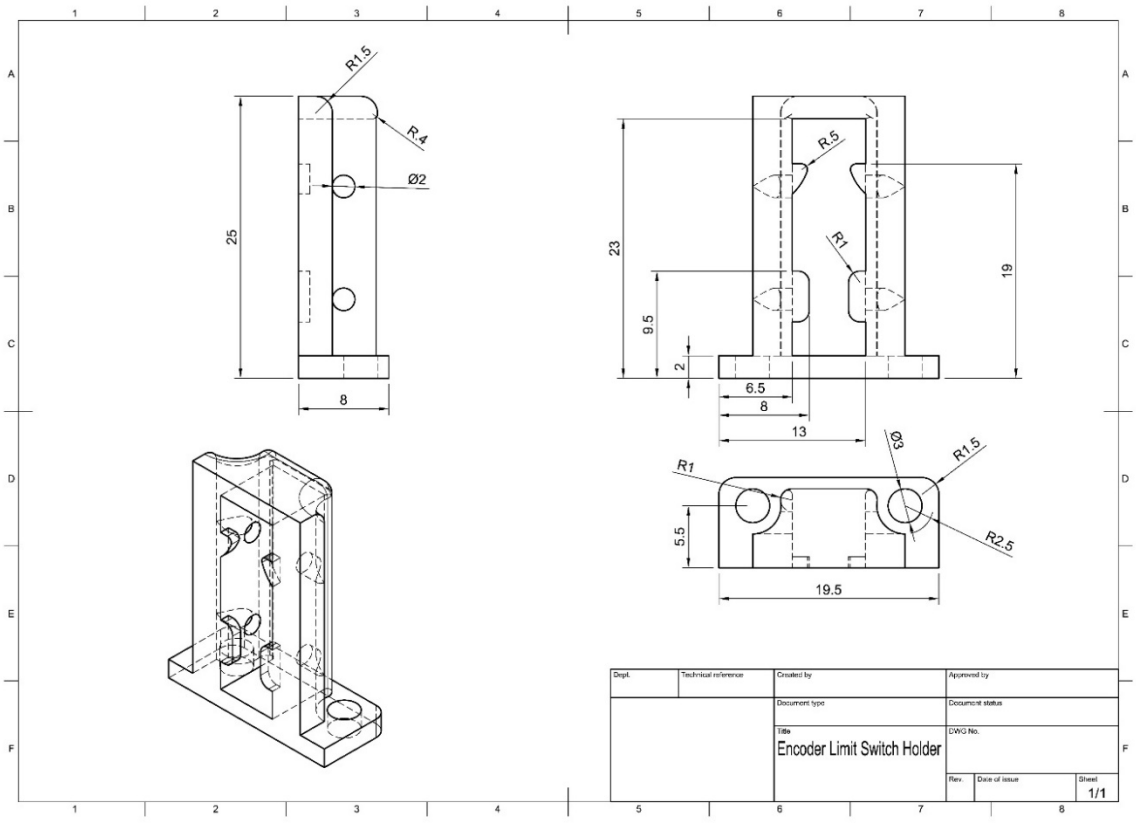


Figure 53: Encoder Limit Switch Holder

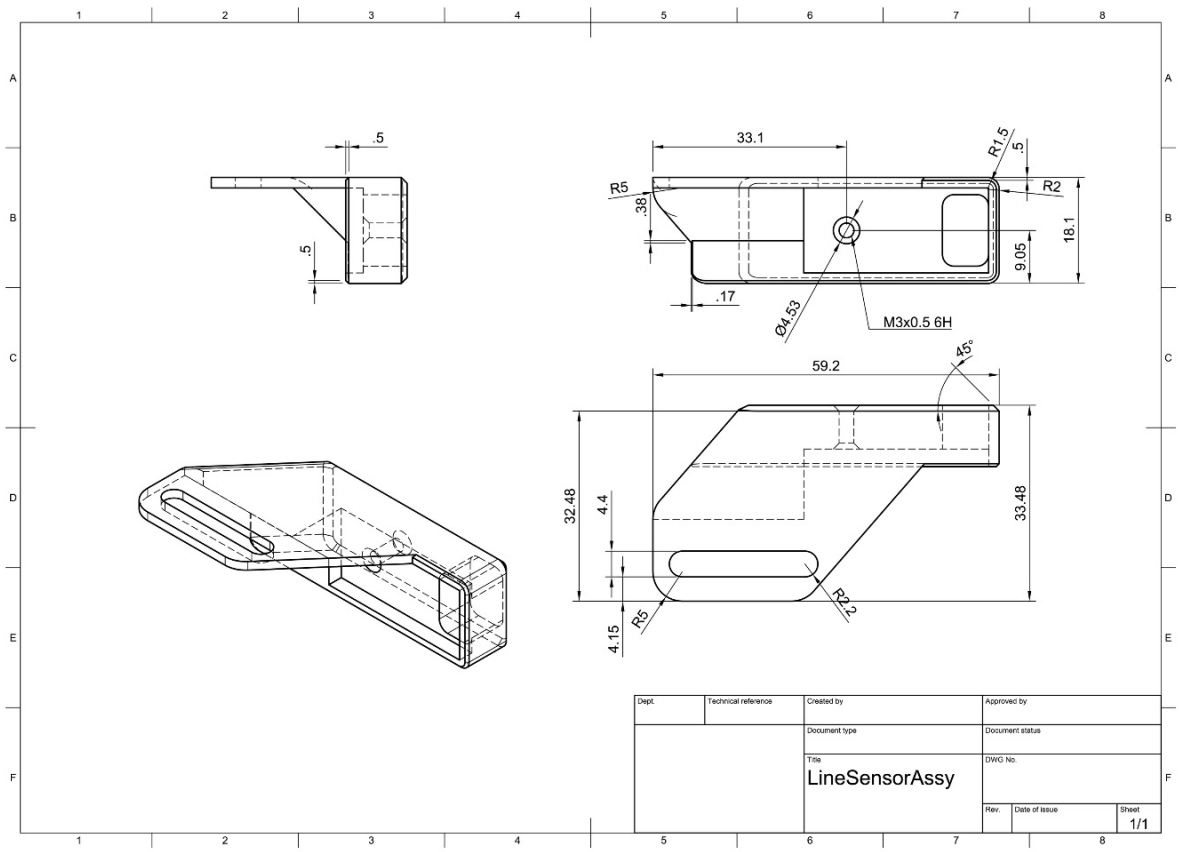


Figure 54: Line Sensor Assembly

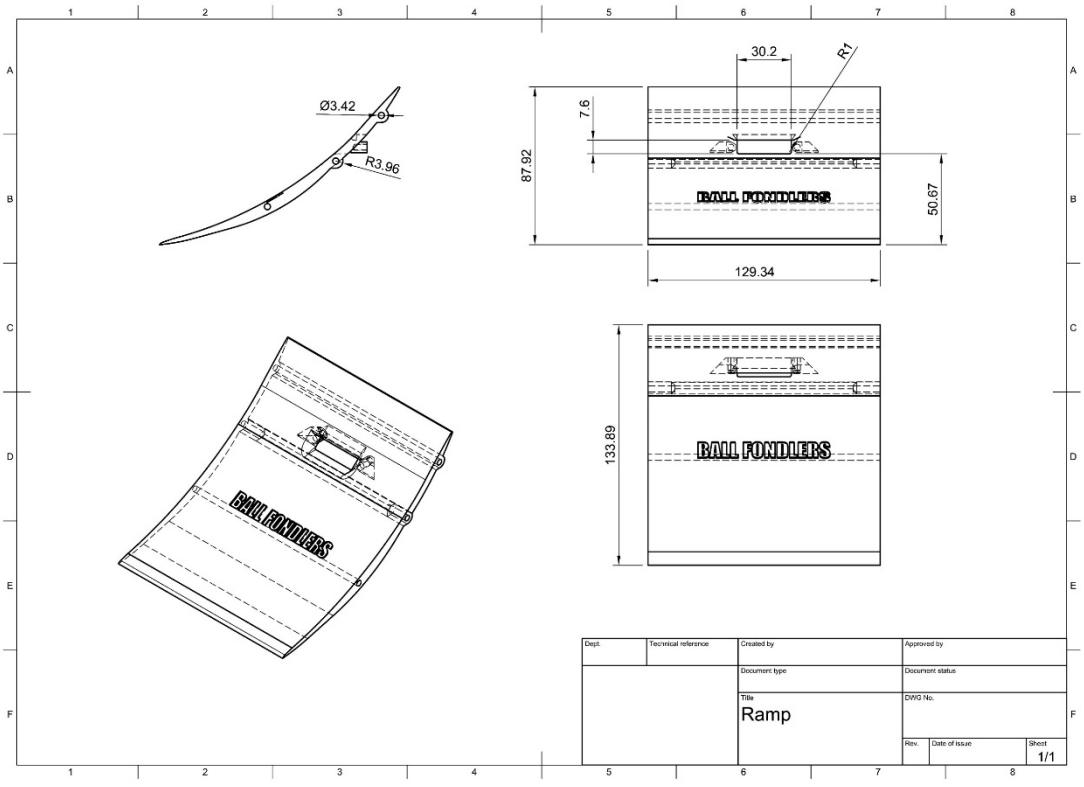


Figure 55: Ramp

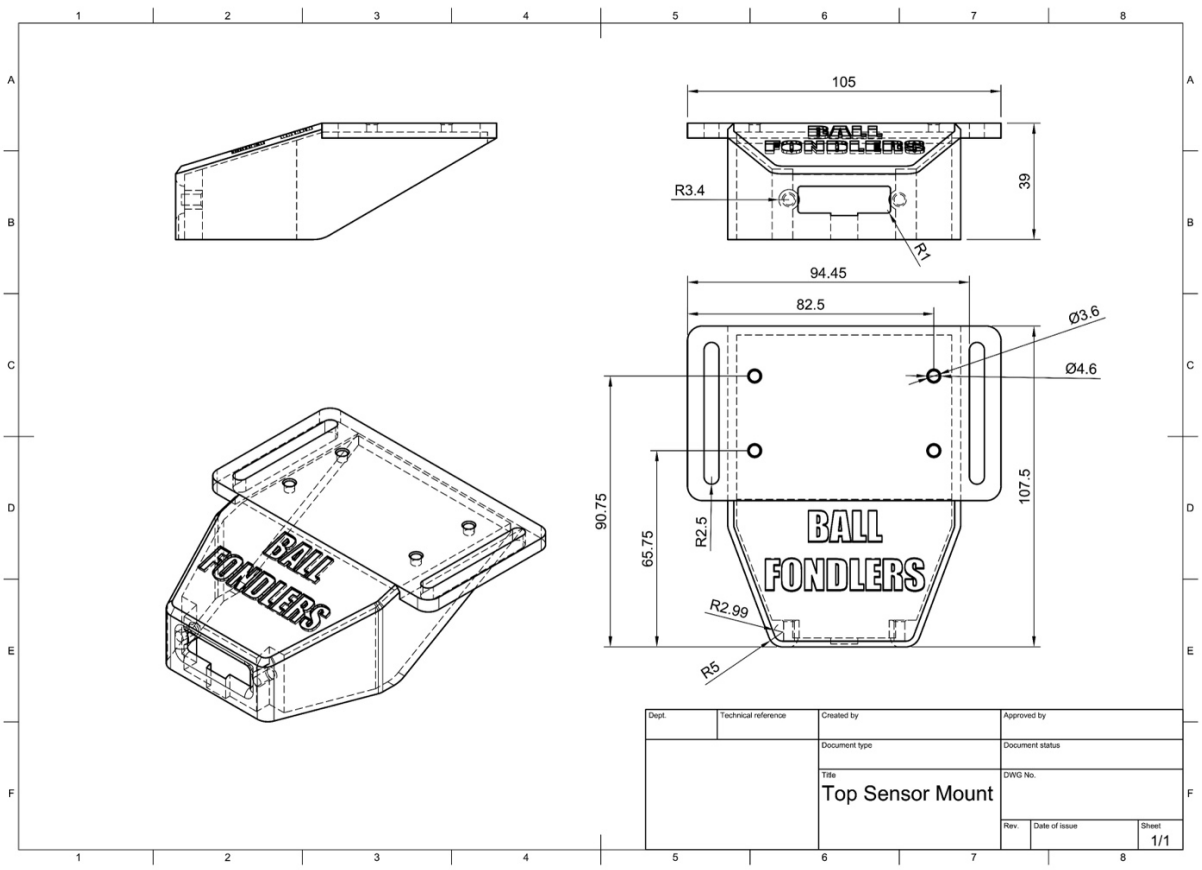


Figure 56: Top Sensor mount

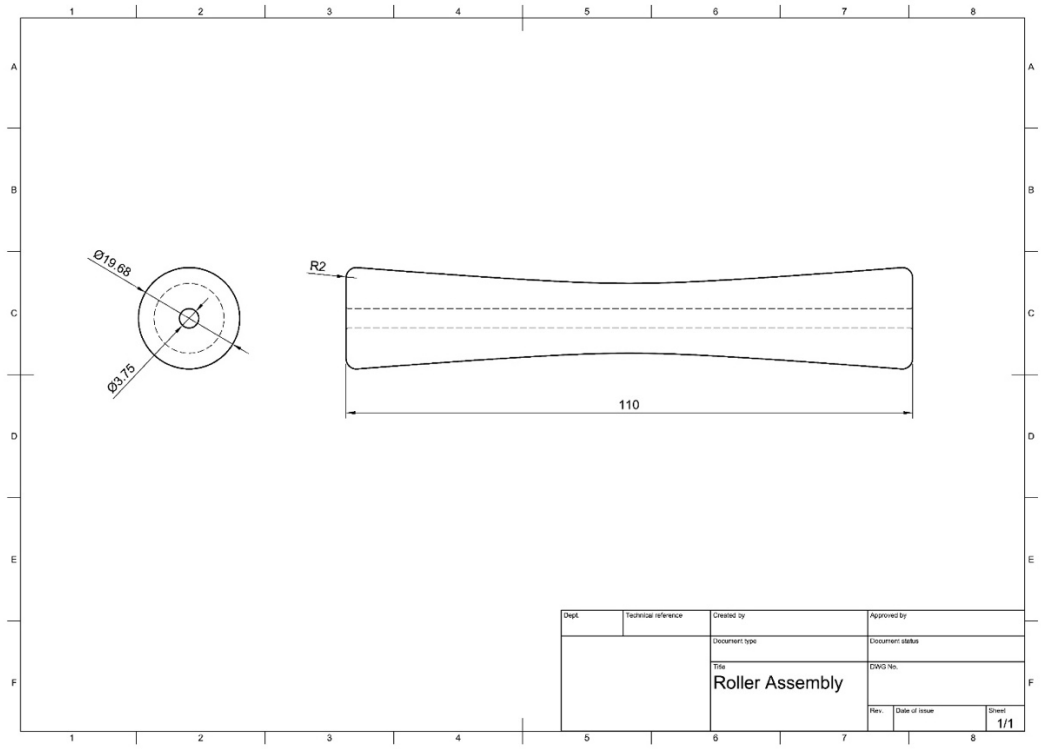


Figure 57: Roller